



Features

- AEC-Q100 qualified for automotive applications
- Full ISO26262 compliant, ASIL-B system ready
- 8 half-bridge outputs
- 4.5V to 28V operating voltage
 - 35V dc, 40V@400ms absolute maximum voltage
- 1-A RMS current for each output(1)
 - 6-A maximum current for paralleled outputs(2)
- Low-power sleep mode (1.5- μ A, $T_A = 25^\circ\text{C}$)
- Supports 3.3-V and 5-V logic inputs
- SPI for configuration and diagnostics
 - 5-MHz, 16-Bit SPI communication
 - Daisy chain functionality
- PWM generators programmable over SPI
 - Individual half-bridge PWM operation
 - Configurable for high-side, low-side, and H-bridge load driving
 - Supports 8-bit duty cycle resolution
- Integrated protection features with per channel detailed diagnostics over SPI
 - nFAULT pin output
 - VM undervoltage lockout (UVLO)
 - VM overvoltage protection (OVP)
 - Logic supply power on reset (POR)
 - Overcurrent protection (OCP)
 - Enhanced open load detection (OLD)
 - Thermal Warning and Shutdown
- Digital Thermal Clusters
- Limp Home Functionality for configuration as HS drivers or LS drivers.
- 2 parallel input pins with Input Mapping functionality



Description

The HE99MD9008 is a pin-to-pin compatible integrated multi-channel half-bridge drivers. These devices can drive brushed-DC (BDC) motors or stepper motors in independent, sequential, or parallel mode. The half-bridges are fully controllable to achieve a forward, reverse, coasting and braking operation of motor.

The integrated Serial Peripheral Interface (SPI) controls all outputs and provides diagnostic information: normal operation, open load in on-state, overcurrent, temperature warning and overtemperature.

Depending on the device, eight programmable PWM generators are integrated to allow for current limiting during motor operation or LED dimming control.

The device includes numerous protection and diagnostic features including an nFAULT pin to alert the system when a fault occurs. The device features a low-current open load detection (OLD) mode to detect open-load conditions when the nominal load current is small and a passive OLD mode for offline OLD. Digital Thermal Clusters are provided for a selective shutdown of a faulty channel; moreover, a limp home feature is included to provide safe operation of the device. All the outputs are fully protected from short-circuit, undervoltage, and over-temperature conditions.

Part number	Package	Marking	Packing
HE99MD9008RGAD	ETSSOP	HE99MD9008	2000 / Tape & Reel

*Not sure if you got the latest datasheet or not. Please kindly contact the sales of KKChips to confirm. Thanks.

(1) It is necessary to ensure that the chip does not trigger over-temperature protection.

(2) Only during the transient state.

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1 Block diagram and pin description

1.1 Block diagram

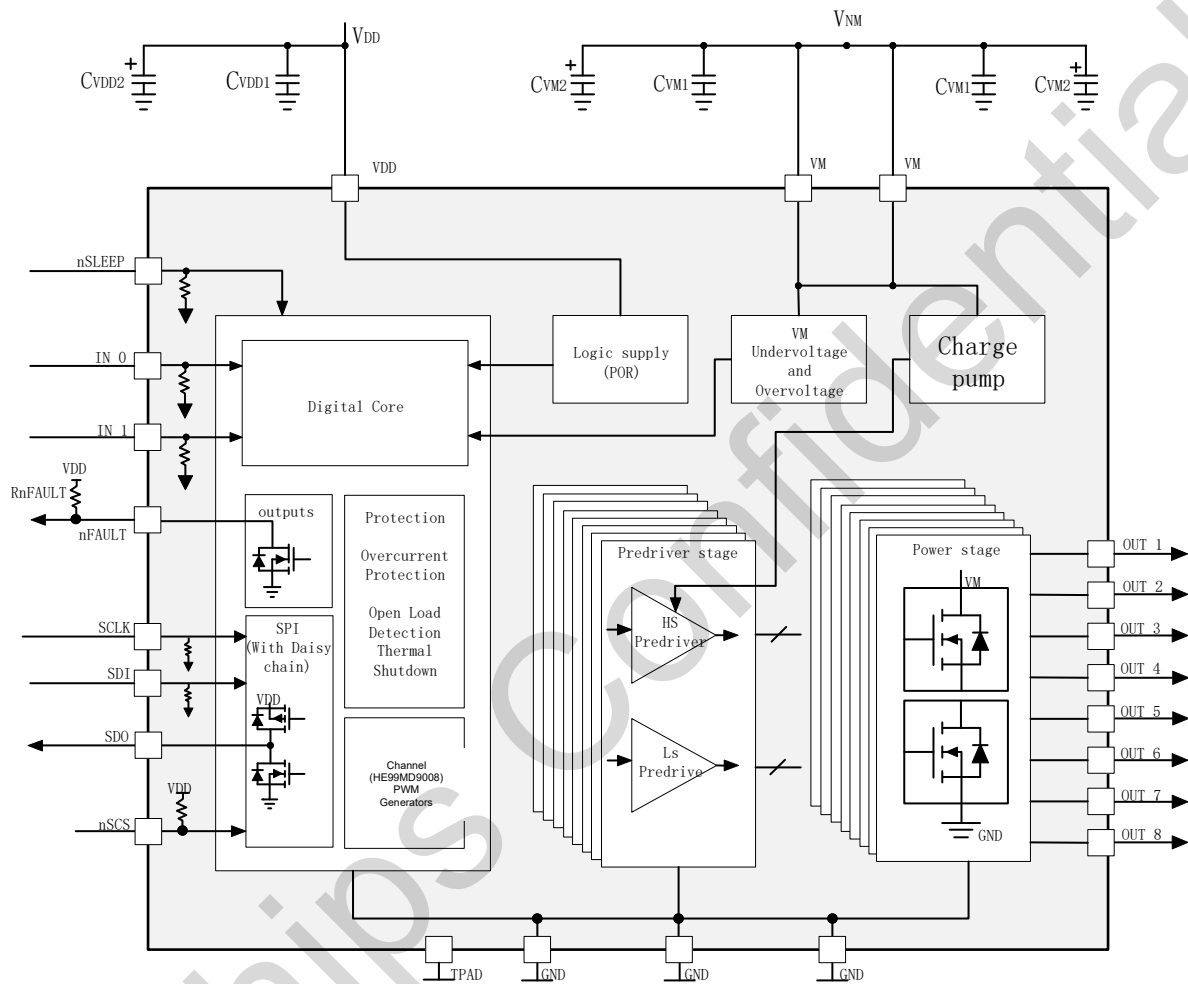


Figure 1-1 Block diagram

1.2 Pin description

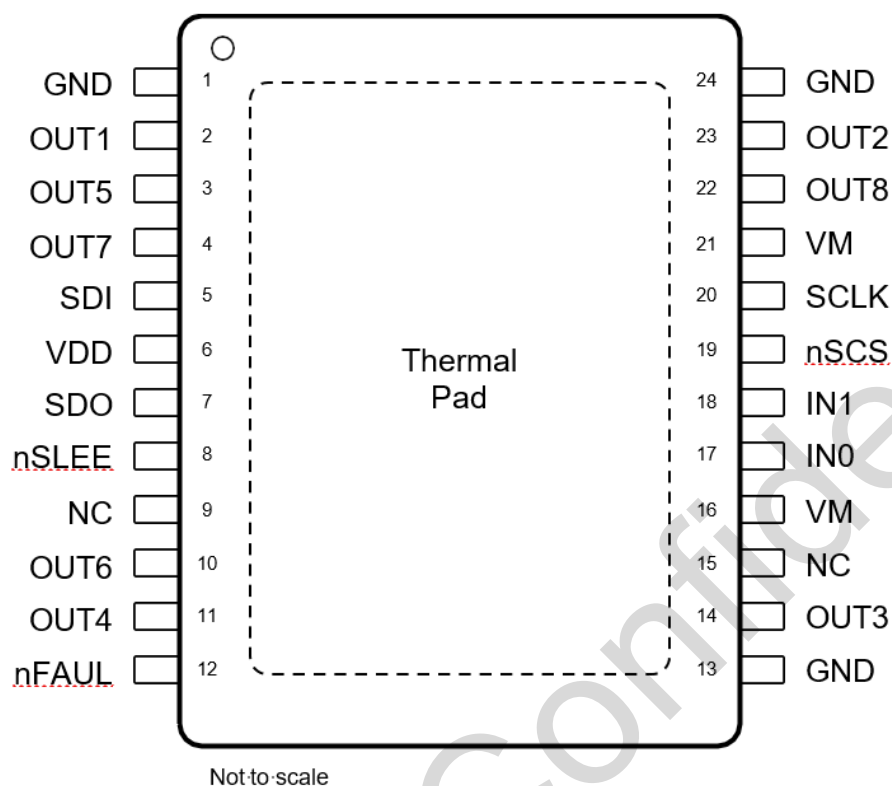


Figure 1-2 HE99MD9008 Thermal Pad Top View

Table 1-1 Pin Functions—HE99MD9008

PIN		TYPE	DESCRIPTION
NAME	NO.		
GND	13	PWR	Device power ground. Connect the GND pin to the system ground.
GND	24	PWR	Device power ground. Connect the GND pin to the system ground.
GND	1	PWR	Device power ground. Connect the GND pin to the system ground.
NC	9	-	Not connected
NC	15	-	Not connected
IN0	17	I	Parallel command input 0
IN1	18	I	Parallel command input 1
nFAULT	12	OD	Fault indicator output. This pin is pulled logic low during a fault condition and requires an external pull-up resistor.
nSCS	19	I	Serial chip select. A logic low on this pin enables serial interface communication. Internal pull-up.
nSLEEP	8	I	Driver enable pin. When this pin is logic low and also IN0, IN1 input pins at logic low, the device goes to a low-power sleep mode. When this pin is logic low while at least one of the input pins is at high state, the device enters limp home mode. Internal pull-down.
OUT1	2	O	Half-bridge 1 output

PIN		TYPE	DESCRIPTION
NAME	NO.		
OUT2	23	O	Half-bridge 2 output
OUT3	14	O	Half-bridge 3 output
OUT4	11	O	Half-bridge 4 output
OUT5	3	O	Half-bridge 5 output
OUT6	10	O	Half-bridge 6 output
OUT7	4	O	Half-bridge 7 output
OUT8	22	O	Half-bridge 8 output
SCLK	20	I	Serial clock input. Serial data is shifted out and captured on the corresponding rising and falling edge on this pin. Internal pull-down.
SDI	5	I	Serial data input. Data is captured on the falling edge of the SCLK pin. Internal pull-down.
SDO	7	O	Serial data output. Data is shifted out on the rising edge of the SCLK pin.
VDD	6	PWR	Logic power supply input. Connect a X5R or X7R, 0.1- μ F, VDD-rated ceramic capacitor and greater than or equal to 1- μ F bulk capacitance between the VDD and GND pins.
VM	16	PWR	Main power supply input. Connect all VM pins together to the motor supply voltage. Connect a X5R or X7R, 0.1- μ F, VM-rated ceramic capacitor and greater than or equal to 10- μ F bulk capacitance between the VM and GND pins.
VM	21	PWR	Main power supply input. Connect all VM pins together to the motor supply voltage. Connect a X5R or X7R, 0.1- μ F, VM-rated ceramic capacitor and greater than or equal to 10- μ F bulk capacitance between the VM and GND pins.

2 Electrical specifications

2.1 Absolute maximum ratings

Table 2-1 Absolute maximum ratings

	MIN	MAX	UNIT
Power supply pin voltage (VM)	-0.3	35 ⁽¹⁾	V
Logic supply pin voltage (VDD)	-0.3	5.75	V
Output pin voltage (OUTx)	-0.7	VM + 0.7	V
Logic pin input voltage (nSCS, nSLEEP, SCLK, SDI)	-0.3	VDD + 0.3	V
Command input pin voltage (IN0, IN1)	-0.3	35	V
Logic pin output voltage (nFAULT, SDO)	-0.3	VDD + 0.3	V
Continuous supply current (VM pins combined)	0	6	A
Peak output current drive (OUTx)	Internally Limited	Internally Limited	A
Continuous sink current (GND pins combined)	0	6	A
Junction temperature, T _J	-40	175	°C
Storage temperature, T _{stg}	-65	150	°C

(1) 35V at dc, 40V for t_≤400ms during load dump transient.

Note: All maximum ratings are absolute ratings. Leaving the limitation of any of these values may cause an irreversible damage of the integrated circuit!

2.2 ESD Ratings

Table 2-2 ESD Ratings

			VALUE	UNIT
V _(ESD)	Electrostatic discharge	Human body model (HBM), per AEC Q100-002 ⁽¹⁾	OUTx and VM pins	±4000
			Other pins	±2000
		Charged device model (CDM), per AEC Q100-011	Corner pins (1, 12, 13, and 24)	±750
			Other pins	±500

(1) AEC Q100-002 indicates that HBM stressing shall be in accordance with the ANSI/ESDA/JEDEC JS-001 specification.

2.3 Recommended Operating Conditions

over operating ambient temperature range (unless otherwise noted)

Table 2-3 Recommended operating conditions

		MIN	NOM	MAX	UNIT
V _{VM}	Power supply voltage (VM)	4.5		28	V
V _{DD}	Logic supply voltage (VDD)	3		5.5	V
V _{IN}	Logic input voltage (nSCS, nSLEEP, SCLK, SDI)	0		5.5	V
V _{OD}	Open drain pullup voltage (nFAULT)	0		5.5	V
I _{OD}	Open drain output current (nFAULT)	0		5	mA

Block diagram and pin description

V _{OP}	Push-pull pullup voltage (SDO)	0		5.5	V
I _{OP}	Push-pull output current (SDO)	0		5	mA
T _A	Operating ambient temperature	-40		125	°C
T _J	Operating junction temperature	-40		175	°C

Note: All parameters are guaranteed in the junction temperature range -40 to 150°C (unless otherwise specified); the device is still operative and functional at higher temperatures (up to 175°C). Note: Parameters limits at junction temperatures higher than 150°C may change with respect to what is specified as per the standard temperature range. Note: Device functionality at high junction temperature is guaranteed by characterization.

2.4 Thermal Information

Table 2-4 Thermal information

THERMAL METRIC(1)		HE99MD9008	UNIT
		(ETSSOP)	
		24 PINS	
R _{θJA}	Junction-to-ambient thermal resistance	31.2	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	25.4	°C/W
R _{θJB}	Junction-to-board thermal resistance	11.2	°C/W

2.5 Electrical Characteristics

At T_J = -40°C to +150°C, V_{VM} = 4.5 to 28 V (Main Supply), V_{VDD} = 3 to 5.5 V (Logic Supply) (unless otherwise noted). Typical limits apply for T_A = 25°C, V_{VM} = 13.5 V, V_{VDD} = 3.3 V

Table 2-5 Electrical characteristics

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
POWER SUPPLIES (VDD, VM)						
I _{VMQ}	VM sleep mode current	V _{VM} = 13.5 V, nSLEEP = 0, T _A = 25 °C		0.35	1	μA
		V _{VM} = 13.5 V, nSLEEP = 0, T _A = 125 °C			1	μA
I _{VDDQ}	VDD sleep mode current	V _{VM} = 13.5 V, V _{VDD} = 3.3 V, nSLEEP = 0, T _A = 25 °C		0.01	0.3	μA
		V _{VM} = 13.5 V, V _{VDD} = 3.3 V, nSLEEP = 0, T _A = 125 °C			2	μA
I _{VM}	VM operating mode current	V _{VM} = 13.5 V, nSLEEP = 1, All High- Side FETs = 'ON', T _A = 25 °C		7	15	mA
		V _{VM} = 13.5 V, nSLEEP = 1, All High- Side FETs = 'ON', T _A = 125 °C			15	mA
I _{VDD}	VDD operating mode current(3)	V _{VM} = 13.5 V, V _{VDD} = 3.3 V, nSLEEP = 1, All High-Side FETs = 'ON', SPI = 'ON' (5 MHz), T _A = 25 °C			1	mA
		V _{VM} = 13.5 V, V _{VDD} = 3.3 V, nSLEEP = 1, All High-Side FETs = 'ON', SPI = 'ON' (5 MHz), T _A = 125 °C			1	mA
t _{WAKE}	Wake-up time	nSLEEP high to SPI ready			300	μs
t _{SLEEP}	Turnoff time	nSLEEP low to device sleep			50	μs
LOGIC-LEVEL INPUTS (IN0, IN1, nSLEEP, SCLK, SDI)						
V _{IL}	Input logic low voltage				0.75	V
V _{IH}	Input logic high voltage		1.75			V
V _{HYS}	Input logic		0.1	0.5		V

Block diagram and pin description

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
	hysteresis					
I _{IL}	Input logic low current	V _{IN} = 0 V	-1		1	μA
RPD	Pull-down resistance		50	100	200	kΩ
C _{ID}	Input capacitance				15	pF
LOGIC-LEVEL INPUTS (nSCS)						
V _{IL}	Input logic low voltage				0.75	V
V _{IH}	Input logic high voltage		1.75			V
V _{HYS}	Input logic hysteresis		0.1	0.5		V
RPU	Pull-up resistance		50	100	200	kΩ
I _{IH}	Input logic high current	V _{IN} = V _{VDD}	-1		1	μA
C _{ID}	Input capacitance				15	pF
OPEN-DRAIN OUTPUTS (nFAULT)						
V _{OL}	Output logic low voltage	I _{OD} = 5 mA	0		0.4	V
I _{OH}	Output logic high current	V _{OD} = 5 V	-1		1	μA
C _{OD}	Output capacitance				15	pF
PUSH-PULL OUTPUTS (SDO)						
V _{OL}	Output logic low voltage	I _{OP} = 5 mA	0		0.4	V
V _{OH}	Output logic high voltage	I _{OP} = 5 mA	V _{DD} -0.6		V _{DD}	V
I _{OL}	Output logic low current	V _{OP} = 0 V	-1		1	μA
I _{OH}	Output logic high current	V _{OP} = V _{VDD}	-1		1	μA
C _{OD}	Output capacitance				30	pF
DRIVER OUTPUTS (OUTx)						
R _{DS(ON)}	High-side MOSFET on resistance	V _{VM} = 13.5 V, I _{OUT} = 0.5 A, T _A = 25°C		0.75	1.1	Ω
		V _{VM} = 13.5 V, I _{OUT} = 0.5 A, T _A = 125°C			1.5	Ω
	Low-side MOSFET on resistance	V _{VM} = 13.5 V, I _{OUT} = 0.5 A, T _A = 25°C		0.75	1.1	Ω
		V _{VM} = 13.5 V, I _{OUT} = 0.5 A, T _A = 125°C			1.5	Ω
SR	Output rise and fall time (high-side and low-side)	V _{VM} = 13.5 V, 10-90%, R _{LOAD} = 27 Ω, HBx SR = 0b		0.6		V/μs
		V _{VM} = 13.5 V, 10-90%, R _{LOAD} = 27 Ω, HBx SR = 1b		2.5		V/μs
t _{DEAD}	Output dead time (high to low / low to high)	V _{VM} = 13.5 V, SR = 0, HS/LS driver OFF to LS/HS driver ON	8	20	32	μs
		V _{VM} = 13.5 V, SR = 0, HS/LS driver OFF to LS/HS driver ON	2	5	15	μs
t _{PD_ON}	Propagation delay (high-side / low-side OFF)	High-side ON or low-side ON command (SPI last transition) to OUTx transition from Hi-Z state, SR = 0	5	12	25	μs
		High-side ON or low-side ON command (SPI last transition) to OUTx transition from Hi-Z state, SR = 1	3	5	10	μs

Block diagram and pin description

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
t_PD_OFF	Propagation delay (high-side / low-side OFF)	High-side OFF or low-side OFF command (SPI last transition) to OUTx transition from ON state, SR = 0		20		μs
		High-side OFF or low-side OFF command (SPI last transition) to OUTx transition from ON state, SR = 1	3	9	15	μs
I_LEAK	Leakage current low-side	V_OUTx = 13.5 V, nSLEEP = 1, SR = 0b		6	10	μA
		V_OUTx = 13.5 V, nSLEEP = 1, SR = 1b		6	10	μA
	Leakage current high-side	V_OUTx = 13.5 V, nSLEEP = 0		4	10	μA
		V_OUTx = 0 V, nSLEEP = 1			2	μA
		V_OUTx = 0 V, nSLEEP = 0			2	μA
PWM MODE						
f_PWM	PWM switching frequency	PWM_CHx FREQ = 00b	56	80	104	Hz
		PWM_CHx FREQ = 01b	70	100	130	Hz
		PWM_CHx FREQ = 10b	140	200	260	Hz
		PWM_CHx FREQ = 11b	1400	2000	2600	Hz
PROTECTION CIRCUITS						
V_UVLO	Supply undervoltage lockout (UVLO)	Supply rising	4.0		4.2	V
		Supply falling	3.8		4.0	V
V_UVLO_HYS	Supply undervoltage lockout hysteresis	Rising to falling threshold		200		mV
t_UVLO	Supply undervoltage deglitch time			10		μs
V_OVP	Supply overvoltage protection (OVP)	Supply rising, EXT_OVP = 0b	21		24	V
		Supply falling, EXT_OVP = 0b	20		23	V
		Supply rising, EXT_OVP = 1b	29		33	V
		Supply falling, EXT_OVP = 1b	28		32	V
V_OVP_HYS	Supply overvoltage protection hysteresis	Rising to falling theshold, EXT_OVP = 0b		1		V
		Rising to falling theshold, EXT_OVP = 1b		0.7		V
t_OVP	Supply overvoltage deglitch time			10		μs
V_VDDUV	Digital output supply undervoltage	Supply rising	2.45		3	V
		Supply falling	2.4		2.95	V
V_VDDUV_HYS	Digital output supply undervoltage hysteresis	Rising to falling threshold		75		mV
I_OCP	Overcurrent protection trip point(1) (2)		1.3	1.8	2.3	A
t_OCP	Overcurrent protection deglitch time	OCP_DEG = 000b	6	10	14	μs
		OCP_DEG = 001b	2.6	5	7.9	μs
		OCP_DEG = 010b	0.4	2.5	5.9	μs
		OCP_DEG = 011b	0.1	1	3.4	μs
		OCP_DEG = 100b	38.2	60	76.6	μs
		OCP_DEG = 101b	23.1	40	51.9	μs
		OCP_DEG = 110b	18.5	30	41.5	μs
		OCP_DEG = 111b	8.4	20	31.6	μs
I_OLD	Open load detection current	Current flowing from VM to OUTx (High- Side = ON) or OUTx to GND (Low-Side = ON)	2	9	18	mA

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
IOLD_NEG	Negative open load detection current	Current flowing from OUTx to VM (High- Side = ON) or GND to OUTx (Low-Side = ON)	2	15	30	mA
IOLD_LOW	Open load detection current in low current OLD mode	Current flowing from VM to OUTx (High- Side = ON) or OUTx to GND (Low-Side = ON)	0.2	0.8	2	mA
IOL_GND	Passive OLD current	HE99MD9008, FETs in Hi-Z state, current from OUTx to GND during OLD trip		113		μA
VOL_GND	Passive OLD voltage threshold	HE99MD9008, FETs in Hi-Z state, voltage at OUTx during OLD trip for GND-connected load		3.3		V
IOL_VM	Passive OLD current	HE99MD9008, FETs in Hi-Z state, current from VM to OUTx for OLD trip, HBX_VM_POLD = 0b		118		μA
VOL_VM	Passive OLD voltage threshold	HE99MD9008, FETs in Hi-Z state, voltage at OUTx during OLD trip for VM-connected load, HBX_VM_POLD = 0b		1.2		V
IOL_VM	Passive OLD current	HE99MD9008, FETs in Hi-Z state, current from VM to OUTx for OLD trip, HBX_VM_POLD = 1b		547		μA
VOL_VM	Passive OLD voltage threshold	HE99MD9008, FETs in Hi-Z state, voltage at OUTx during OLD trip for VM-connected load, HBX_VM_POLD = 1b		1.8		V
ROL	Passive OLD detect resistance threshold	HE99MD9008, FETs in Hi-Z state, Full bridge connection	5		100	kΩ
ROL	Passive OLD detect resistance threshold	HE99MD9008, FETs in Hi-Z State, Load connected to GND	5		100	kΩ
ROL	Passive OLD detect resistance threshold	HE99MD9008, FETs in Hi-Z State, Load connected to VM, HBX_VM_POLD = 0b	5		400	kΩ
ROL	Passive OLD detect resistance threshold	HE99MD9008, FETs in Hi-Z State, Load connected to VM, HBX_VM_POLD = 1b	5		100	kΩ
tOLD	Open load deglitch time	Active OLD (Continuous Mode)	2	3	4	ms
tOLD	Open load deglitch time	Active OLD (PWM Mode)	150	200	300	μs
TOTW	Thermal warning temperature	Die temperature (T _j)	120	140	170	°C
TOTW_HYS	Thermal warning hysteresis	Die temperature (T _j)		20		°C
TOTSD	Thermal shutdown temperature	Die temperature (T _j)	150	175	200	°C
TOTSD_HYS	Thermal shutdown hysteresis	Die temperature (T _j)		20		°C

Block diagram and pin description

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
V_{HSLS_TH}	Auto-configuration HS/LS detection voltage threshold		1.6		2	V
T_{HSLS}	Auto-configuration HS/LS detection timing window during start up			100		μs
T_{HSLS_DEG}	Auto-configuration HS/LS detection deglitch filter time				20	μs

(1) For $20V < V_{VM} < 28V$, the OCP deglitch time must be limited to 10- μs (Default Deglitch Value, OCP_DEG = 000b).

(2) For $V_{VM} > 28V$, the OCP deglitch time must be limited to 1- μs (Lowest Deglitch Value, OCP_DEG = 011b).

(3) For the definition of SPI=ON (5MHz): SDO = low, SCLK frequency = 5MHz, SDI = low, nSCS = low.

2.6 Timing Requirements

Table 2-6 Timing requirements

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
SPI (nSCS, SCLK, SDI, SDO)					
t _{READY}	SPI ready after after enable VM > UVLO, nSLEEP = 3.3 V			1	ms
t _{CLK}	SCLK minimum period	200			ns
t _{CLKH}	SCLK minimum high time	100			ns
t _{CLKL}	SCLK minimum low time	100			ns
t _{SU_SDI}	SDI input data setup time	40			ns
t _{HD_SDI}	SDI input data hold time	60			ns
t _{DLY_SDO}	SDO output data delay time SCLK high to SDO valid			60	ns
t _{SU_nSCS}	nSCS input setup time	100			ns
t _{HD_nSCS}	nSCS input hold time	100			ns
t _{HI_nSCS}	nSCS minimum high time before active low	600			ns
t _{DIS_nSCS}	nSCS disable delay time nSCS high to SDO high impedance		30		ns
t _{SC_SPI}	Successive SPI write gaps		2.5		μs

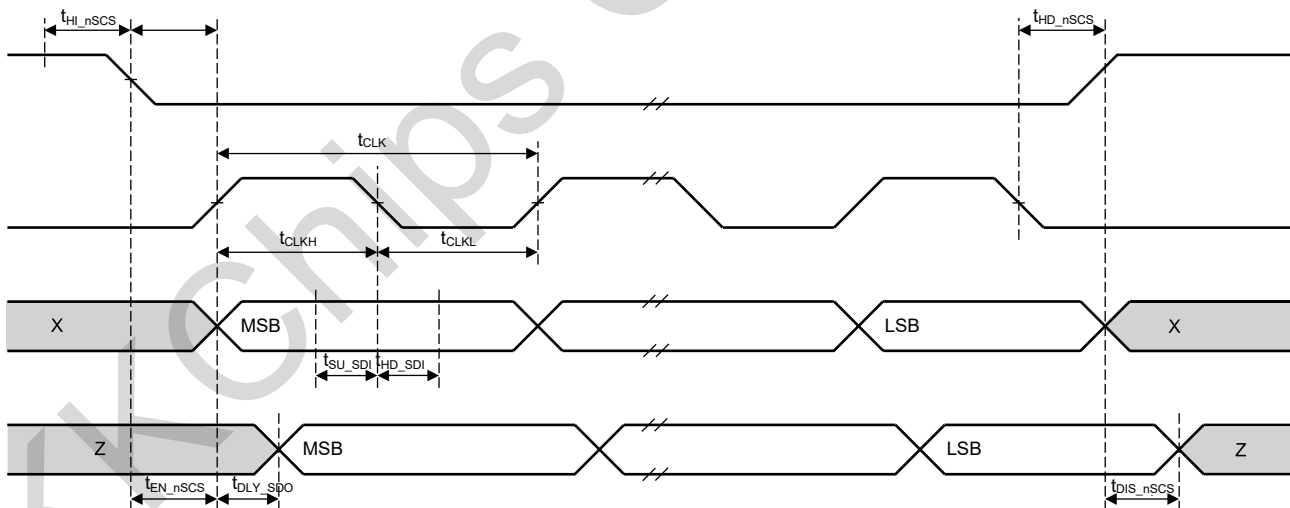


Figure 2-1 SPI Timing

2.7 Typical Characteristics

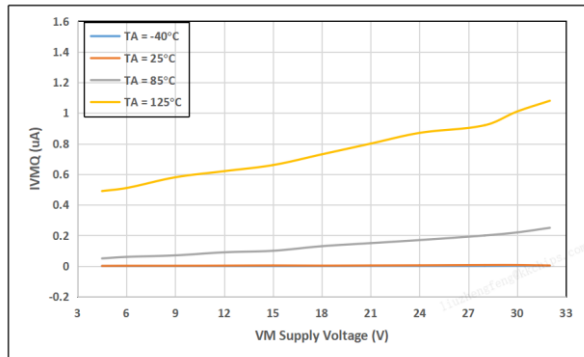


Figure 2-2 VM Sleep Mode Current (VVMQ) vs Supply Voltage(VVM)

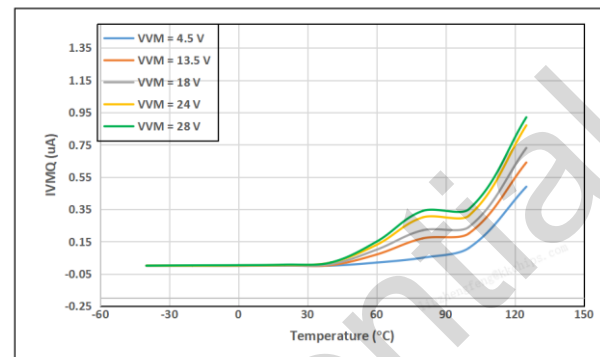


Figure 2-3 VM Sleep Mode Current (VVMQ) vs Ambient Temperature (TA)

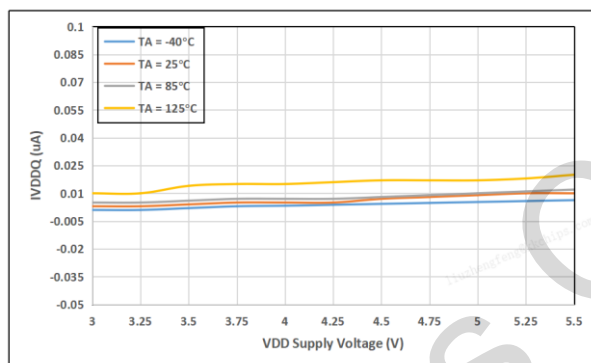


Figure 2-4 VDD Sleep Mode Current (IVDDQ) vs Supply Voltage (VVDD)

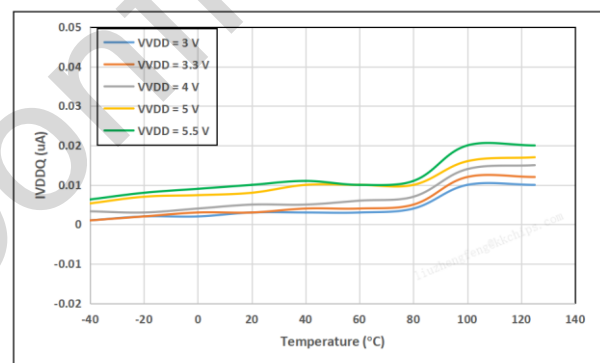


Figure 2-5 VDD Sleep Mode Current (IVDDQ) vs Ambient Temperature (TA)

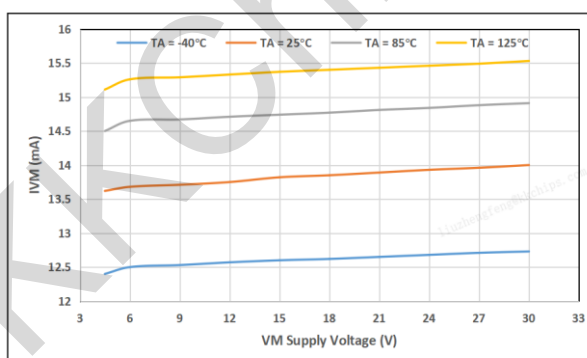


Figure 2-6 VM Operating Mode Current (I_{VM}) vs Supply Voltage (V_{VM})

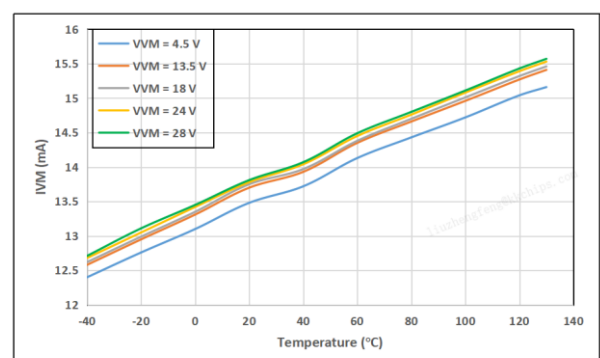


Figure 2-7 VM Operating Mode Current (I_{VM}) vs Ambient Temperature (T_A)

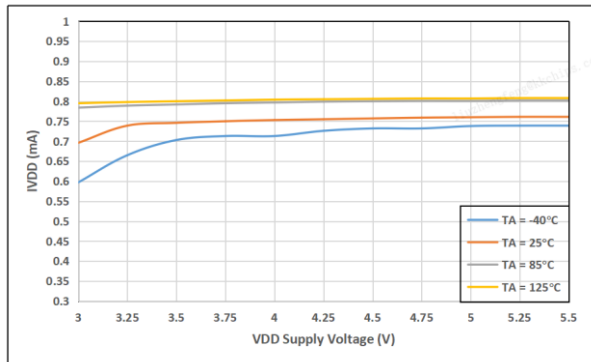


Figure 2-8 VDD Operating Mode Current (I_{VDD}) vs Supply Voltage (V_{VDD})

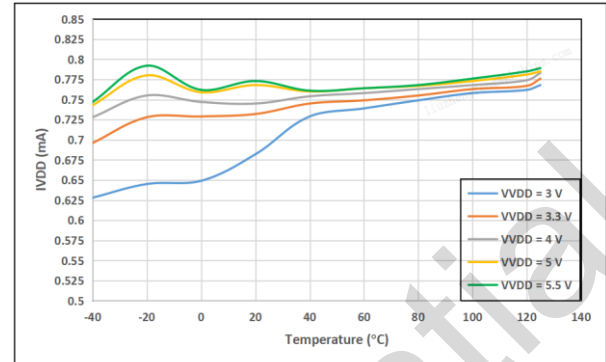


Figure 2-9 VDD Operating Mode Current (I_{VDD}) vs Ambient Temperature (T_A)

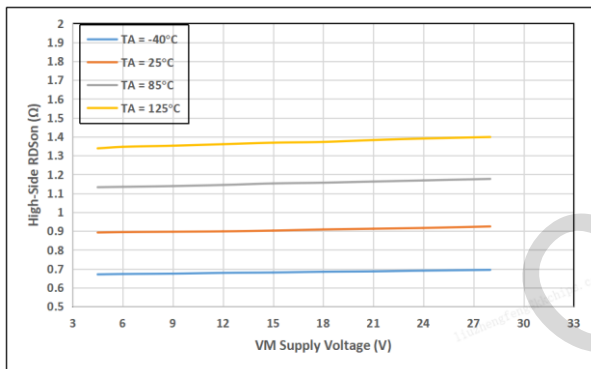


Figure 2-10 High Side On-State Resistance ($R_{DS(on)}$) vs Supply Voltage (V_{VM})

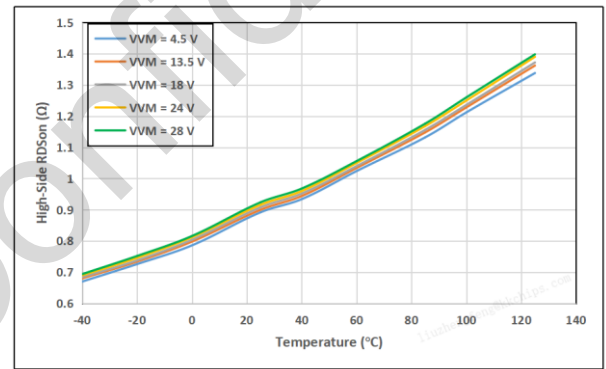


Figure 2-11 High Side On-State Resistance ($R_{DS(on)}$) vs Ambient Temperature (T_A)

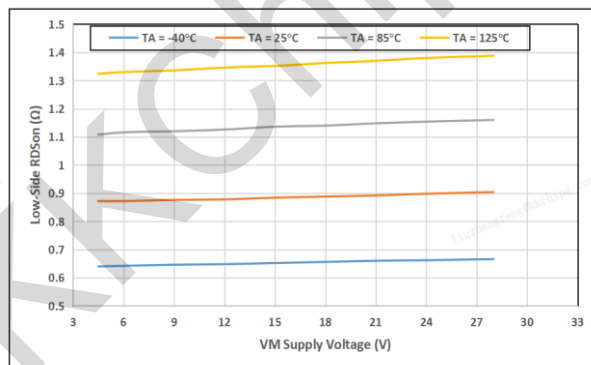


Figure 2-12 Low Side On-State Resistance ($R_{DS(on)}$) vs Supply Voltage (V_{VM})

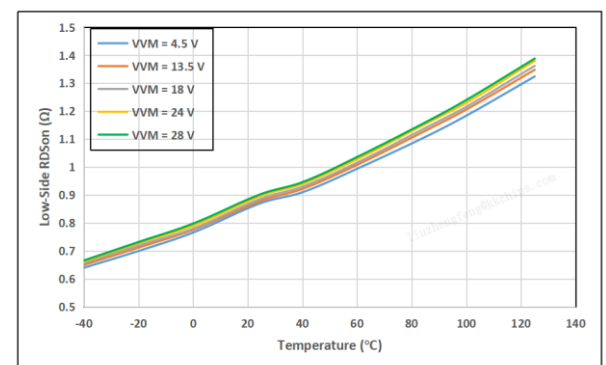


Figure 2-13 Low Side On-State Resistance ($R_{DS(on)}$) vs Ambient Temperature (T_A)

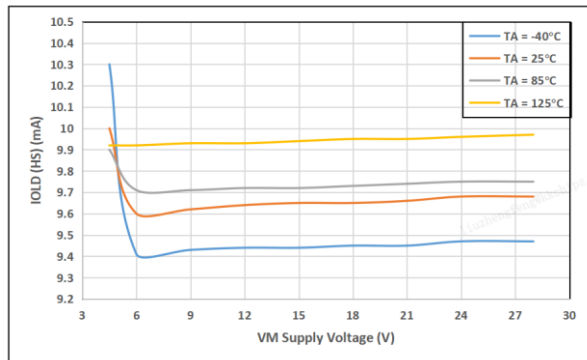


Figure 2-14 High-Side Open-Load Detection Current (I_{OL_D}) vs Supply Voltage (V_{VM})

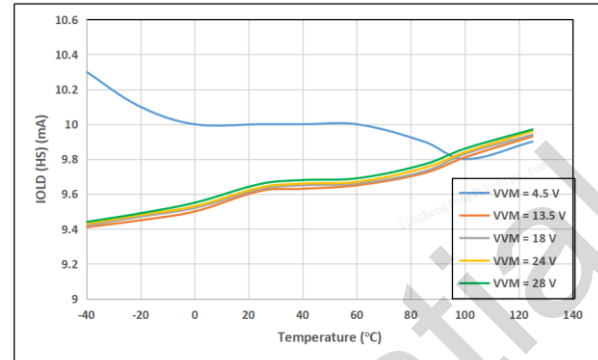


Figure 2-15 High-Side Open-Load Detection Current (I_{OL_D}) vs Ambient Temperature (T_A)

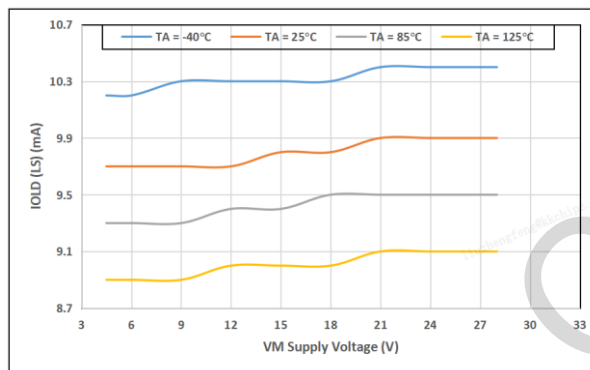


Figure 2-16 Low-Side Open-Load Detection Current (I_{OL_D}) vs Supply Voltage (V_{VM})

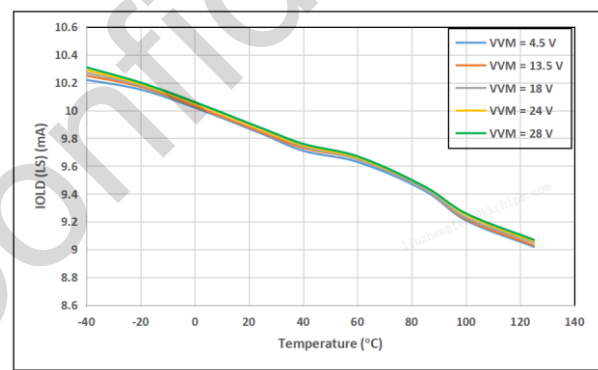


Figure 2-17 Low-Side Open-Load Detection Current (I_{OL_D}) vs Ambient Temperature (T_A)

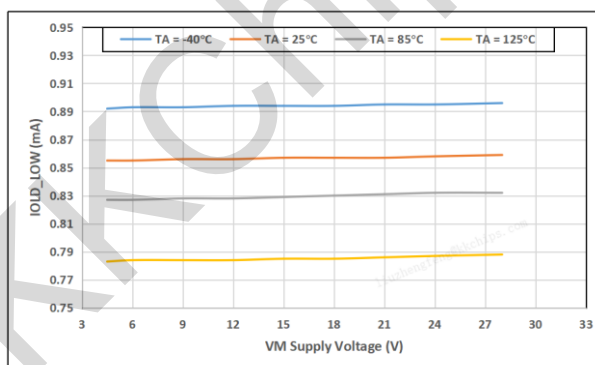


Figure 2-18 Low Open-Load detection Current ($I_{OL_D_LOW}$) vs Supply Voltage (V_{VM})

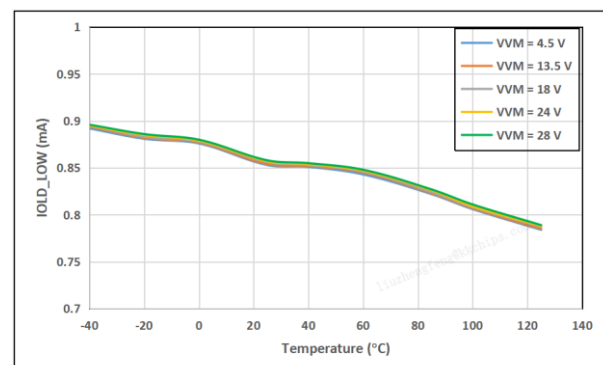


Figure 2-19 Low Open-Load detection Current ($I_{OL_D_LOW}$) vs Ambient Temperature (T_A)

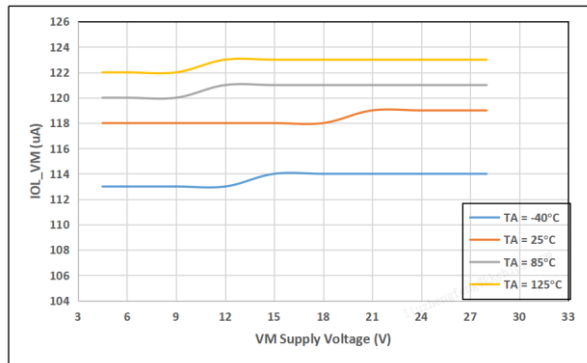


Figure 2-20 Open-load pull-down current (I_{OL_VM}) when HBX_VM_POLD = 0b vs supply voltage (V_{VM})

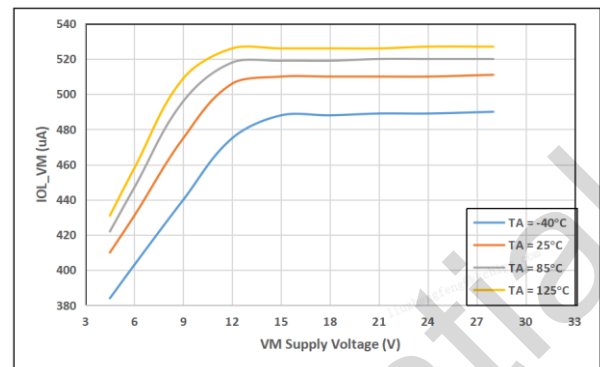


Figure 2-21 Open-load pull-down current (I_{OL_VM}) when HBX_VM_POLD = 1b vs supply voltage (V_{VM})

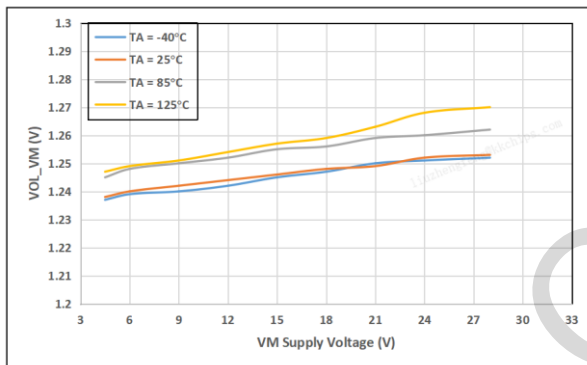


Figure 2-22 Open-load low-side threshold voltage (V_{OL_VM}) when HBX_VM_POLD = 0b vs supply voltage (V_{VM})

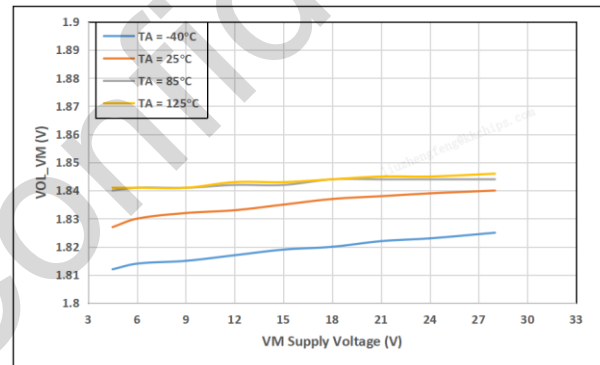


Figure 2-23 Open-load low-side threshold voltage (V_{OL_VM}) when HBX_VM_POLD = 1b vs supply voltage (V_{VM})

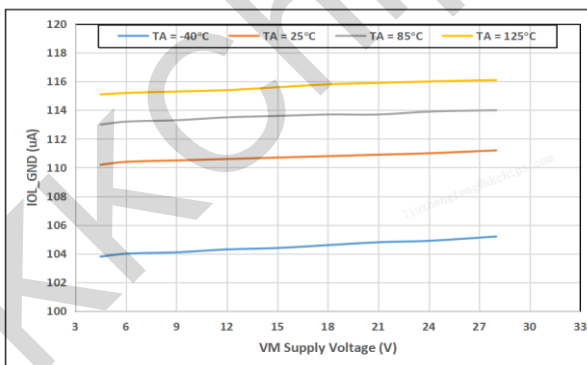


Figure 2-24 Open-load pull-up current (I_{OL_GND}) vs supply voltage (V_{VM})

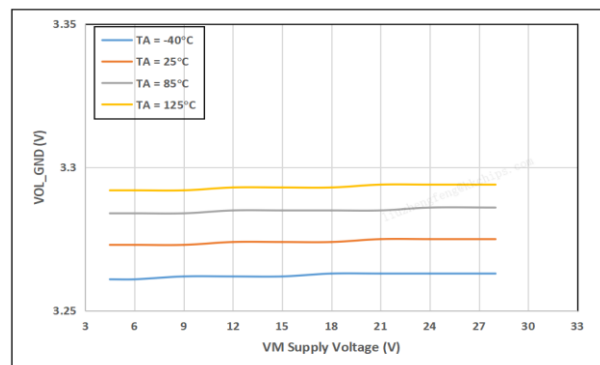


Figure 2-25 Open-load low-side threshold voltage (V_{OL_GND}) vs supply voltage (V_{VM})

3 Detailed Description

3.1 Overview

The HE99MD9008 are 4.5V to 28V integrated multi half-bridge drivers which supports a maximum voltage of 40V for 400ms in case of load-dump scenario. The half-bridges are designed to support 1-A per half-bridge and 6-A from the VM/GND pins. The HE99MD9008 offers drivers from 8 half-bridge outputs.

A standard 16-bit, 5-MHz serial peripheral interface (SPI) provides a simple method for configuring the various device settings and reading fault diagnostic information through an external controller. The device is also equipped with a daisy-chain functionality which allows connecting multiple devices using a single nSCS line and saving on multiple resources.

This device has 8 internal PWM generators which can be mapped to any of the half-bridge through SPI registers. The PWM frequency (4 options) and duty (8-bit resolution) for each channel can be selected using the SPI registers. This PWM mode is useful for implementing the current control of motor or dimming control of LEDs. Alternatively, the devices can be driven by external PWM signals. Limp home functionality is also featured, which allows using 2 selected drivers in particularly faulty conditions, such as SPI fault, micro fault or supply UV.

The device also has numerous integrated protection features which protects the device in case of any abnormal scenario. The over-current protection (OCP) ensures the device protection in any short scenarios like the phase short, phase to ground short and phase to supply short conditions. Undervoltage lockout (UVLO) and overvoltage protection (OVP) ensures the driver operation in fluctuating voltages to support the crank-start and load-dump scenario in automotive applications; the OVP thresholds can be configured via two thresholds, and via bit OVP_OP is possible to configure device action in case of an OV event→all outputs off with a status bit flag or just a status bit flag. In addition to this, the open-load detection (OLD) feature ensure the proper load connection. All devices support active OLD, low-current OLD, and negative-current OLD. Passive OLD is only supported on HE99MD9008 devices. Device faults are indicated on the nFAULT pin, and detailed information is available in the device SPI registers.

The device integrates a spread spectrum clocking feature for both the internal digital oscillator and internal charge pump. This feature combined with programmable output slew-rate control minimizes the radiated emissions from the device.

The device is available in a 24-pin ETSSOP package with a thermal pad.

3.2 Feature Description

Table 3-1 lists the recommended values of the external components for the driver.

Table 3-1 HE99MD9008 driver external components

COMPONENTS	PIN 1	PIN 2	RECOMMENDED
CVM1	VM	GND	X5R or X7R, 0.1- μ F, VM-rated capacitor
CVM2	VM	GND	$\geq 10 \mu$ F, VM-rated capacitor
CVDD1	VDD	GND	X5R or X7R, 0.1- μ F, 6.3-V capacitor
CVDD2	VDD	GND	$\geq 1 \mu$ F, 6.3-V capacitor
RnFAULT	VDD	nFAULT	Pullup resistor

3.2.1 Input/Output

3.2.1.1 Parallel inputs (IN0, IN1)

Device has two input pins available. Each input pin is connected by default to one channel (IN0 to channel 1, IN1 to channel 2). Input Mapping Registers MAP_IN0 and MAP_IN1 can be programmed to connect different channels to each input pin. Control according to the status of IN0 and IN1.

The logic level of the input pins can be monitored via the status register ADD_STAT_1.

The Input Status Monitor is operative also when device is in Limp Home mode. If one of the Input pins is set to “high” and the nSLEEP pin is set to “low”, the device switches into Limp Home mode and activates the channel mapped by default to the input pins. HS or LS configuration of the limp home channels are automatically detected depending on connection of the external load.

3.2.2 Half Bridge Drivers

3.2.2.1 Control Modes

The half-bridge drivers can be programmed to drive loads (motor, solenoids, LEDs) continuously (without PWM) or in chopping mode (with PWM) and in parallel operation for driving high current.

3.2.2.1.1 Continuous Mode (Without PWM)

The half-bridges are configured to operate in the continuous mode without using any PWM switching by default. Any high-side or low-side switch is switched on by individually setting the high-side enable bits (HBX_HS_EN) and low-side enable bits (HBX_LS_EN) in operation control registers (OP_CTRL_1, OP_CTRL_2 and OP_CTRL_3).

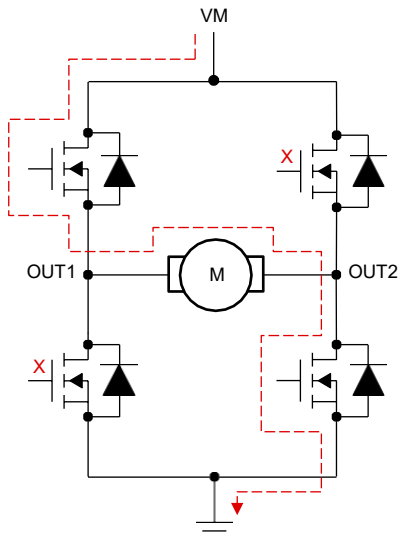
Note: If the high-side enable bit (HBX_HS_EN) and low-side enable bit (HBX_LS_EN) of a particular half-bridge is set high (shoot-through configuration), then the particular half-bridge driver will remain in Hi-Z state until the shoot-through condition is cleared

The high-side and low-side enable bits of a particular half-bridge are configured to drive the motor in forward mode, reverse mode, brake mode and coast mode as shown in [Table 3-2](#).

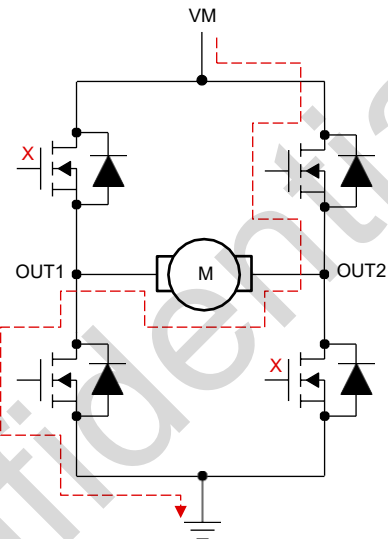
Table 3-2 Motor Operation in Continuous Mode (Motor Connected between HB1 and HB2)

nSLEEP	HALF-BRIDGE-1	HALF-BRIDGE-2	OUT1	OUT2	BRIDGE OPERATION (DC MOTOR)
0	HB1_HS_EN = Don't Care HB1_LS_EN = Don't Care	HB2_HS_EN = Don't Care HB2_LS_EN = Don't Care	Z	Z	Sleep Mode
1	HB1_HS_EN = 0 HB1_LS_EN = 0	HB2_HS_EN = 0 HB2_LS_EN = 0	Z	Z	Motor Coast
1	HB1_HS_EN = 1 HB1_LS_EN = 0	HB2_HS_EN = 0 HB2_LS_EN = 1	H	L	Forward Direction
1	HB1_HS_EN = 0 HB1_LS_EN = 1	HB2_HS_EN = 1 HB2_LS_EN = 0	L	H	Reverse Direction
1	HB1_HS_EN = 0 HB1_LS_EN = 1	HB2_HS_EN = 0 HB2_LS_EN = 1	L	L	Motor Brake (Low-Side)
1	HB1_HS_EN = 1 HB1_LS_EN = 0	HB2_HS_EN = 1 HB2_LS_EN = 0	H	H	Motor Brake (High-Side)
1	HB1_HS_EN = 1 HB1_LS_EN = 1	HB2_HS_EN = 1 HB2_LS_EN = 1	Z	Z	Motor Coast

Figure 3-1 shows the bridge configuration for motor operation in forward direction with high-side FET of OUT1 and low-side FET of OUT2 in conducting state with current flowing from OUT1 to OUT2. Similarly, the motor operation in reverse direction is achieved by switching ON the high-side FET of OUT2 and low-side FET of OUT1 such that current flows from OUT2 to OUT1 as shown in Figure 3-2.

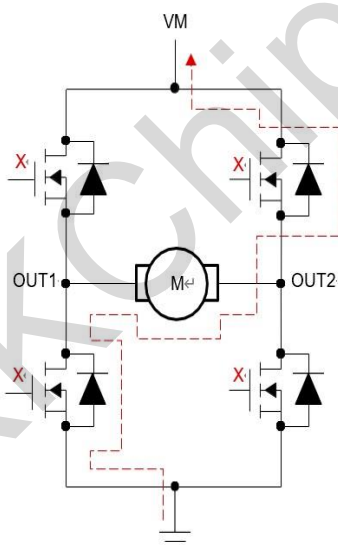


**Figure 3-1 Continuous Mode
(Forward Direction)**

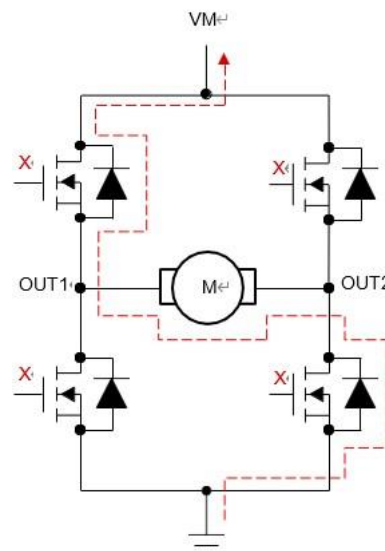


**Figure 3-2 Continuous Mode
(Reverse Direction)**

Figure 3-4 and Figure 3-3 shows the bridge operation in coast mode with motor initially running in forward and reverse direction respectively. As shown in these figures, due to the energy stored in motor's inductance, the current will continue to flow in motor and take the path flow through the body diodes of FETs.

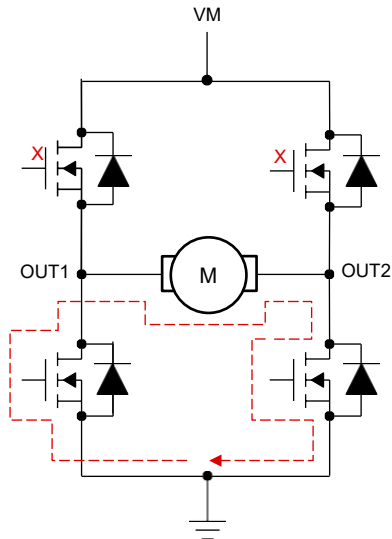


**Figure 3-4 Continuous Mode
(Coast - From Forward
Direction)**

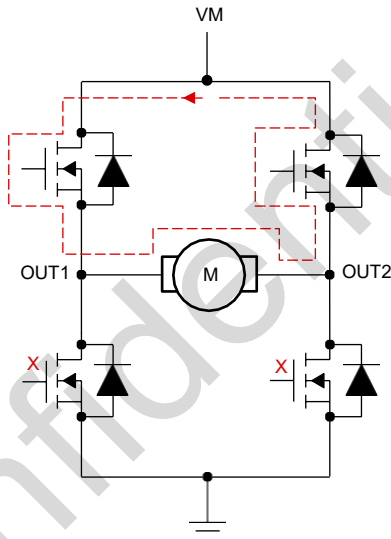


**Figure 3-3 Continuous Mode
(Coast - From Reverse
Direction)**

Figure 3-5 shows the low-side braking of the motor when both low-side FET's of the driver are turned ON. In this case, the motor is considered to be operating in forward direction (current flow from OUT1 to OUT2) and then braking is applied. Similarly, for the high-side braking, both high-side FET's of the driver are turned ON as shown in Figure 3-6.



**Figure 3-5 Continuous Mode
(Brake - Low-Side)**



**Figure 3-6 Continuous Mode
(Brake - High-Side)**

3.2.2.1.2 Chopping Mode (With PWM)

The half-bridges can be configured in the chopping mode by enabling the PWM switching on any particular half- bridge or both half-bridges. Each half-bridge can be mapped to any of the 4 PWM channels for which frequency and duty can be controlled independently. User has the flexibility to select the PWM frequency of channels out of 4 settings of 80-Hz, 100-Hz, 200-Hz and 2-kHz. Moreover, duty (8-bit resolution) of the 8 PWM generators can be adjusted independently.

The PWM chopping mode operation is done in five steps as follows and explained in detail below.

1. PWM Configuration
2. Free-Wheeling Mode (Synchronous Rectification) Disable / Enable
3. PWM Channels Mapping
4. PWM Channels Configuration (PWM Frequency and PWM Duty)
5. Half-Bridge Enable

3.2.2.1.2.1 PWM Configuration

The operation of selected half-bridge to operate in continuous mode or chopping mode (PWM mode) is selected using the PWM control register (PWM_CTRL_1 and PWM_CTRL_2). The HBX_PWM bit in PWM control register is set to enable the PWM switching in half-bridge.

Note: The default mode of any half-bridge is continuous mode. If the corresponding HBx_PWM bit in PMW_CTRL_X register is not set, then the particular half-bridge will operate in continuous mode.

3.2.2.1.2.2 Free-Wheeling Mode (Synchronous Rectification) Disable / Enable

The synchronous rectification of the half-bridge operating in PWM can be enabled by setting the HBX_FW bit in free-wheeling control registers (FW_CTRL_1 and FW_CTRL_2). Figure 3-7 shows the operation of the driver when the synchronous rectification mode is disabled. As shown in this figure, during the PWM off time, the high-side diode of the OUT2 conducts to close the current path required for motor.

When synchronous rectification mode is enabled, if either of the low-side or high-side of the half-bridge operates in the PWM switching, the other switch of the same half-bridge operates in complementary fashion. Figure 3-8 shows such example of the synchronous rectification, where the high-side FET of OUT2 half-bridge is turned ON when the low-side FET of same half-bridge is turned off in a PWM cycle.

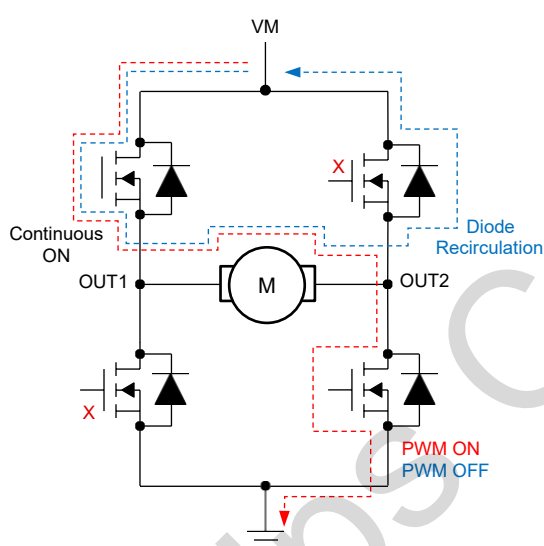


Figure 3-7 PWM Mode (Synchronous Rectification=OFF)

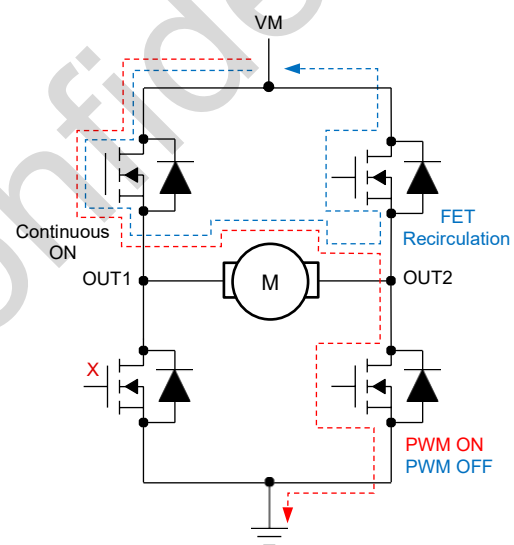


Figure 3-8 PWM Mode (Synchronous Rectification=ON)

Note: The default mode of any half-bridge is asynchronous rectification mode. If the corresponding bit in FW_CTRL_X register is not set, then the particular half-bridge will operate in asynchronous rectification mode.

3.2.2.1.2.3 PWM Channels Mapping

HE99MD9008 devices includes 8 PWM generators which can be mapped to any of the OUTX half bridge outputs using the PWM map control registers. The HBx_PWM_MAP bits in the PWM_MAP_CTRL_X registers are used to map any of the 8 channels to the OUTX outputs as shown in Table 3-3.

Table 3-3 PWM Mapping of HE99MD9008

HBX_PWM MAP BITS	PWM CHANNEL
HBX_PWM_MAP = 000b	Channel 1 Selected for OUTX
HBX_PWM_MAP = 001b	Channel 2 Selected for OUTX
HBX_PWM_MAP = 010b	Channel 3 Selected for OUTX
HBX_PWM_MAP = 011b	Channel 4 Selected for OUTX
HBX_PWM_MAP = 100b	Channel 5 Selected for OUTX
HBX_PWM_MAP = 101b	Channel 6 Selected for OUTX
HBX_PWM_MAP = 110b	Channel 7 Selected for OUTX
HBX_PWM_MAP = 111b	Channel 8 Selected for OUTX

Note: Any half-bridge is mapped to PWM channel 1 by default.

3.2.2.1.2.4 PWM Channels Configuration (PWM Frequency and PWM Duty)

The frequency and duty of each PWM generator can be controlled independently. The PWM_CHx_FREQ bits of PWM frequency control register (PWM_FREQ_CTRL) is used to select the frequency of PWM generator as shown in Table 3-4. The PWM duty of each channel is controlled by the PWM duty control register (PWM_DUTY_CTRL_X).

Table 3-4 PWM Frequency

HBX_PWM MAP BITS	PWM CHANNEL
PWM_CHx_FREQ = 00b	80 Hz
PWM_CHx_FREQ = 01b	100 Hz
PWM_CHx_FREQ = 10b	200 Hz
PWM_CHx_FREQ = 11b	2000 Hz

3.2.2.1.2.5 Half-bridge Enable

The four steps of PWM mode enable, free-wheeling mode configuration, PWM channel mapping and PWM channels configuration ensure the proper configuration of PWM mode. Once the half-bridge is configured for the PWM generation, the half-bridge is enabled by enabling either of the high-side or low-side switch by individually setting the high-side enable bits (HBX_HS_EN) or low-side enable bits (HBX_LS_EN) in operation control registers (OP_CTRL_1, OP_CTRL_2 and OP_CTRL_3).

Note: The PWM is applicable to either of the high-side or low-side switch depending upon the HBX_HS_EN and HBX_LS_EN bits in OP_CTRL_X registers. In synchronous rectification mode, the opposite side switch will conduct in PWM off time.

3.2.2.1.3 Parallel Mode (Continuous Operation)

Parallel mode in HE99MD9008 device is implemented to support higher current loads which cannot be supported by a single channel. This mode can also be used for reducing the effective on-state resistance ($R_{DS(ON)}$) for achieving a better thermal performance of the device.

The configuration of various mode is very similar to the single half-bridge operation as explained in Continuous Mode (Without PWM) section. Considering six half-bridges for the parallel operation (OUT1, OUT2, OUT3 as group – 'X' and OUT4, OUT5, OUT6 as group 'Y'), various modes can be summarized in [Table 3-5](#).

Table 3-5 Motor Operation in Parallel Mode (Continuous Operation) (with Motor Connected between OUT1/2/3 and OUT4/5/6)

nSLEEP	HALF-BRIDGE-1 HALF-BRIDGE-2 HALF-BRIDGE-3 (X)	HALF-BRIDGE-4 HALF-BRIDGE-5 HALF-BRIDGE-6 (Y)	OUT1 OUT2 OUT3	OUT4 OUT5 OUT6	BRIDGE OPERATION (DC MOTOR)
0	HBX_HS_EN = Don't Care HBX_LS_EN = Don't Care	HBX_HS_EN = Don't Care HBX_LS_EN = Don't Care	Z	Z	Sleep Mode
1	HBX_HS_EN = 0 HBX_LS_EN = 0	HBX_HS_EN = 0 HBX_LS_EN = 0	Z	Z	Motor Coast
1	HBX_HS_EN = 1 HBX_LS_EN = 0	HBX_HS_EN = 0 HBX_LS_EN = 1	H	L	Forward Direction
1	HBX_HS_EN = 0 HBX_LS_EN = 1	HBX_HS_EN = 1 HBX_LS_EN = 0	L	H	Reverse Direction
1	HBX_HS_EN = 0 HBX_LS_EN = 1	HBX_HS_EN = 0 HBX_LS_EN = 1	L	L	Motor Brake (Low-Side)
1	HBX_HS_EN = 1 HBX_LS_EN = 0	HBX_HS_EN = 1 HBX_LS_EN = 0	H	H	Motor Brake (High-Side)
1	HBX_HS_EN = 1 HBX_LS_EN = 1	HBX_HS_EN = 1 HBX_LS_EN = 1	Z	Z	Motor Coast

Note: For parallel mode operation, the device operation under safe operating area (SOA) is recommended for supply voltage, $V_{VM} \leq 20V$, $HBX_{SR} = HBX_{SR} = 1b$, $t_{OCP} \leq 10\mu s$ and $PL_MODE_EN = 01b$.

Figure 3-9 shows three half-bridges (OUT1, OUT2 and OUT3) operating as a parallel high-side switch and other three half-bridges (OUT4, OUT5 and OUT6) are operating as a parallel low-side switch for achieving a forward motor operation. Similarly the reverse direction of motor is achieved by operation of OUT1, OUT2 and OUT3 as a parallel low-side switch and other three half-bridges (OUT4, OUT5 and OUT6) as parallel high-side switch as shown in Figure 3-10.

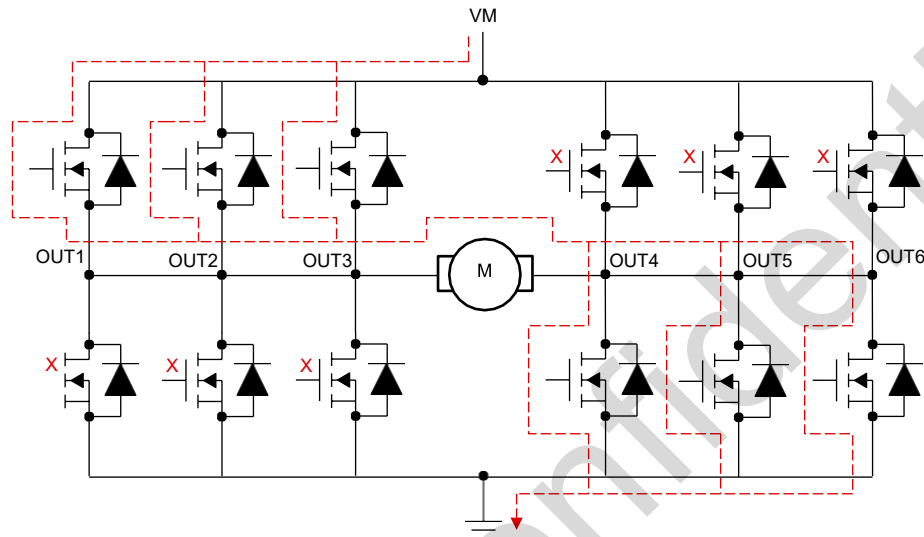


Figure 3-9 Parallel Mode (Forward Direction)

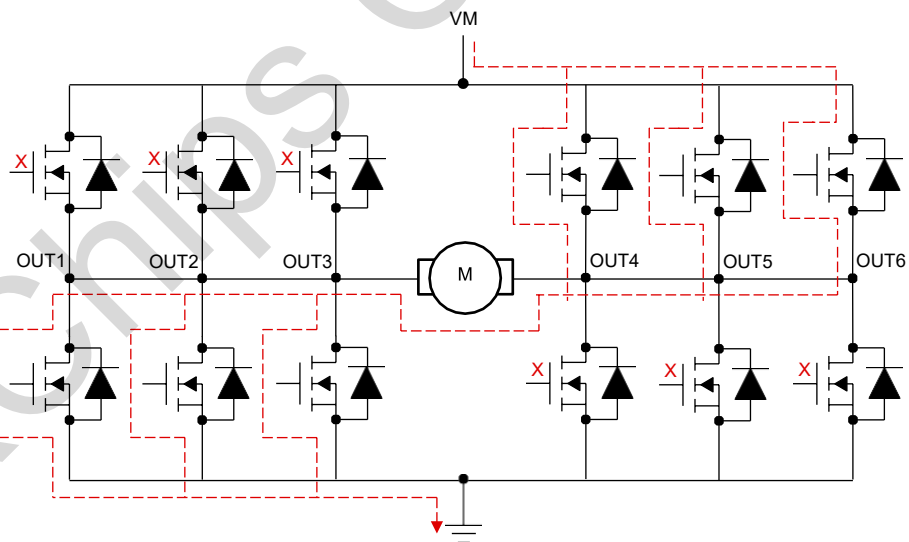


Figure 3-10 Parallel Mode (Reverse Direction)

Figure 3-11 and Figure 3-12 shows the bridge operation in coast mode with motor initially running in forward and reverse direction respectively. As shown in these figures, the body diodes of the FETs conducts to continue the current flow path due to energy stored in motor's inductance.

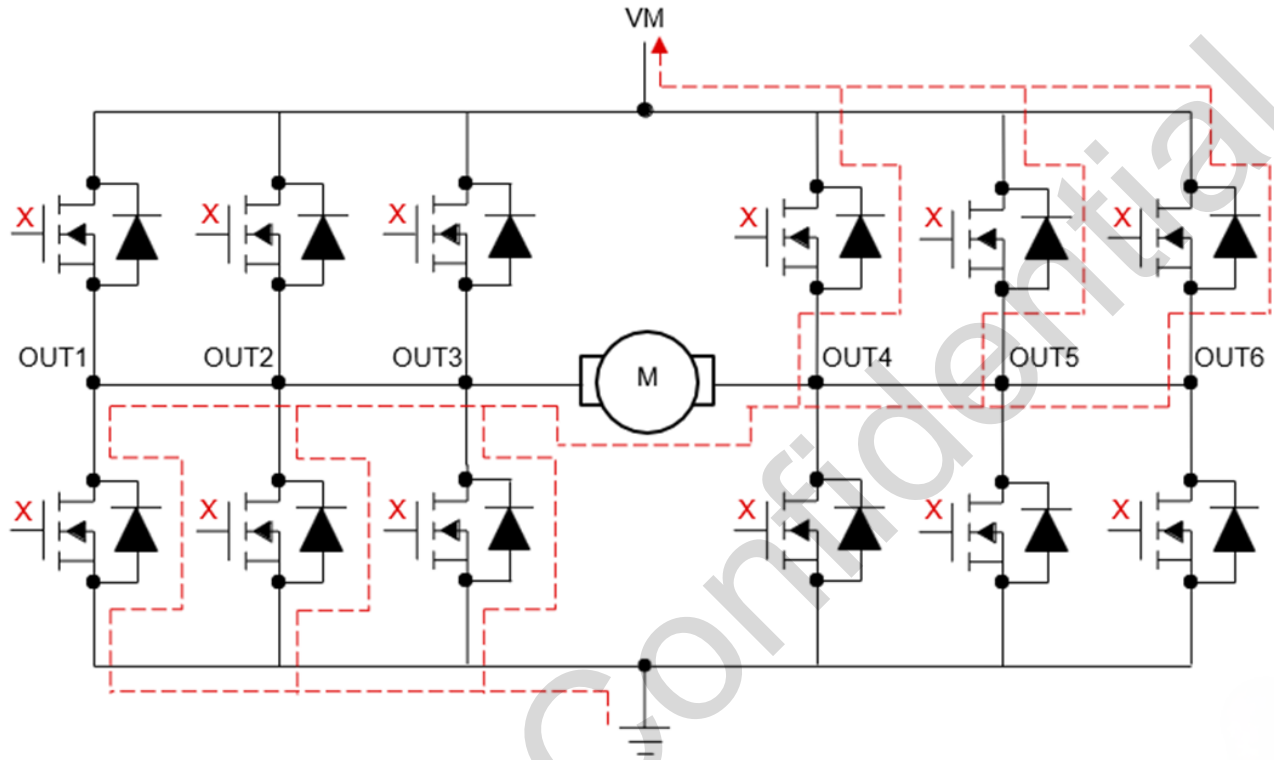


Figure 3-11 Parallel Mode (Coast from Forward Direction)

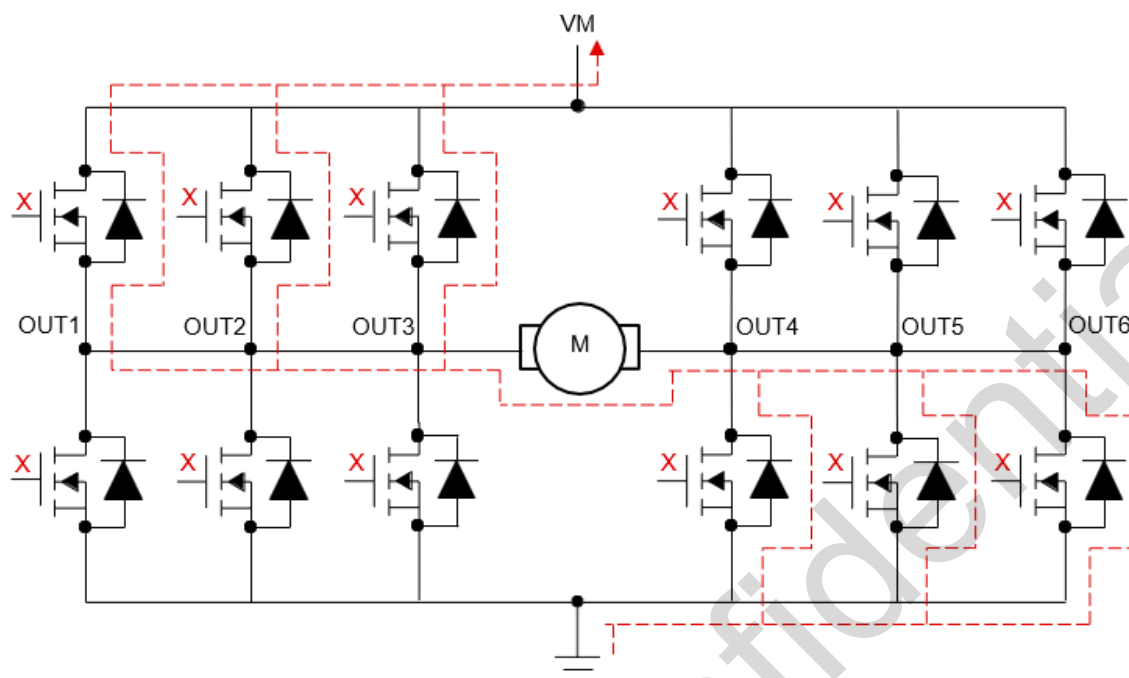


Figure 3-12 Parallel Mode (Coast from Reverse Direction)

The low-side braking of the motor during all the low-side FET's of the driver turning ON is shown in [Figure 3-13](#). In this case, the motor is considered to be operating in forward direction (current flow from OUT1/2/3 to OUT4/5/6) and then braking is applied. Similarly, for the high-side braking, all high-side FET's of the driver are turned ON as shown in [Figure 3-14](#).

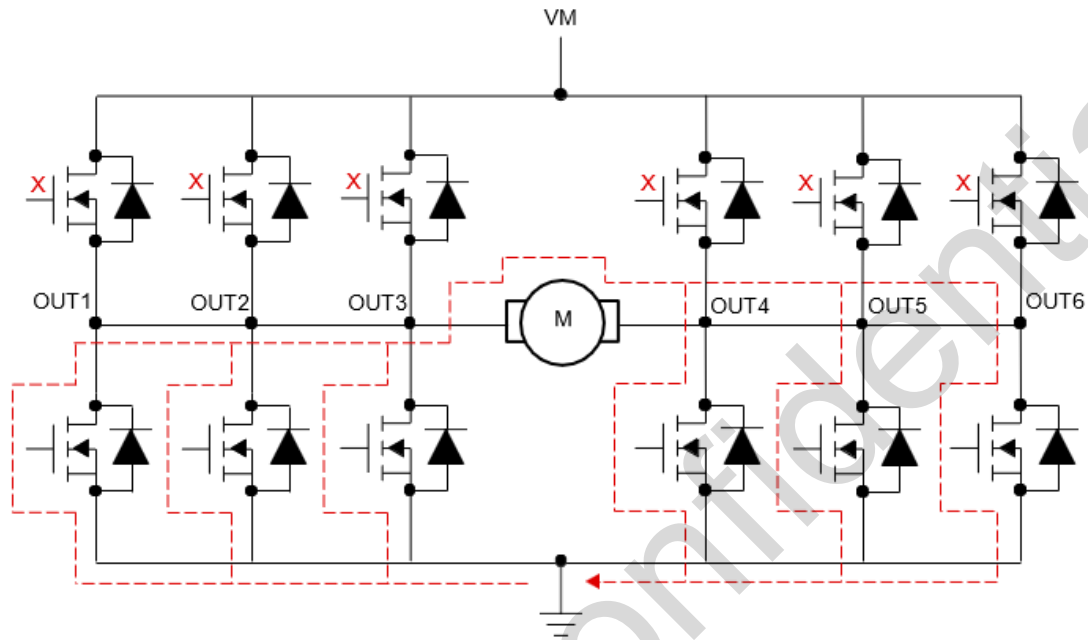


Figure 3-13 Parallel Mode (Brake - Low-Side)

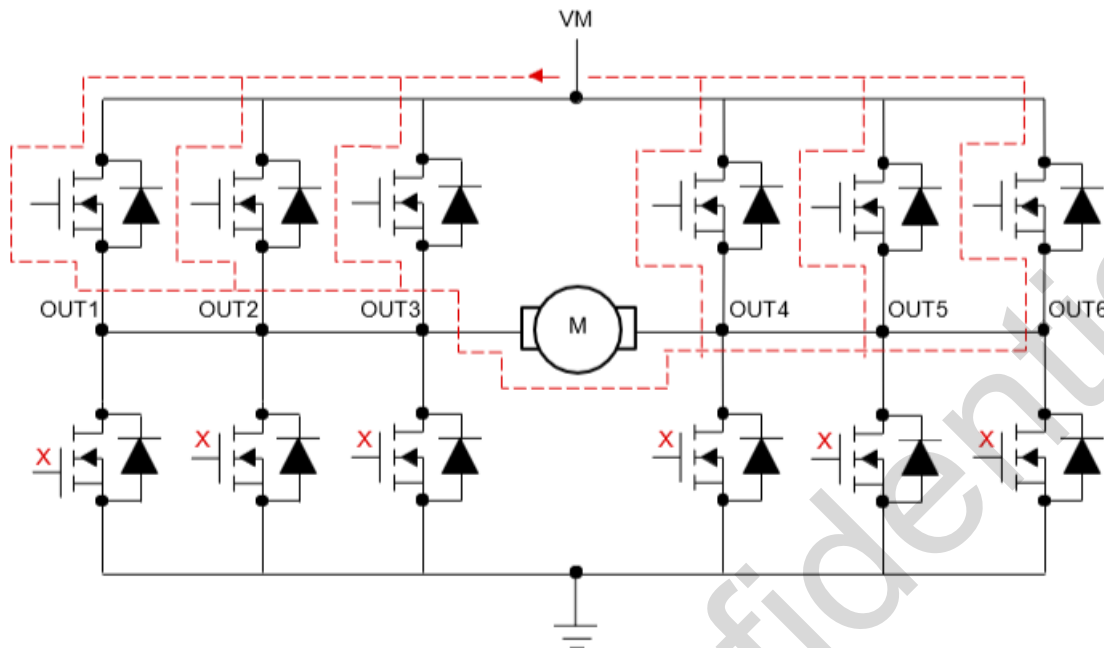


Figure 3-14 Parallel Mode (Brake - High-Side)

3.2.2.1.4 Parallel Mode (PWM Operation)

The half-bridges connected in parallel mode can be configured in the chopping mode by enabling the PWM switching on any particular group of high-side or low-side half-bridges or both group of half-bridges. For PWM operation in parallel mode, all half-bridges is to be mapped to a single PWM channel selected from any of the 4 PWM channels to avoid any delay in the PWM durations which can lead to undesired OCP condition. The user has the flexibility to select the PWM frequency of channels out of 4 settings of 80-Hz, 100-Hz, 200-Hz and 2-kHz and the duty adjustment which supports 8-bit resolution. Following steps enable the PWM operation with driver connected for parallel mode and are explained below.

1. PWM Configuration
2. Free-Wheeling Mode (Synchronous Rectification) Disable / Enable
3. PWM Channels Mapping
4. PWM Channels Configuration (PWM Frequency and PWM Duty)
5. PWM Generators Disable
6. Half-Bridge Enable
7. PWM Generators Enable

3.2.2.1.4.1 PWM Configuration

The PWM control register (PWM_CTRL_1 and PWM_CTRL_2) are used to select the operation of particular half-bridges in the PWM mode. Considering a case for the motor movement in forward direction as shown in [Figure 3-11](#), with low-side FETs of OUT4, OUT5 and OUT6 operating in PWM mode. The HBX_PWM bit in PWM control register is set to enable the PWM switching in selected half-bridges as shown below:

- HB4_PWM = 1b

- HB5_PWM = 1b
- HB6_PWM = 1b

3.2.2.1.4.2 Free-Wheeling Mode (Synchronous Rectification) Disable / Enable

The synchronous rectification of the half bridges operating in PWM mode (OUT4, OUT5 and OUT6) are enabled by setting the corresponding HBX_FW bits in free-wheeling control register (FW_CTRL_1 and FW_CTRL_2). By default, the synchronous rectification mode is disabled.

- HB4_FW = 1b
- HB5_FW = 1b
- HB6_FW = 1b

Figure 3-15 shows the parallel operation of half-bridges in PWM mode with synchronous rectification disabled. As shown in this figure, during the PWM off time, the high-side diode of the OUT4, OUT5 and OUT6 conducts to close the current path required for motor.

When synchronous rectification mode is enabled, the high-side FETs of OUT4, OUT5 and OUT6 starts conducting during the PWM OFF time to close the motor current path as shown in Figure 3-16.

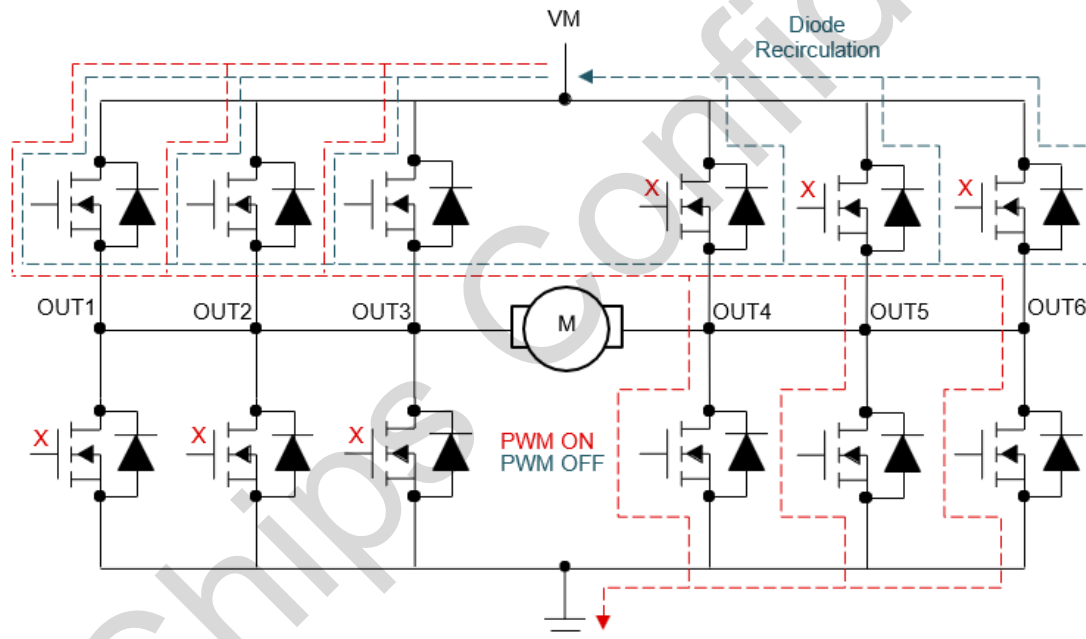


Figure 3-15 Parallel Mode (PWM with Synchronous Rectification = OFF)

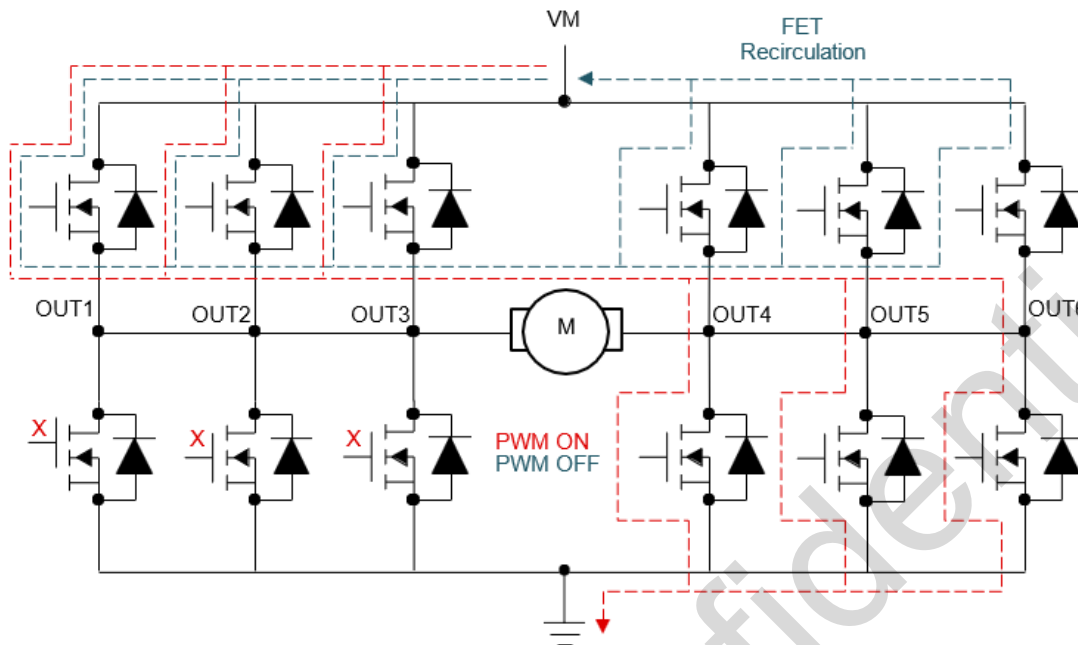


Figure 3-16 Parallel Mode (PWM with Synchronous Rectification = ON)

3.2.2.1.4.3 PWM Channel Mapping

The low-side FET's of half-bridges OUT4, OUT5 and OUT6 are mapped to any of the PWM generator by using the HBX_PWM_MAP bits in PWM mapping control registers. For parallel operation, all the half-bridges operating in PWM mode is mapped to a single PWM generator. Considering that PWM generator-4 is used for the mapping of half-bridges, following bits of the PWM_MAP_CTRL_X registers are affected:

- HB4_PWM_MAP = 11b
- HB5_PWM_MAP = 11b
- HB6_PWM_MAP = 11b

Note: If the PWM of any channel is enabled, then it is mapped to PWM generator-1 by default.

3.2.2.1.4.4 PWM Channels Configuration (PWM Frequency and PWM Duty)

The PWM_CHx_FREQ bits of PWM frequency control registers (PWM_FREQ_CTRL_X) is used to select the frequency of PWM generator. Moreover, the PWM duty of each channel is controlled by the PWM duty control register (PWM_DUTY_CTRL_X). Considering a frequency of 2-kHz is selected for the PWM operation (for PWM Generator-4), following frequency control and duty control registers are effected:

- PWM_CH4_FREQ = 11b
- PWM_CH4_FREQ = '8-bit duty'

3.2.2.1.4.5 PWM Generators Disable

The PWM generators are disabled to ensure that all the half-bridges are turned-on at same time to avoid false OCP conditions for supporting higher current operation. The false OCP condition can arise due to the minimum time required for the SPI delay to switch on various half-bridges available in different registers. This can cause higher current (OCP condition) in one of the paralleled half-bridge while other half-bridge turning ON is delayed to the SPI register write delay and the propagation delay. Therefore, this sequence includes disabling the PWM generators initially, then enabling half-bridges and followed by enabling the PWM generators to avoid such issue. The PWM generator-4 is disabled by using the following command in the PWM_CTRL_X registers:

- PWM_CH4_DIS = 1b

Note: All PWM generators are enabled by default (Default value of PWM_CTRL_X registers is 00h).

3.2.2.1.4.6 Half-Bridge Enable

Once the PWM generators are disabled, the high-side and low-side FETs in half-bridges to be paralleled are enabled. High-side switches (connected in parallel) operating in continuous mode are enabled using the following bits in the OP_CTRL_X registers:

- HB1_HS_EN = 1b
- HB2_HS_EN = 1b
- HB3_HS_EN = 1b

Moreover, the low-side switches (connected in parallel) operating in PWM mode are enabled using the following bits in the OP_CTRL_X:

- HB4_LS_EN = 1b
- HB5_LS_EN = 1b
- HB6_LS_EN = 1b

3.2.2.1.4.7 PWM Generators Enable

After the half-bridges are enabled, the PWM generators are also enabled for tuning-on the respective FETs operating in PWM mode. For this case, the low-side FETs of OUT4, OUT5 and OUT6 are turned ON for PWM operation connected to PWM generator-4. The PWM generator is enabled by the bits in the PWM_CTRL_X registers as shown:

- PWM_CH4_DIS = 0b

3.2.2.2 Half-Bridge Drive Architecture

3.2.2.2.1 Slew Rate

An adjustable gate-drive current control to the MOSFETs of half-bridges is implemented to achieve the slew rate control. The MOSFET VDS slew rates are a critical factor for optimizing radiated emissions, energy and duration of diode recovery spikes and switching voltage transients related to parasitics. These slew rates are predominantly determined by the rate of gate charge to internal MOSFETs as shown in [Figure 3-17](#).

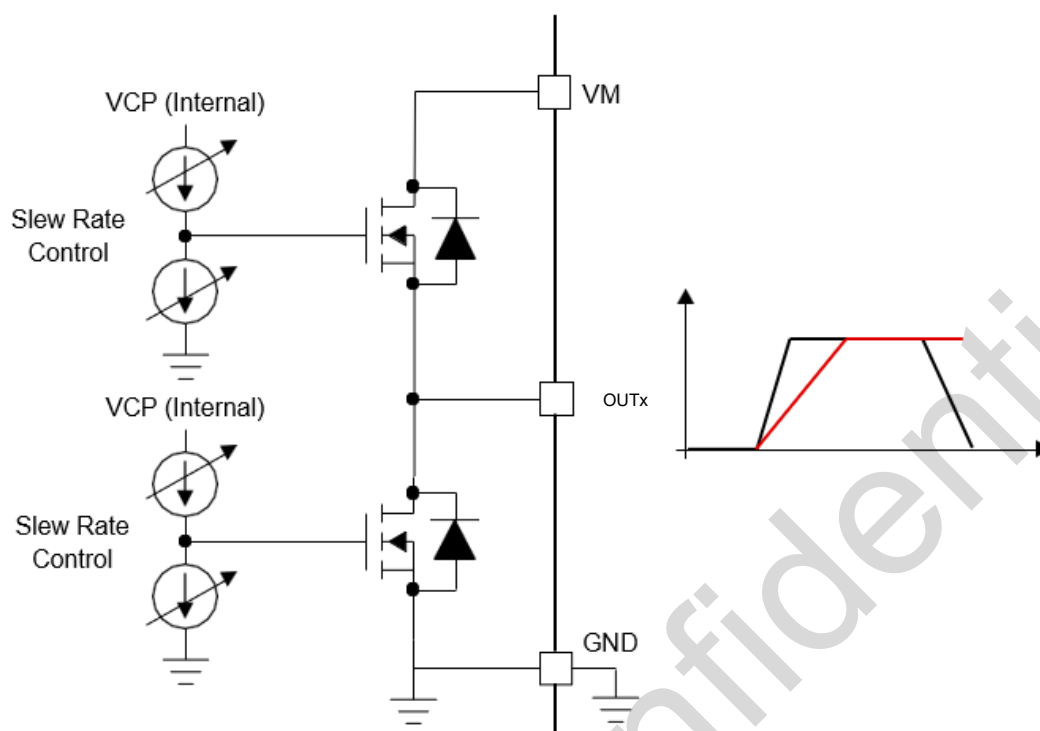


Figure 3-17 Slew Rate Circuit Implementation

The slew rate of each half-bridge can be adjusted by HBX_SR bits in Slew Rate control register (SR_CTRL_1 and SR_CTRL_2). Each half-bridge can be selected to a slew rate of 0.6-V/ μ s or 2.5-V/ μ s. The slew rate is calculated by the rise-time and fall-time of the voltage on OUTx pin as shown in Figure 3-18. The slew rate (SR) is calculate as shown in Figure 3-18.

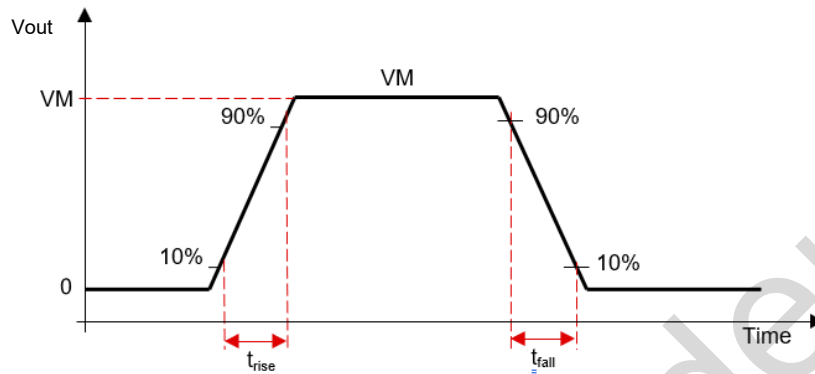


Figure 3-18 Slew Rate Timings

3.2.2.2.2 Cross Conduction (Dead Time)

The device is fully protected for any cross conduction of MOSFETs. In half-bridge configuration, the operation of high-side and low-side MOSFETs are ensured to avoid any shoot through currents by inserting a dead time (t_{dead}). This is implemented by sensing the gate-source voltage (VGS) of the high-side and low-side MOSFETs and ensured that VGS of high-side MOSFET has reached below turn-off levels before switching on the low-side MOSFET of same half-bridge as shown in Figure 3-19 and Figure 3-20.

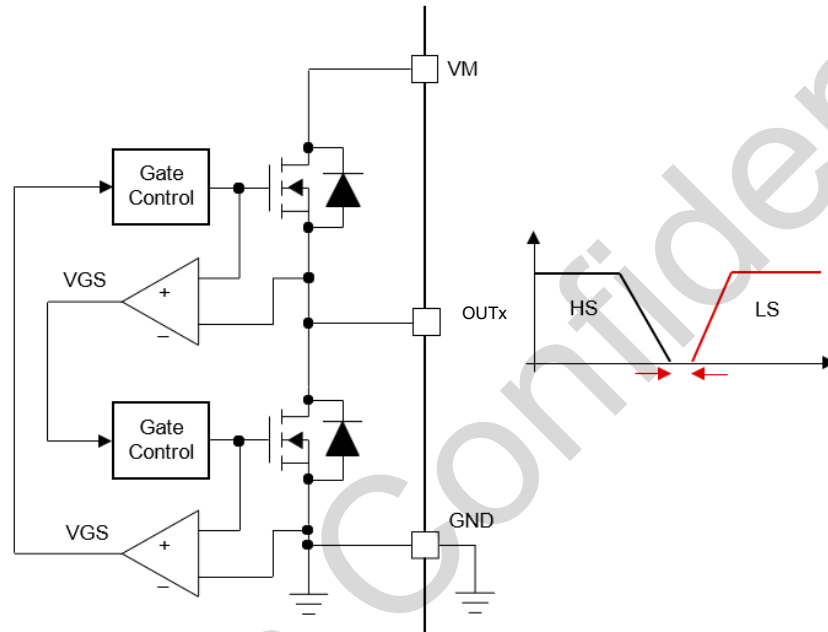


Figure 3-19 Cross Conduction Protection

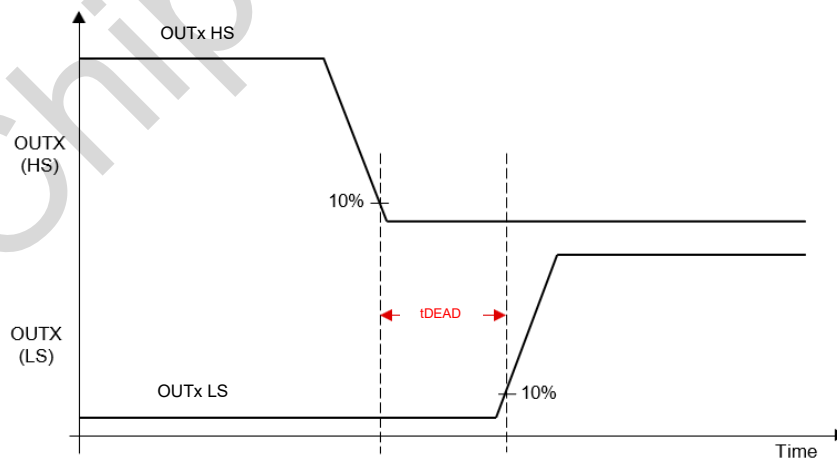


Figure 3-20 Dead Time

3.2.2.2.3 Propagation Delay

Propagation delay refers to the delay time from SPI valid condition to OUTx going high (10% level) as shown in Figure 3-21. The propagation constitutes of three major parameters.

1. Digital delay for SPI command decodes.
2. Analog delay for driver switch-on and gate current charging delay.
3. Slew rate delay for OUTx node to reach 10% of the final settling value.

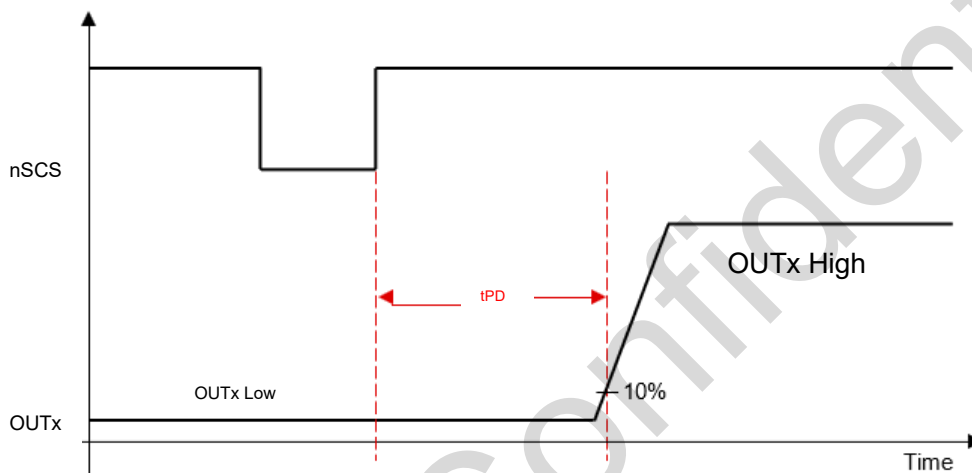


Figure 3-21 Propagation Delay

3.2.3 Pin Diagrams

This section presents the I/O structure of all digital input and output pins.

3.2.3.1 Logic Level Input Pin (IN0, IN1, nSLEEP, SCLK and SDI)

Figure 3-22 shows the input structure for the logic levels pins, nSLEEP, SCLK and SDI. The input can be with a voltage or external resistor. It is recommended to put SCLK and SDI pin low in device sleep mode to reduce leakage current through internal pull-down resistors.

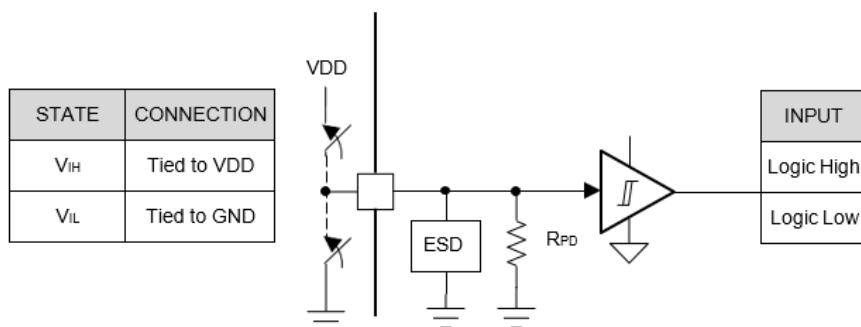


Figure 3-22 Logic Level Input Pin Structure (IN0, IN1, nSLEEP, SCLK and SDI)

3.2.3.2 Logic Level Input Pin (nSCS)

Figure 3-23 shows the input structure for the logic levels pin, nSCS. The input can be with a voltage or external resistor.

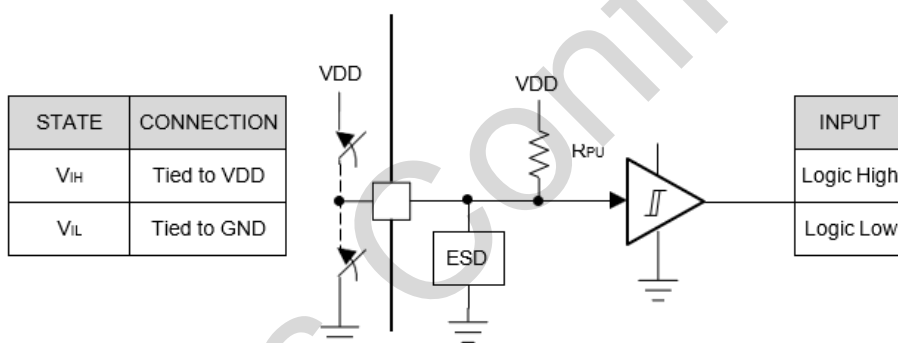


Figure 3-23 Logic Level Input Pin Structure (nSCS)

3.2.3.3 Open Drain Output Pin (nFAULT)

Figure 3-24 shows the structure of the open-drain output pin, nFAULT. The open-drain output requires an external pullup resistor to function properly.

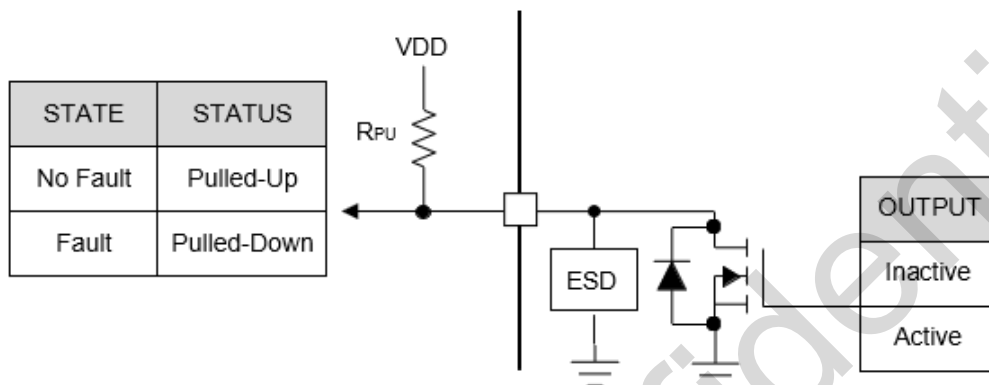


Figure 3-24 Open Drain Output Pin Structure (nFAULT)

3.2.3.4 Push Pull Output Pin (SDO)

Figure 3-25 shows the structure of push-pull pin, SDO.

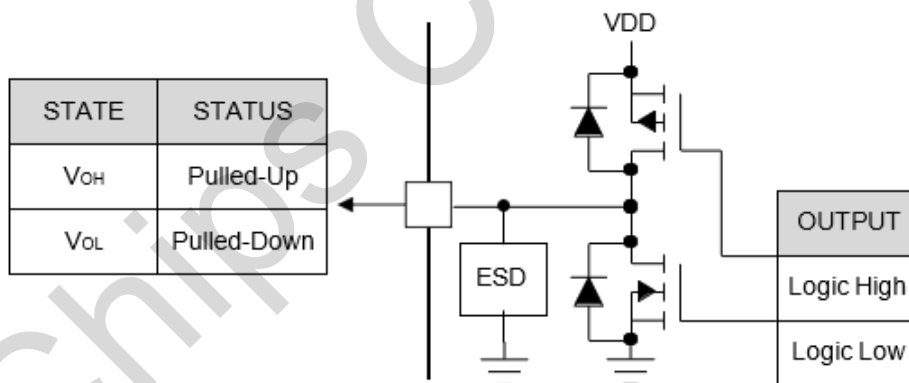


Figure 3-25 Push Pull Output Pin (SDO) Structure

3.2.4 Protection Circuits

The HE99MD9008 device is fully protected against undervoltage, overcurrent, and over-temperature events.

3.2.4.1 VM Supply Undervoltage Lockout (UVLO)

If at any time the input supply voltage on the VM pin falls below the VUVLO threshold, all of the half-bridges are disabled, the charge pump is disabled, and the nFAULT pin is driven low as shown in [Figure 3-26](#). The UVLO bit is also latched high in the IC status (IC_STAT) register. Normal operation resumes (driver operation and the nFAULT pin is released) when the VM undervoltage condition is removed. The UVLO bit remains set until cleared through the CLR_FLT bit.

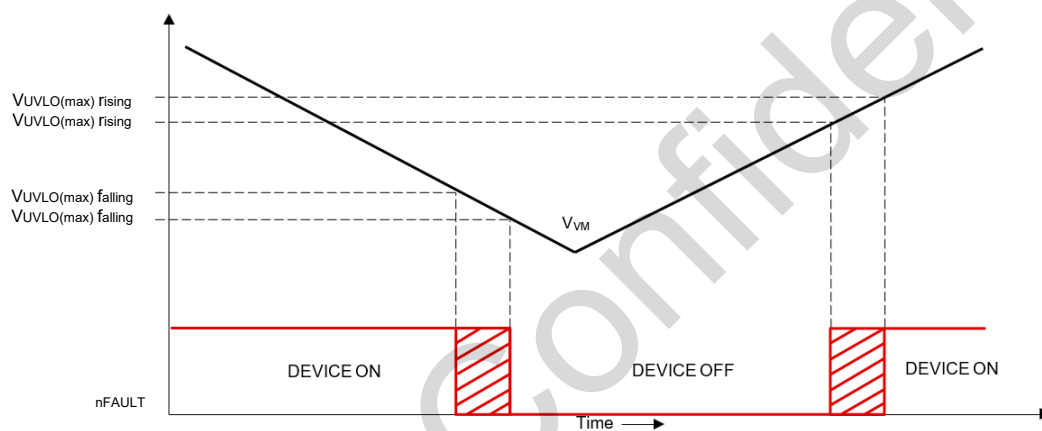


Figure 3-26 VM UVLO Operation

3.2.4.2 VM Supply Overvoltage Protection (OVP)

If at any time the input supply voltage on the VM pin rises above the VOVP threshold, all of the half-bridges are disabled, the charge pump is disabled, and the nFAULT pin is driven low as shown in [Figure 3-27](#). The OVP bit is also latched high in the IC status (IC_STAT) register. Normal operation resumes (driver operation and the nFAULT pin is released) when the VM overvoltage condition is removed. The OVP bit remains set until cleared through the CLR_FLT bit.

An extended overvoltage operation is also supported in this device for higher over-voltage range up to 32-V. This operation is enabled by setting the EXT_OVP bit in the configuration (CONFIG_CTRL) register.

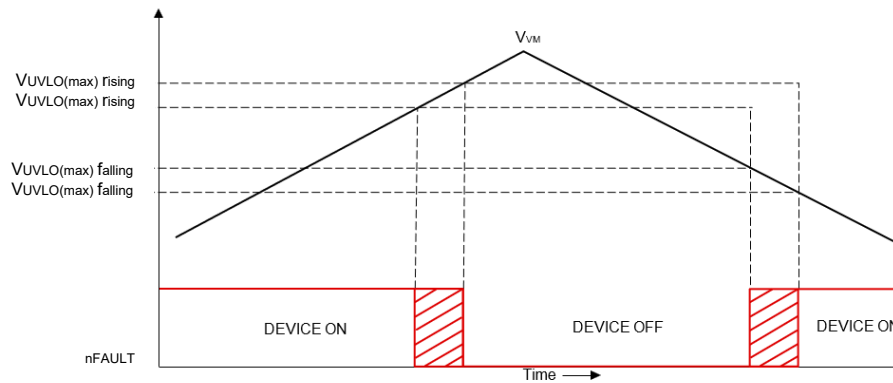


Figure 3-27 Over Voltage Protection

3.2.4.3 Digital Output Supply Power Undervoltage (VDDUV)

If at any time the supply voltage on the VDD pin falls below the VDDUV threshold, SPI function will be disabled and all input command will be ignored. If VDDUV happens when device is at normal operating mode, channels will be operating at previous state. Channels controlled by SPI will stay in ON/OFF state according to last SPI command. Channels mapped to IN0/IN1 or PWM generators keep operating according to corresponding command. Once recovered, SPI function will be available again, as shown in Figure 3-28. If VDDUV happens when device is at limp home mode, channels will still be operating according to IN0/IN1 status. Once recovered from VDDUV, SPI with read-only function will be available.

Note: VDDUV is not reported to nFAULT pin.

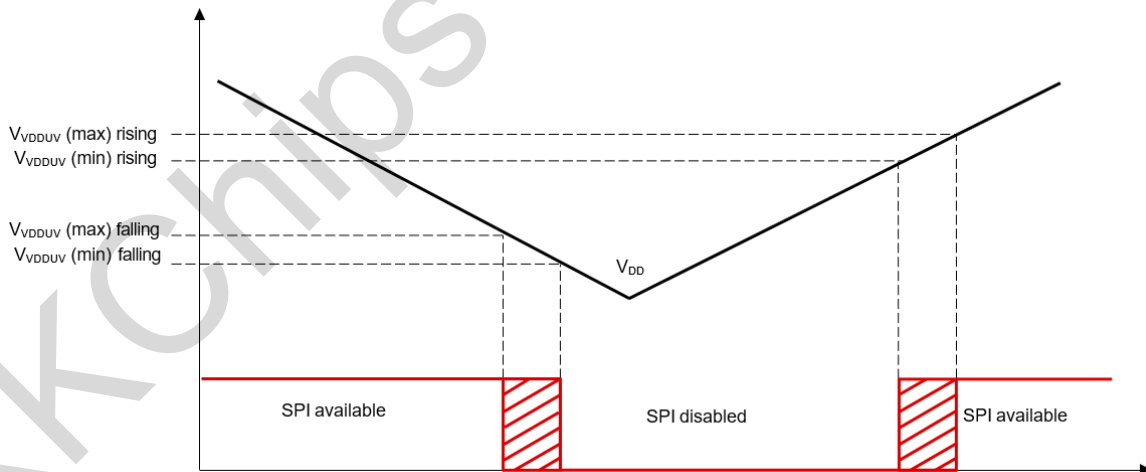


Figure 3-28 VDD UVLO Operation

3.2.4.4 Overcurrent Protection (OCP)

A overcurrent protection circuit is implemented in each driver. If FET current increase higher than I_{OCP} threshold for longer than the t_{OCP} deglitch time, the high-side and the low-side FETs in the corresponding half bridge are disabled and the nFAULT pin is driven low. The OCP bit in the IC status (IC_STAT) register and corresponding bit in overcurrent protection status register (OCP_STAT_X) register is latched high. The charge pump remains active during this condition. The OCP bit in the IC status (IC_STAT) register and corresponding bits (HBX_HS_OCP / HBX_LS_OCP) in overcurrent protection status (OCP_STAT_X) register remains set until cleared through the CLR_FLT bit.

User also has the programmability of disabling the OCP fault on the nFAULT pin by setting the OCP_REP bit in the CONFIG_CTRL register.

The device also provides two slew-rate options for the device turn-off during an OCP event which can be programmed via the PL_MODE_EN bits in OLD_CTRL_2 register. The default option (PL_MODE_EN = 00b) is the faster slew rate option (typical around 1μs) which can be used for the single bridge operation. The slower option (PL_MODE_EN = 01b) provides a slower slew rate (half-bridge slew rate, HBX_SR) which can be used for the higher current applications in device parallel mode operation.

Table 3-6 Overcurrent Protection

Drive Current	BRIDGE CONFIGURATION	REGISTER SETTINGS	BRIDGE STATE	nFAULT PIN		BITS AFFECTED	RECOVERY
				OCP_REP = 0	OCP_REP = 1		
$I_{LOAD} < I_{OCP}$	OUT1 High-Side ON	HB1_HS_EN = 1	ENABLED	HIGH	HIGH	N/A	N/A
	OUT2 Low-Side ON	HB1_LS_EN = 1	ENABLED	HIGH	HIGH		
$I_{LOAD} < I_{OCP}$	Full Bridge (OUT1/2) Forward Direction	HB1_HS_EN = 1 HB2_LS_EN = 1	ENABLED	HIGH	HIGH		
	Full Bridge (OUT1/2) Reverse Direction	HB1_LS_EN = 1 HB2_HS_EN = 1	ENABLED	HIGH	HIGH		
I_{SHORT} or $I_{LOAD} > I_{OCP}$	OUT1 High-Side ON OUT1 Short to GND	HB1_HS_EN = 1	Hi-Z	LOW	HIGH	OCP = 1 (IC_STAT) HB1_HS_OCP = 1	OCP Condition Removed CLR_FLT = 1
	OUT1 Low-Side ON OUT1 Short to VM	HB1_LS_EN = 1	Hi-Z	LOW	HIGH	OCP = 1 (IC_STAT) HB1_LS_OCP = 1	
I_{SHORT} or $I_{LOAD} > I_{OCP}$	Full Bridge (OUT1/2) Forward Direction OUT1 / OUT2 Short	HB1_HS_EN = 1 HB2_LS_EN = 1	Hi-Z	LOW	HIGH	OCP = 1 (IC_STAT) HB1_HS_OCP = 1 or HB2_LS_OCP = 1 ⁽¹⁾	
	Full Bridge (OUT1/2)	HB1_LS_EN = 1				OCP = 1	

Drive Current	BRIDGE CONFIGURATION	REGISTER SETTINGS	BRIDGE STATE	nFAULT PIN		BITS AFFECTED	RECOVERY
				OCP_REP = 0	OCP_REP = 1		
	Reverse Direction OUT1 / OUT2 Short	HB2_HS_EN = 1	Hi-Z	LOW	HIGH	(IC_STAT) HB1_LS_OCP = 1 or HB2_HS_OCP = 1 ⁽²⁾	

(1) Either of the HB1_HS_OCP or HB2_LS_OCP will set depending upon which half-bridge OCP trigger first.

(2) Either of the HB1_LS_OCP or HB2_HS_OCP will set depending upon which half-bridge OCP trigger first.

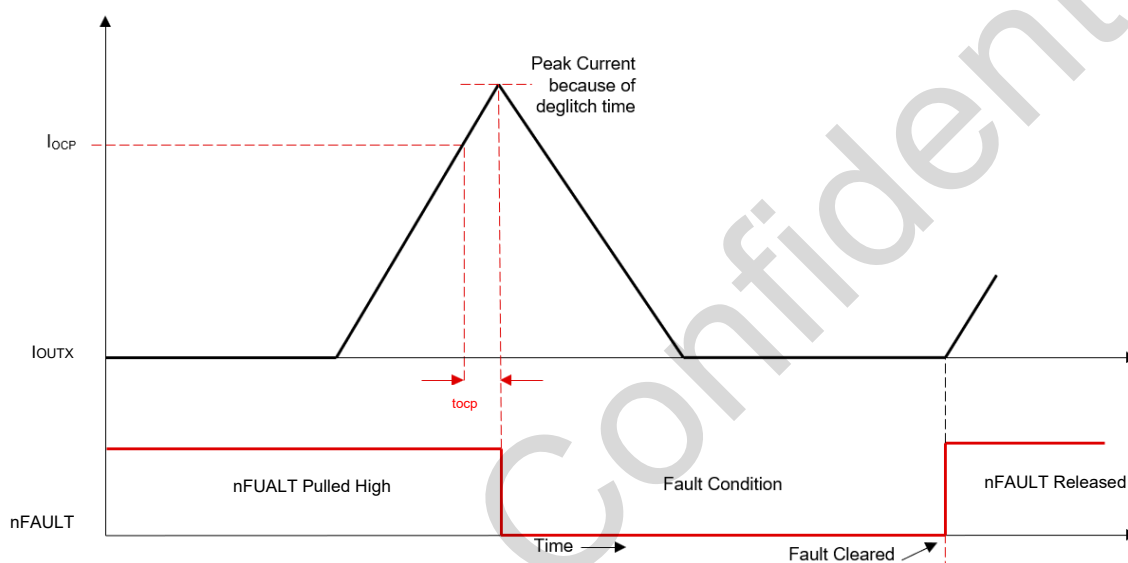


Figure 3-29 Over Current Protection

3.2.4.5 Open-load detection (OLD)

The HE99MD9008 devices implement multiple open-load detection schemes to allow the controller to determine if a load is connected to the OUTX terminals. The OLD schemes available in HE99MD9008 are listed below. Table 3-7 summarizes the use-cases for each OLD scheme.

- Active OLD
- Low-Current Active OLD
- Negative-Current Active OLD
- Passive OLD (only available in HE99MD9008)

Table 3-7 Summary of OLD features

	PASSIVE OLD	ACTIVE OLD	LOW-CURRENT ACTIVE OLD	NEGATIVE-CURRENT ACTIVE OLD
Part number	HE99MD9008	HE99MD9008		

s				
When is it used	Detects open-load condition prior to enabling the outputs	Detects the open-load condition while driving a load	Detects the open-load condition while driving a load with small operating current	Detects the open-load condition during current re-circulation (on high-side or low-side) when synchronous rectification is enabled for both half-bridge and full-bridge configurations
State of OUTx for valid OLD	Hi-Z (outputs disabled)	H/L	L	H/L
OLD Trigger Condition	$VOLx_HS(+) > VOLx_HS$ or $VOLx_LS(-) < VOLx_LS$	$IOUTX < IOLD$	$IOUTX < IOLD_LOW$	$IOUTX > -IOLD_NEG$ $IOUTX < IOLD$
Tradeoffs	Passive OLD sequence is not enabled if any other fault other than OCP/OLD is present.	A false flag can occur during current re-circulation if synchronous rectification is ON. Enabling Negative-current OLD will solve this. A false flag can also occur if the operating current for the load is small (below $IOLD$ threshold). See Low-current OLD for solution.	Only applicable for the current flowing in the low-side FETs. The I_{OCP} for the low-side FET is also reduced by 11 times. The $R_{DS(ON)}$ of the low-side FET will increase by 11 times, hence the thermal performance has to be monitored.	Only functional during current re-circulation, however it works in conjunction with active OLD. This feature is not needed if synchronous rectification is disabled.
Can be used with other OLD schemes	No	Negative-current OLD and low-current OLD	Active OLD and negative-current OLD	Active OLD and low-current OLD

3.2.4.5.1 Active OLD

Active OLD can identify an open-load condition on the OUTX pins while driving a load. As shown in Figure 3-30, the HE99MD9008 identifies an open-load fault condition when the current through the MOSFET ($IOUTX$) is lower than the open-load current threshold ($IOLD$) for longer than the open-load deglitch time ($tOLD$). At that point the device takes the following actions.

- OLD bit in the IC status (IC_STAT) register sets to 1
- HBX_HS_OLD or HBX_LS_OLD bit in the open-load status register (OLD_STAT_X) sets to 1 (depending if the fault is on the high-side MOSFET or low-side MOSFET, respectively).
- nFAULT pin is driven low to indicate a fault to the controller.

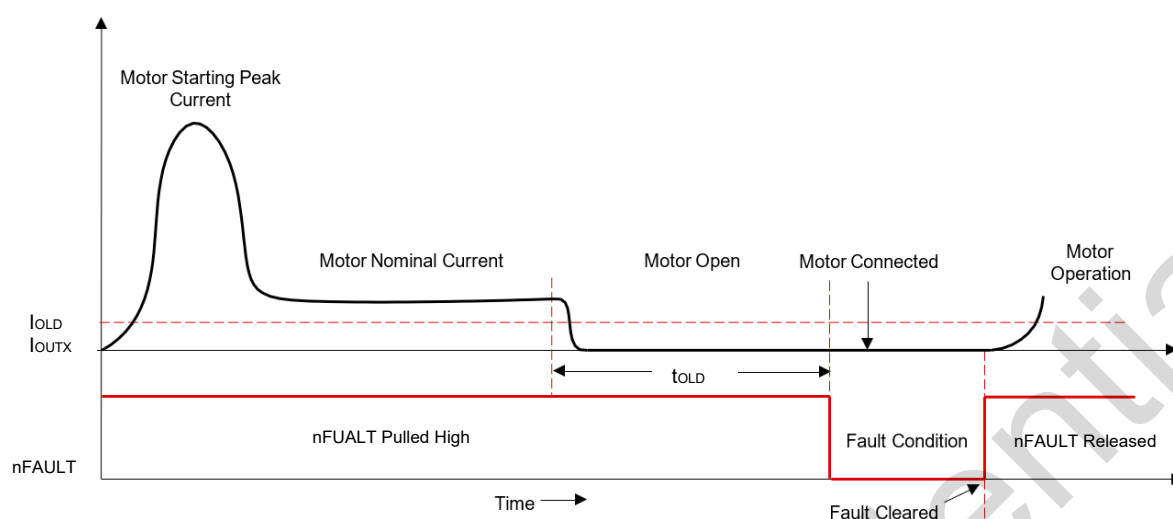


Figure 3-30 Active open-load detection

Normal operation resumes (driver operation resumes, the nFAULT pin goes high, OLD bit is reset to 0) when the open-load condition is removed (the user reconnects the load to the OUTX connection) and the controller writes the CLR_FLT bit to 1.

Note: After the open-load fault condition is removed, the nFAULT pin will be driven high and the fault status are removed when,

- CLR_FLT command is issued after deglitch time (tOLD) while OUTX is set high.
- CLR_FLT command is issued after OUTX is set to Hi-Z.
- CLR_FLT command is issued after HBX_OLD_DIS bit is set.

By default, OLD on the HE99MD9008 devices is enabled. The OLD control registers (OLD_CTRL_1 and OLD_CTRL_2) allow the user to disable OLD on the OUTX pins with the HBX_OLD_DIS bits. The OLD_OP bit in the OLD_CTRL_2 register determines the response of the device to an active OLD fault. If OLD_OP = 0, the OUTX pins go to the Hi-Z state to stop driving the outputs. If OLD_OP = 1, the OUTX pins stay in their previous state and do not react to the OLD fault unless the user takes action. Similarly, the OLD_REP bit determines if the OLD fault will report on the nFAULT pin or only in the IC_STAT register. Table 3-8 summarizes the open-load detection feature and conditions.

Note: By default the OLD feature is enabled, the outputs disable (go Hi-Z) when the OLD flags, and the Nfault pin will report the OLD.

Table 3-8 OLD Configuration

LOAD / OPEN	REGISTER SETTINGS	OLD_OP	OLD_REP	OUT1	OUT2	nFAULT	BITS EFFECTED	RECOVERY
Half-Bridge Load Connected	HB1_HS_EN = 1	X	X	H	X	HIGH	N/A	N/A
	HB1_LS_EN = 1	X	X	L	X	HIGH		
Full-Bridge Load Connected	HB1_HS_EN = 1 HB2_LS_EN = 1	X	X	H	L	HIGH		
	HB1_LS_EN = 1	X	X	L	H	HIGH		

Detailed Description

LOAD / OPEN	REGISTER SETTINGS	OLD_OP	OLD_REP	OUT1	OUT2	nFAULT	BITS EFFECTED	RECOVERY
	HB2_HS_EN = 1							
Half-Bridge Open	HB1_HS_EN = 1	0	0	Hi-Z	X	LOW	OLD = 1 (IC_STAT) HB1_HS_O LD = 1	OLD Condi ti on Remov ed CLR_F LT = 1
		0	1	Hi-Z	X	HIGH		
		1	0	H	X	LOW		
		1	1	H	X	HIGH		
	HB1_LS_EN = 1	0	0	Hi-Z	X	LOW	OLD = 1 (IC_STAT) HB1_LS_O LD = 1	
		0	1	Hi-Z	X	HIGH		
		1	0	L	X	LOW		
		1	1	L	X	HIGH		
Full-Bridge Open	HB1_HS_EN = 1 HB2_LS_EN = 1	0	0	Hi-Z	Hi-Z	LOW	OLD = 1 (IC_STAT) HB1_HS_O LD = 1 or HB2_LS_O LD = 1 ⁽¹⁾	
		0	1	Hi-Z	Hi-Z	HIGH		
		1	0	H	L	LOW		
		1	1	H	L	HIGH		
	HB1_LS_EN = 1 HB2_HS_EN = 1	0	0	Hi-Z	Hi-Z	LOW	OLD = 1 (IC_STAT) HB1_LS_O LD = 1 or HB2_HS_O LD = 1 ⁽²⁾	
		0	1	Hi-Z	Hi-Z	HIGH		
		1	0	L	H	LOW		
		1	1	L	H	HIGH		

(1) Either of the HB1_HS_OLD or HB2_LS_OLD will set depending upon which half-bridge OLD triggers first.

(2) Either of the HB1_LS_OLD or HB2_HS_OLD will set depending upon which half-bridge OLD triggers first.

3.2.4.5.2 Negative-current OLD

The HE99MD9008 device also includes a negative-current OLD mode. The negative current can flow either through the body diode of high-side FET or the FET itself depending on whether or not the channel is configured for synchronous rectification. Figure 3-32 shows the current re-circulation through the body diode of the high-side FET when the synchronous rectification mode is OFF (i.e. HB2_FW = 0). In this case, the active OLD will not falsely report an open-load condition since the OLD circuit only enables when the FET is ON. Negative-current OLD will also work during re-circulation through the low-side FETs.

Figure 3-31 shows the negative current re-circulation through the high-side FET when synchronous rectification is ON (i.e. HB2_FW = 1). In this scenario, for default operation (OLD_NEG_EN = 0), the device can show a false open-load fault since the FET current is lower than the positive OLD threshold. However, when negative-current OLD mode is enabled, the device will only flag an open-load fault if $I_{OUTX} < I_{OLD_NEG}$. This mode is enabled by setting the OLD_NEG_EN bit in OLD_CTRL3 register.

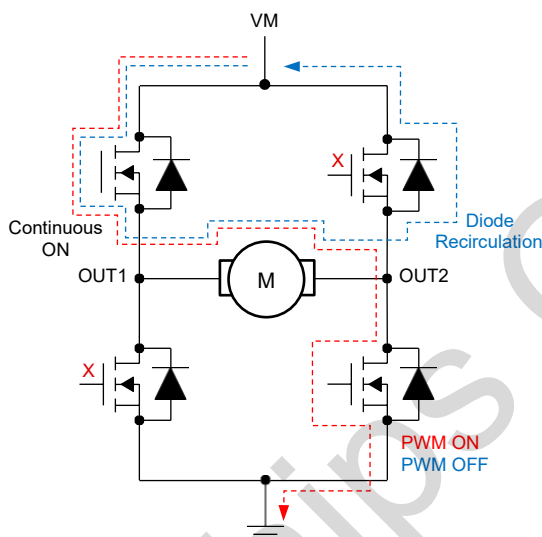


Figure 3-31 Negative Current Flow in OUT2 by Body Diode of High-Side FET

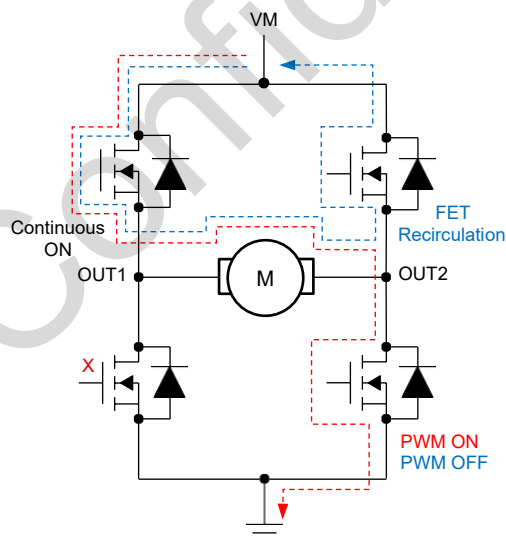


Figure 3-32 Negative Current Flow in High-Side FET of OUT2 Channel

Figure 3-33 shows the waveforms of false open-load detection when the negative-current OLD setting is disabled (OLD_NEG_EN = 0). As shown in this figure, the high-side FET of the OUT1 channel is always switched ON and the low-side and high-side FET of the OUT2 channel are operating in complimentary way (i.e. synchronous rectification mode is enabled). In synchronous rectification, the current flows in negative direction from OUT2 to VM (i.e. FET Source to Drain) during the high-side FET conduction. Initially, for the first PWM cycle, the OLD mode is disabled to show the currents in different FETs during the motor operation. When OLD is enabled in second PWM cycle, then the device registers a false open-load detect during the high-side FET conduction

as shown in

Figure 3-33. The nFAULT pin is pulled low and both high-side and low-side FET of OUT2 channels are disabled. The body diode of the high side FET (OUT2) conducts to complete the motor current path.

This false detection of open load is eliminated by enabling the negative-current OLD setting (OLD_NEG_EN = 1). As shown in **Figure 3-34**, the negative OLD current setting (IOLD_NEG) is enabled for the high-side FET of OUT2 channel. This setting allows the negative current path (from source to drain) in high-side FET. The nFAULT pin is latched high and OUT2 channel is not disabled when OLD is enabled in second PWM cycle.

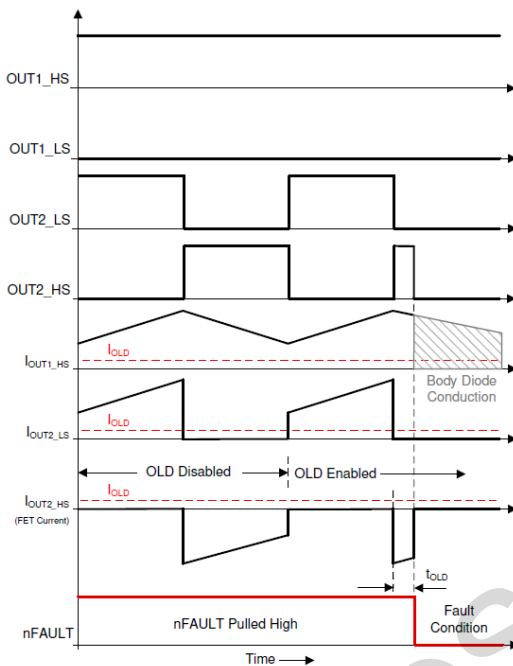


Figure 3-33 Waveforms Showing False OLD With Negative-Current OLD Disabled

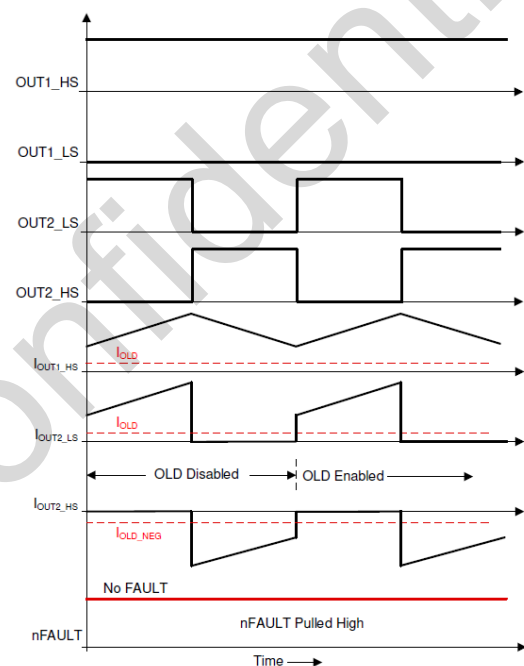


Figure 3-34 Waveforms Showing Operation with Negative-Current OLD Enabled

3.2.4.5.3 Low-current OLD

Low-current open-load detection is another type of active open-load detection in the HE99MD9008 devices. In low-current open-load detection, the current detection threshold is around 10x lower than the active open-load detection scheme. This feature gives the user flexibility to detect a valid open-load condition when driving loads that require low current.

As shown in **Figure 3-35**, if the low-side MOSFET is in operating condition (switched-ON) and the current flowing in the particular MOSFET is lower than the low-current active open-load current threshold (IOLD_LOW) for at least open-load detection deglitch time (tOLD), then an open-load condition is detected. The OLD bit in the IC status (IC_STAT) register is set, the HBX_LS_OLD bit in the open-load status register (OLD_STAT_X) is set and nFAULT pin is driven low during an open-

load detect. Normal operation resumes (driver operation and the nFAULT pin is released) when the open-load condition is removed and CLR_FLT command is issued. The OLD bit remains set until cleared through the CLR_FLT bit.

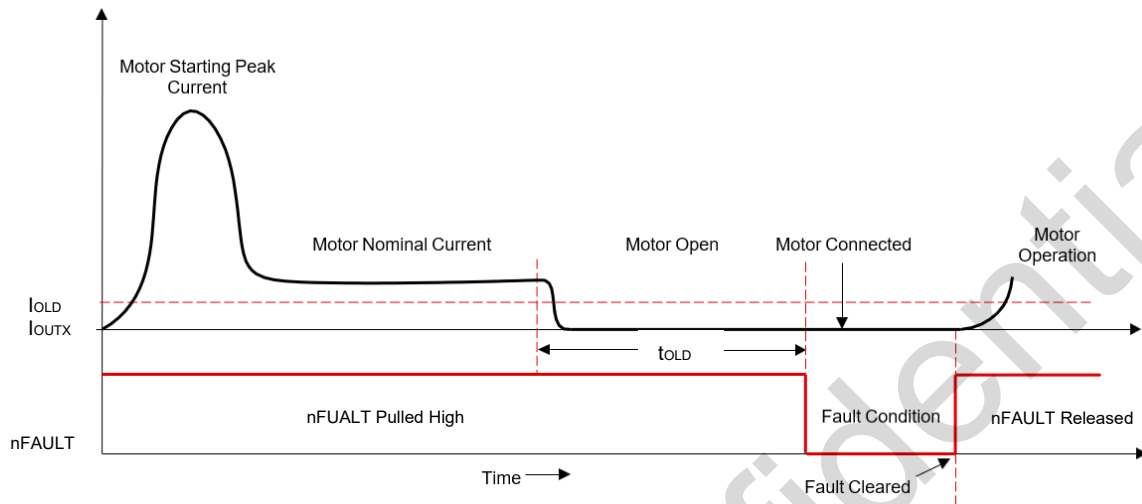


Figure 3-35 Low-Current OLD

Note: The low-current OLD has following limitations

This feature is only applicable for the current flowing in low-side FET.

Once this mode is enabled the corresponding over-current threshold for the low-side FET is also reduced by 11 times (~120 mA min.).

The RDS(on) of the low side FET will increase by 11 times (~7.5 Ω typical), hence the thermal performance has to be monitored. However, for the lower current the thermal dissipation is limited.

Figure 3-36 shows the flowchart for implementing the low-current active OLD in continuous mode of operation. Following are the steps to configure and detect the low-current active OLD in the HE99MD9008 device.

1. Enable OLD by setting the OLD_OP bit in OLD_CTRL_2 register. This setting will ensure that the OUTX outputs continue to operate when and OLD fault occurs.
2. Enable the full-bridge operation by setting the individual HBX_HS_EN/HBX_LS_EN bits in operation control (OP_CTRL_1, OP_CTRL_2 and OP_CTRL_3) registers.
3. Check for the nFAULT pin status and the OLD fault in the IC_STAT register.
4. If the nFAULT pin is low and the OLD fault is high, then check for the individual HBX_HS_OLD/HBX_LS_OLD bits in OLD status (OLD_STAT_1, OLD_STAT_2 and OLD_STAT_3) registers.
5. Disable OLD using the HBX_OLD_DIS bits for the OUTX pins acting as high-side drivers.
6. Enable the low-current OLD mode for the half-bridge which low-side is operating by using the HBX_LOLD_EN bit in OLD control (OLD_CTRL_3 and OLD_CTRL_4) registers. This will also disable the high-side OLD for the particular half-bridge.
7. Wait for the open-load deglitch time (tOLD).
8. Issue the clear fault command (CLR_FLT) to release the nFAULT pin and clear the OLD bits if low-current OLD is not detected.
9. If the OLD bit is high and the nFAULT pin is not released (low), then low-current OLD fault is detected.

Note: The low-current OLD is applicable only for low-side FETs. The user has to enable the low-

current OLD mode for the corresponding low-side FET.

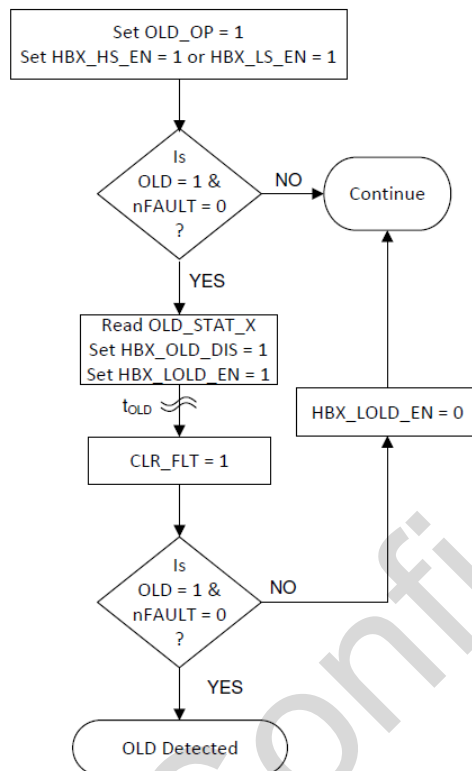


Figure 3-36 Flowchart for Enabling Low-Current OLD (Continuous Mode)

3.2.4.5.4 Passive OLD

In passive OLD, the detection of open load is carried before the driver is turned on. The state of all FETs remains in Hi-Z state, while a minimal amount of current flows through motor for short amount of time to test the motor connection. The diagnostic current is very small to avoid causing the motor rotation.

Figure 3-37, Figure 3-38, and Figure 3-39 show the circuit implementations of passive OLD. As shown in these figures, a constant current source pulls the OUT1 pin to the AVDD (internal fixed voltage) which allows current flow from OUT1 to OUT2 terminal. The current drawn is completely dependent on the motor resistance between OUT1 and OUT2 and limited by the internal current sourcing (IOL_PU) and sinking (IOL_PD) capability of the passive OLD circuitry.

Depending on this current and the comparator threshold voltage (VOL_HS and VOL_LS), the comparator output OL1_HS and OL2_LS are either set or reset which determines the open-load status. When an open load is detected, the OLD bit in the IC status (IC_STAT) register is set, the HBX_LS_OLD bit in the open-load status register (OLD_STAT_X) is set and nFAULT pin is driven low. The OLD bit remains set until cleared through the CLR_FLT bit. This implementation is applicable for half-bridge driving as shown in Figure 3-38, and Figure 3-39.

*Note: Passive OLD sequence is not disabled by the HBX_OLD_DIS bits.
In Full Bridge Operation, diagnosis information of passive OLD is only valid for HS.*

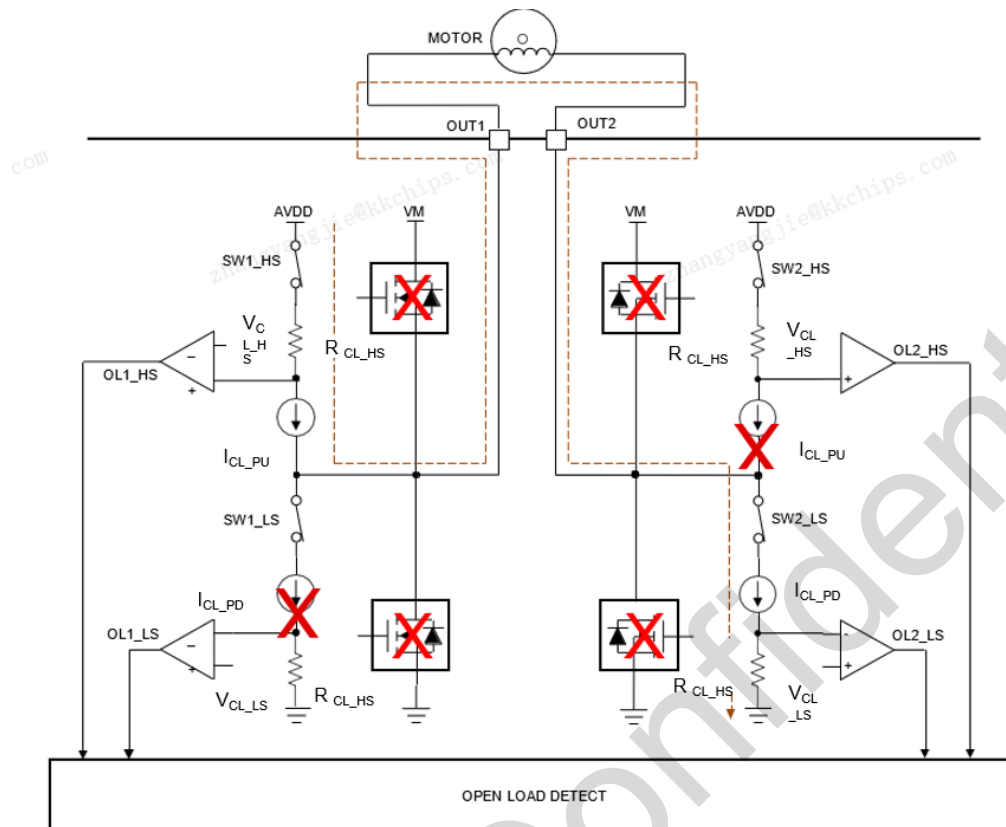


Figure 3-37 Passive OLD Circuit for a Load Driven in Full-Bridge Operation

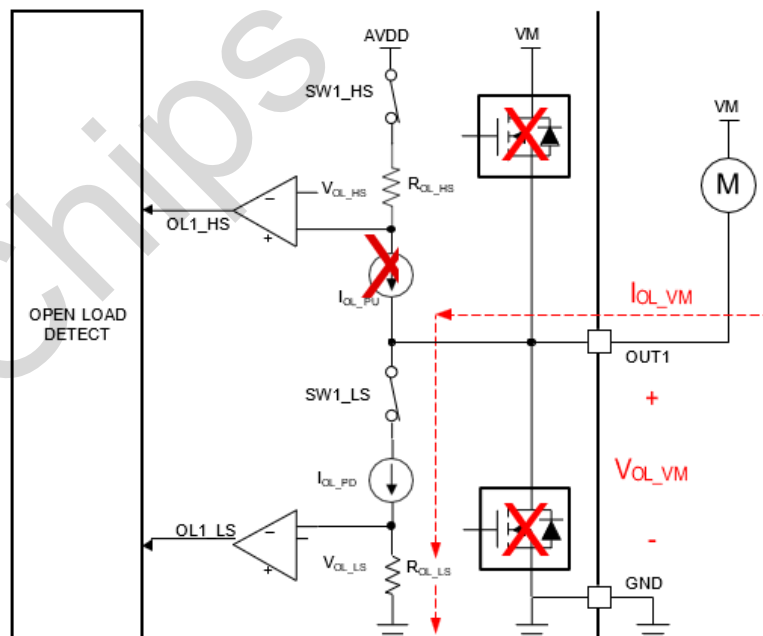


Figure 3-38 Passive OLD Circuit for Half-Bridge Operation with Load Connected to VM

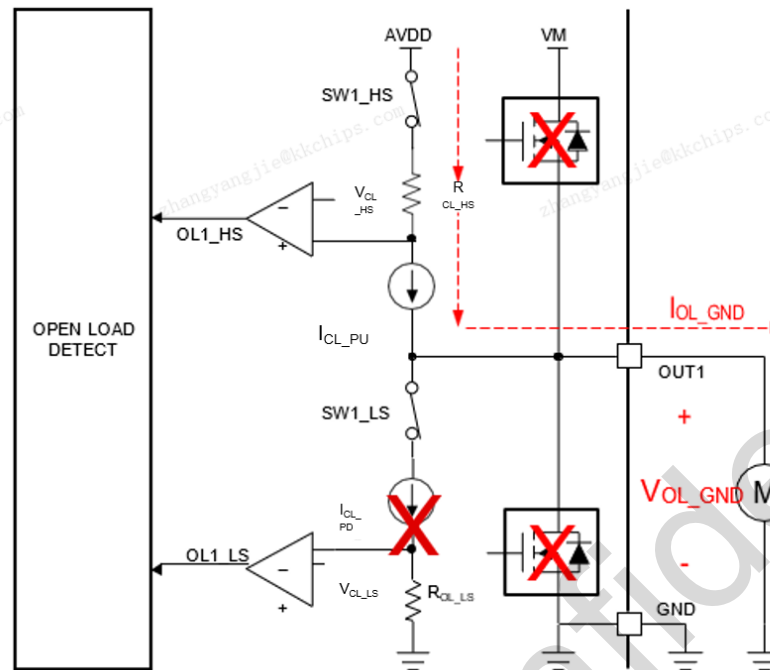


Figure 3-39 Passive OLD Circuit for Half-Bridge Operation with Load Connected to GND

Following are the steps to configure and detect the passive OLD in the HE99MD9008 device.

1. Enable the passive OLD mode for the individual half-bridges which is to be diagnosed using the Half-bridge passive OLD enable bits (HBX_POLD_EN) in OLD_CTRL_5 register.
2. Enable the passive OLD using the Passive OLD Enable bit (POLD_EN) in CONFIG_CTRL register. Setting the POLD_EN bit enables the passive OLD detection circuit on all OUTx pin for which the corresponding HBX_HS_EN or HBX_LS_EN are set to 1.
3. Configure the half-bridge operation control bits (HBX_HS_EN/HBX_LS_EN) in operation control register (OP_CTRL_X) to determine the high-side and low-side OLD check. Note that these bits are now used for the passive OLD configuration. If anytime, the HBX_POLD_EN is reset then the bridge starts operating.
4. Wait for the passive OLD time as determined by the user.
5. After the completion of passive OLD time, disable the passive OLD enable bit (POLD_EN).
6. Monitor the nFAULT pin / OLD bit in Status register (IC_STAT) and the HBX_HS_OLD/HBX_LS_OLD bit in the OLD status registers (OLD_STAT_X) for any open-load detection.
7. Restart the sequence for other half-bridges / full-bridges.

During normal driving, HBX_HS_EN and HBX_LS_EN bits control the state of OUTX. However, when POLD_EN and HBX_POLD_EN are 1, the OUTX channel is disabled, and HBX_HS_EN and HBX_LS_EN control SWX_HS and SWX_LS used for passive OLD (see schematic representation in [Figure 3-37](#)). [Table 3-9](#) shows the truth table for this operation.

Table 3-9 Truth Table for Passive OLD

POLD_EN	HBX_POLD_EN	HBX_HS_EN	HBX_LS_EN	OUTX	SWX_HS	SWX_LS	OPEN LOAD SEQUENCE
0	0	X	X	Follows HBX_HS_EN and HBX_LS_EN	Open	Open	Passive OLD for the channel is disabled and HBX_HS_EN/HBX_LS_EN set output OUTx state
1	1	0	0	Z	Open	Open	Off state - no passive OLD
1	1	0	1	Z	Open	Closed	Valid passive OLD for VM-connected load
1	1	1	0	Z	Closed	Open	Valid passive OLD for GND-connected load
1	1	1	1	Z	Open	Open	Invalid state

Note:

1. The OLD_REP bit works in a similar way as for the active OLD. The OLD_OP bit is not applicable for passive OLD operation since the outputs are already disabled.
2. Passive OLD sequence is not enabled if any other fault (other than OCP/OLD) is present.

Table 3-10 shows an example for configuring passive OLD for various loads. The HBX_VM_POLD bits can be enabled for any loads that connect directly to VM. In cases where VM is low, the passive OLD current may need to be larger so passive OLD does not falsely indicate an open load. Setting HBX_VM_POLD = 1 chooses a smaller ROL so more current flows and the device can properly detect an open load.

Table 3-10 Passive OLD Configurations

CONNCTIO N	HB 1_V M_ PO LD	HB2_VM _POLD	HB1_H S_EN	HB1_L S_EN	HB2_H S_EN	HB2_L S_EN	OPEN-LOAD DETECTION
Full-Bridge Operation (Motor Connected Between OUT1 and OUT2)	0	0	1	0	0	1	Detection based on resistance threshold (Forward Connection)
	0	0	0	1	1	0	Detection based on resistance threshold (Reverse Connection)
	0	0	1	0	1	0	Invalid case (both high-side OLD circuitry operating)
	0	0	0	1	0	1	Invalid case (both low-side OLD circuitry operating)
Half-Bridge Operation (Load Connected Between OUT1/2 and VM)	1	X	0	1	0	0	Detection only for OUT1 channel based on resistance threshold
	X	1	0	0	0	1	Detection only for OUT2 channel based on resistance threshold
	1	1	0	1	0	1	Detection for both outputs based on resistance threshold
	X	X	1	0	0	0	Invalid case (OUT1 high-side OLD circuitry is operating for VM connected load)
	X	X	0	0	1	0	Invalid case (OUT2 high-side OLD circuitry is operating for VM connected load)
	X	X	1	0	1	0	Invalid case (both high-side OLD circuitry is operating for VM connected load)
Half-Bridge Operation (Load Connected Between OUT1/2 and GND)	0	0	1	0	0	0	Detection only for OUT1 channel based on resistance threshold
	0	0	0	0	1	0	Detection only for OUT2 channel based on resistance threshold
	0	0	1	0	1	0	Detection for both outputs based on resistance threshold
	0	0	0	1	0	0	Invalid case (OUT1 low-side OLD circuitry is operating for GND connected load)
	0	0	0	0	0	1	Invalid case (OUT2 low-side OLD circuitry is operating for GND connected load)
	0	0	0	1	0	1	Invalid case (both low-side OLD circuitry is operating for GND connected load)

3.2.4.6 Digital Thermal Clusters (to be defined after Floor Plan is completed)

In order to provide an Advanced On-chip Temperature Control, the power outputs are grouped in clusters with dedicated thermal sensors. The sensors are suitably located on the device (add picture). In case the temperature of an output cluster reaches the thermal shutdown threshold, the outputs assigned to this cluster are shut down (all other outputs remain active). Each output cluster has a dedicated temperature warning and shutdown flag (add SR). Thermal clusters can be configured using bit TSD_Cluster_EN (CRxx):

- Standard Mode (default): as soon as any cluster reaches thermal threshold the device is switched off.
- Cluster Mode: only the cluster that has reached shutdown temperature is switched off

3.2.4.6.1 Thermal Warning (OTW) – Per Cluster

If the die temperature exceeds the trip point of the thermal warning (T_{OTW}), the OTW bit is set in the IC status (IC_STAT) register. The reporting of OTW on the nFAULT pin can be enabled by setting the over-temperature warning reporting (OTW_REP) bit in the configuration control (CONFIG_CTRL) register. The device performs no additional action and continues to function. In this case, the nFAULT pin releases when the die temperature decreases below the hysteresis point of the thermal warning (T_{OTW_HYS}). The OTW bit remains set until cleared through the CLR_FLT bit and the die temperature is lower than thermal warning trip (T_{OTW}).

Note: Over Temperature warning is not reported on nFAULT pin by default.

3.2.4.6.2 Thermal Shutdown (OTSD) – Per Cluster or per IC

If the die temperature exceeds the trip point of the thermal shutdown limit (T_{OTSD}), all half-bridge drivers are disabled, the charge pump is shut down, and the nFAULT pin is driven low. In addition, the OTSD bit is latched high in IC status (IC_STAT) register. Normal operation resumes (driver operation and the nFAULT pin is released) when the overtemperature shutdown condition is removed and die temperature decreases below the hysteresis point of the thermal warning (T_{OTSD_HYS}). The OTSD bit remains latched high indicating that a thermal event occurred until a clear fault command is issued through the CLR_FLT bit. This protection feature cannot be disabled.

3.3 Device Functional Modes

3.3.1 State Diagram

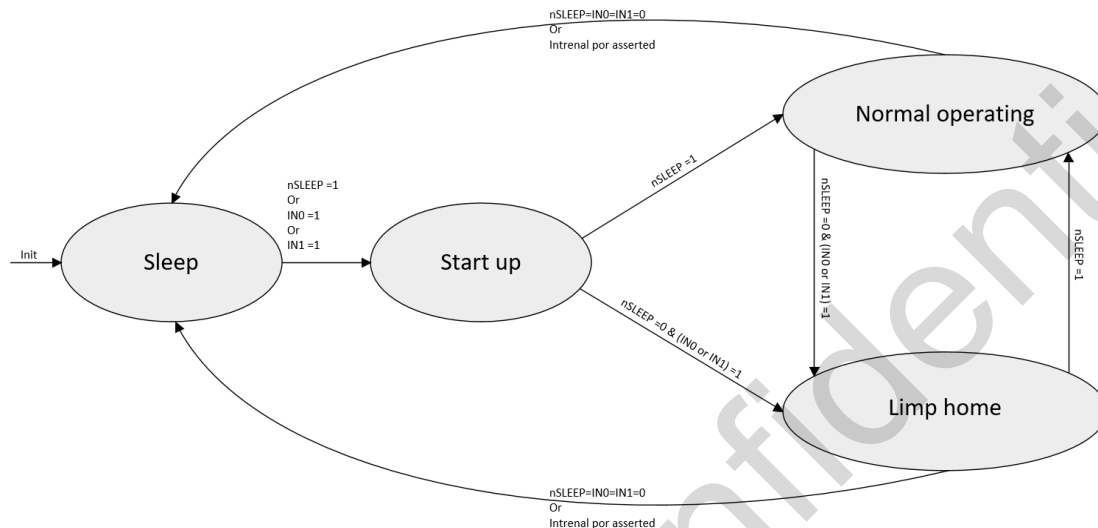


Figure 3-40 HE99MD9008 state diagram

3.3.2 Sleep Mode (nSLEEP = IN0 = IN1 = 0)

The nSLEEP pin manages the state of the HE99MD9008 device. When the nSLEEP pin is low, the device enters a low-power sleep mode. In sleep mode, all half-bridge drivers are disabled, the internal charge pump is disabled, the internal regulators are disabled, and the SPI bus is disabled. The tSLEEP time must elapse after a falling edge on the nSLEEP pin before the device enters sleep mode. The device comes out of sleep mode automatically if the nSLEEP pin is pulled high. The tWAKE time must elapse before the device is ready for inputs.

3.3.3 Start Up

Start-up mode is the transition state from sleep mode to normal operating mode or limp home. Some initial tests are executed during this state. Start-up takes less than 0.5ms. When start up finished device is ready to operate in normal operating mode or limp home mode.

During this state, device will detect OUT1 & OUT2 load connection, to decide the channel works as HS or LS in limp home mode. The automatic detection result is reported in OUTx_HSLS bits in ADD_STAT_1 register.

3.3.4 Normal Operating Mode (nSLEEP = 1)

When the nSLEEP pin is high and $V_{VM} > V_{UVLO}$, the device enters operating mode. The tWAKE time must elapse before the device is ready for inputs. In this mode the half bridge drivers, charge pump, internal regulators, and SPI bus are active. Table 3-11 summarizes the different operating modes of HE99MD9008 device.

Table 3-11 Functional Modes

MODE	CONDITION	HALF-BRIDGES	INTERNAL CIRCUITS
Normal Operating	4.5-V < V_{VM} < 24-V (EXT_OVP = 0b) 4.5-V < V_{VM} < 28-V (EXT_OVP = 1b) nSLEEP Pin = High	Operating	Operating
Sleep	4.5-V < V_{VM} < 28-V nSLEEP Pin = Low IN0 IN1 = Low	Disabled	Disabled
Limp home	4.5-V < V_{VM} < 28-V nSLEEP Pin = Low IN0 IN1 = High	Operating	Operating

3.3.5 Limp Home Mode (nSLEEP = 0, (IN0 | IN1) = 1)

The Limp is intended to handle limited functionality of the device in critical faulty situations. A typical application scenario is reported in [Figure 3-41](#), with a system which is reconfigured during Limp Home as shown in [Figure 3-42](#).

Should the micro that controls the device fail, an external safety micro takes over controlling parallel inputs IN0 and IN1. The SBC will force the nSLEEP signal low which brings the device in Limp Home. Once the switching activity on IN0, IN1 is over, being the nSLEEP pin still de-asserted, the device will reach Sleep mode. During this mode active, HE99MD9008 is only capable of driving two outputs i.e., OUT1 through IN0 and OUT2 through IN1; such a mapping is fixed and overcomes an eventual different mapping programmed via SPI once the Limp Home is reached. In limp home mode, HS or LS configuration of the limp home channels are automatically detected depending on connection of the external load. When INx is high, the corresponding driver FET is switched on, and the other half FET is always off. The result of auto-configuration is reported in OUTx_HSLS bits in ADD_STAT_1 register. In case "load error" reported or user prefer to reconfigure the HS or LS configuration in limp home mode, uC can rewrite OUTx_HSLS bits in normal operating mode.

Limp home mode is available as long as $V_{VM} > V_{UVLO}$. If $V_{VDD} > V_{VDDUV}$, SPI is available but in read-only mode. If $V_{VDD} < V_{VDDUV}$, SPI function will be disabled.

Detailed Description

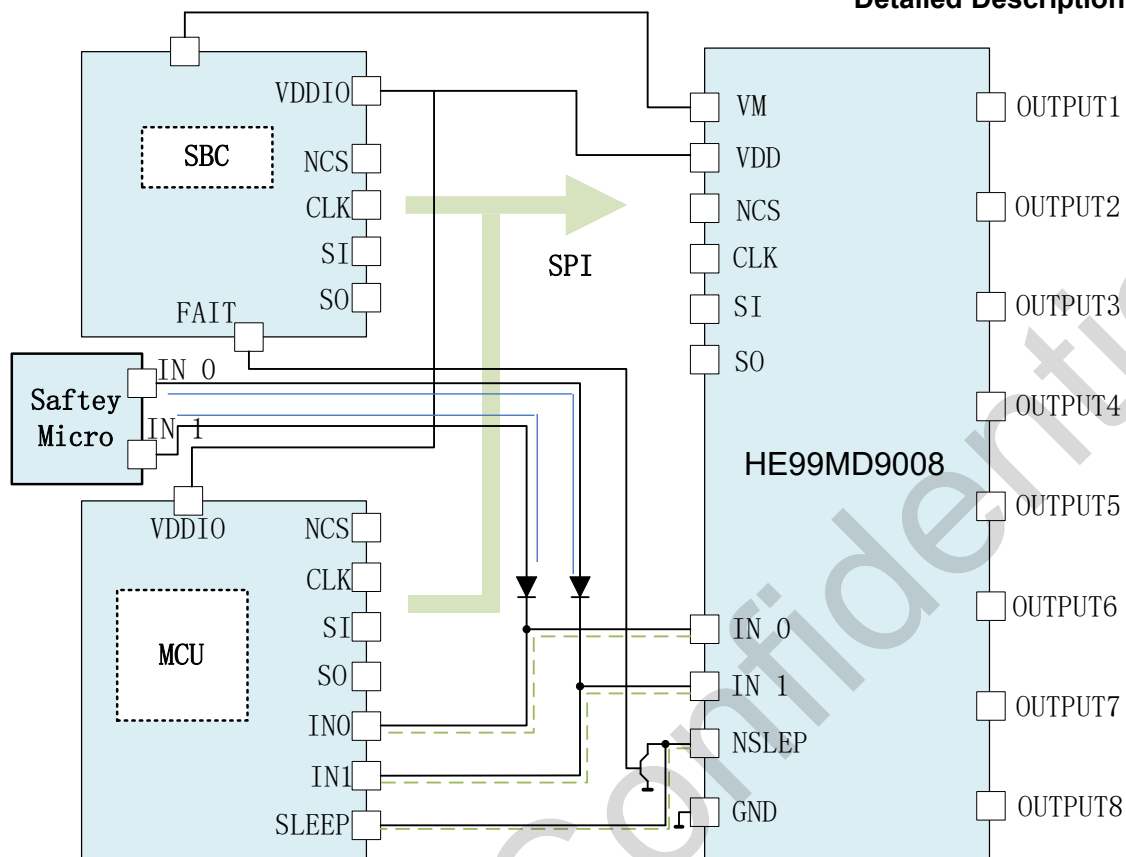


Figure 3-41 Possible system configuration during normal operating conditions

In [Figure 3-41](#), the dotted green lines mark the signals path when the device is operating in normal node, nSLEEP, IN0, IN1 are provided by the main MCU. [Figure 3-42](#) shows a possible Limp Home configuration where the dotted bluelines mark the signals path.

Detailed Description

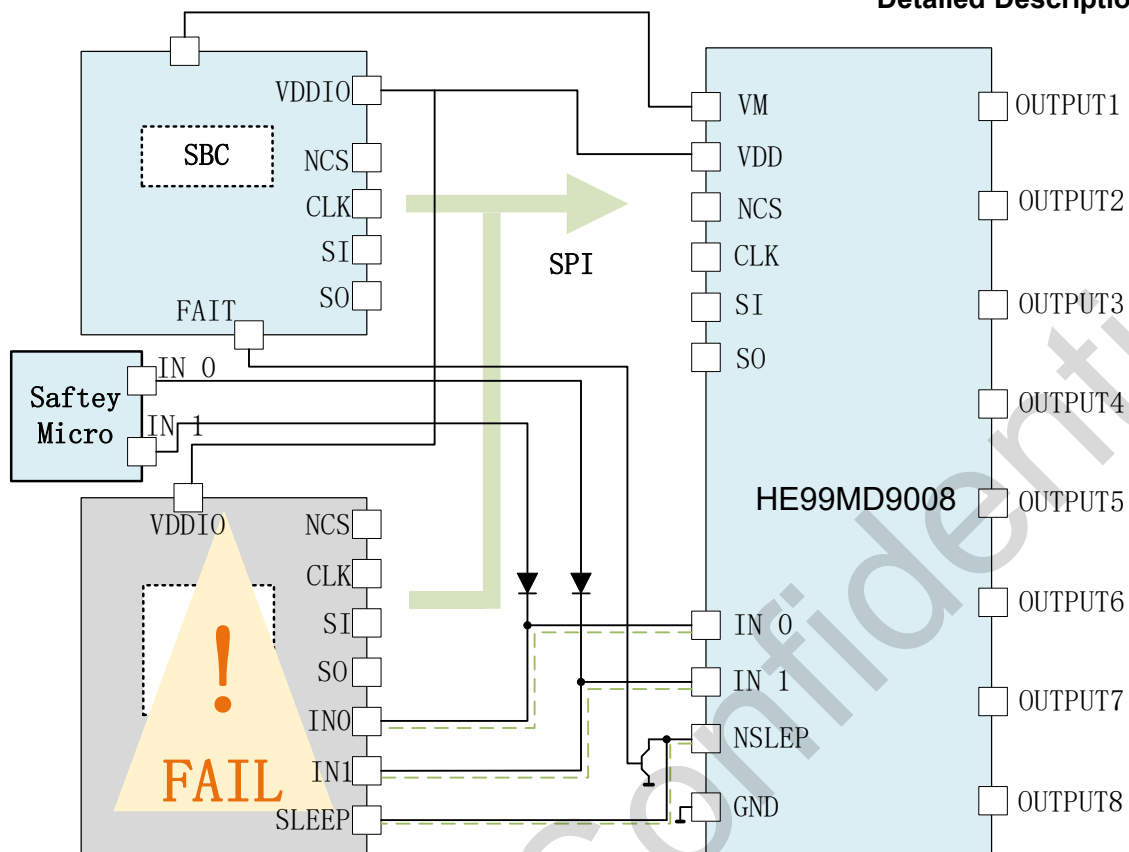


Figure 3-42 Possible system configuration during Limp mode

3.3.6 Fault Table

The HE99MD9008 is protected against various faults as summarized in [Table 3-12](#).

Table 3-12 Fault Action and Response

FAULT	CONDITION	CONFIGURATION	REPORT	HALF-BRIDGE	LOGIC	RECOVERY
VM Undervoltage (UVLO)	$V_{VM} < V_{UVLO}$ (Max. 4.5-V)	—	nFAULT Pin IC_STAT Register	Hi-Z	Active	Automatic: $V_{VM} > V_{UVLO}$
VDD Undervoltage (UVLO)	$V_{VDD} < V_{POR}$ (Max 3-V)	—	IC_STAT Register	Hi-Z	Reset	Automatic: $V_{VDD} > V_{POR}$
VM Overvoltage (OVP)	$V_{VM} > V_{OVP}$ (Min. 24-V)	EXT_OVP = 0	nFAULT Pin IC_STAT Register	Hi-Z	Active	Automatic: $V_{VM} < V_{OVP}$
	$V_{VM} > V_{OVP}$ (Min. 32-V)	EXT_OVP = 1				
Over Current Protection (OCP)	$I_{OUT} > I_{OCP}$ (Min. 1.3-A)	OCP_REP = 0	IC_STAT Register	Hi-Z	Active	CLR_FLT = 1 & $I_{OUT} < I_{OCP}$
		OCP_REP = 1	nFAULT Pin IC_STAT Register	Hi-Z	Active	
Open-Load Detect (OLD)	$I_{OUT} < I_{OLD}$ (Max. 15-mA)	OLD_OP = 0 OLD_REP = 0	nFAULT Pin IC_STAT Register	Hi-Z	Active	CLR_FLT = 1 & $I_{LOAD} > I_{OLD}$
		OLD_OP = 0 OLD_REP = 1	IC_STAT Register	Hi-Z	Active	
		OLD_OP = 1 OLD_REP = 0	nFAULT Pin IC_STAT Register	Operating	Active	
		OLD_OP = 1 OLD_REP = 1	IC_STAT Register	Operating	Active	
	$R_{LOAD} > R_{OLD}$ (Max. 100-k Ω)	OLD_REP = 0	nFAULT Pin IC_STAT Register	N/A	Active	CLR_FLT = 1 & OLD Sequenced & $R_{LOAD} < R_{OLD}$
		OLD_REP = 1	IC_STAT Register	N/A	Active	
Over-Temperature Warning (OTW)	$T_J > T_{OTW}$ (Min. 120°C)	OTW_REP = 0	IC_STAT Register	Operating	Active	No Action
		OTW_REP = 1	nFAULT Pin IC_STAT Register	Operating	Active	Automatic: $T_J < T_{OTW} - T_{OTW_HYS}$
Over-Temperature Shutdown (OTSD)	$T_J > T_{OTSD}$ (Min. 150°C)	—	nFAULT Pin IC_STAT Register	Hi-Z	Active	Automatic: $T_J < T_{OTSD} - T_{OTSD_HYS}$

3.3.7 Functional Table

At $T_J = -40^{\circ}\text{C}$ to $+150^{\circ}\text{C}$, $V_{VM} > V_{UVLO}^{(1)}$

Table 3-13 HE99MD9008 Functional Table

	Normal Operating Mode		Limp Home Mode	
	VDD>VDDUV	VDD<VDDUV	VDD>VDDUV	VDD<VDDUV
SPI	available	not available	available(read-only)	not available
Channel on/off	according to SPI command/ PWM generators/ INx	according to previous command ⁽²⁾	according to INx	according to INx
Parallel mode	√	√	x	x
VM undervoltage	√	√	√	√
VM overvoltage	√	√	√	√
Overcurrent	√	√	√	√
Open load	√	√	√	√
Thermal warning	√	√	√	√
Thermal shutdown	√	√	√	√
Fault pin	√	√	√	√

3.4 Programming

3.4.1 SPI

SPI bus is used to set device configurations, operating parameters, and read out diagnostic information on the HE99MD9008 device. The SPI operates in slave mode and connects to a master controller. The SPI input data (SDI) word consists of a 16-bit word, with an 8-bit command and 8 bits of data. The SPI output data (SDO) word consists of 8-bit register data and the first 8 bits make up the Status Register with Fault Status indication. The data sequence between the MCU and the SPI slave driver is shown in Figure 3-43.



Figure 3-43 SPI Data Frame

A valid frame must meet the following conditions:

- The SCLK pin should be low when the nSCS pin transitions from high to low and from low to high.
- The nSCS pin should be pulled high for at least 400 ns between words.
- When the nSCS pin is pulled high, any signals at the SCLK and SDI pins are ignored and the SDO pin is placed in the Hi-Z state.
- Data is captured on the falling edge of SCLK and data is propagated on the rising edge of SCLK.
- The most significant bit (MSB) is shifted in and out first.
- A full 16 SCLK cycles must occur for transaction to be valid.
- If the data word sent to the SDI pin is less than or more than 16 bits, a frame error occurs and the data word is ignored.
- For a write command, the existing data in the register being written to is shifted out on the SDO pin following the 8-bit command data.

3.4.2 SPI Format

The SDI input data word is 16 bits long and consists of the following format:

- 1 read or write bit, W (bit B14)
- 6 address bits, A (bits B13 through B8)
- 8 data bits, D (bits B7 through B0)

The SDO output data word is 16 bits long and the first 8 bits makes up the IC status register. The

Detailed Description

report word is the content of the register being accessed.

For a write command ($W0 = 0$), the response word on the SDO pin is the data currently in the register being written to.

For a read command ($W0 = 1$), the response word is the data currently in the register being read.

Table 3-14 SDI Input Data Word Format

		R/ W	Address						Data							
Bit	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
Data	0	W0	A5	A4	A3	A2	A1	A0	D7	D6	D5	D4	D3	D2	D1	D0

Table 3-15 SDO Output Data Word Format

	IC Status								Report							
Bit	B15	B14	B13	B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
Data	1	1	OT	OLD	OC	UVLO	OV	NPOR	D7	D6	D5	D4	D3	D2	D1	D0

3.4.3 SPI Interface for Multiple Slaves

Multiple HE99MD9008 devices can be connected to the master controller with and without the daisy chain. For connecting a 'n' number of HE99MD9008 to a master controller without using a daisy chain, 'n' number of I/O resources from master controller has to be utilized for nSCS pins as shown [Figure 3-44](#). Whereas, if the daisy chain configuration is used, then a single nSCS line can be used for connecting multiple HE99MD9008 devices as shown in [Figure 3-45](#).

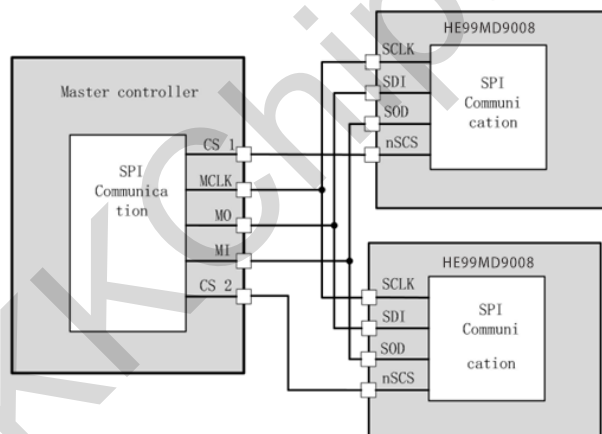


Figure 3-44 SPI Operation Without Daisy Chain

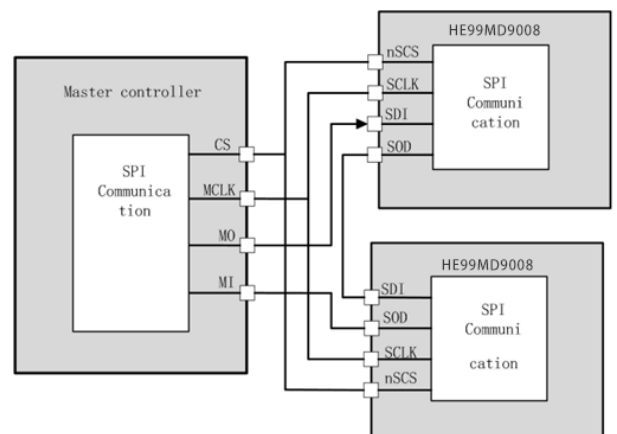


Figure 3-45 SPI Operation With Daisy Chain

3.4.3.1 SPI Interface for Multiple Slaves in Daisy Chain

The HE99MD9008 device can be connected in a daisy chain configuration to save GPIO ports when multiple devices are communicating to the same MCU. Figure 3-46 shows the topology when 3 devices are connected in series with waveforms.

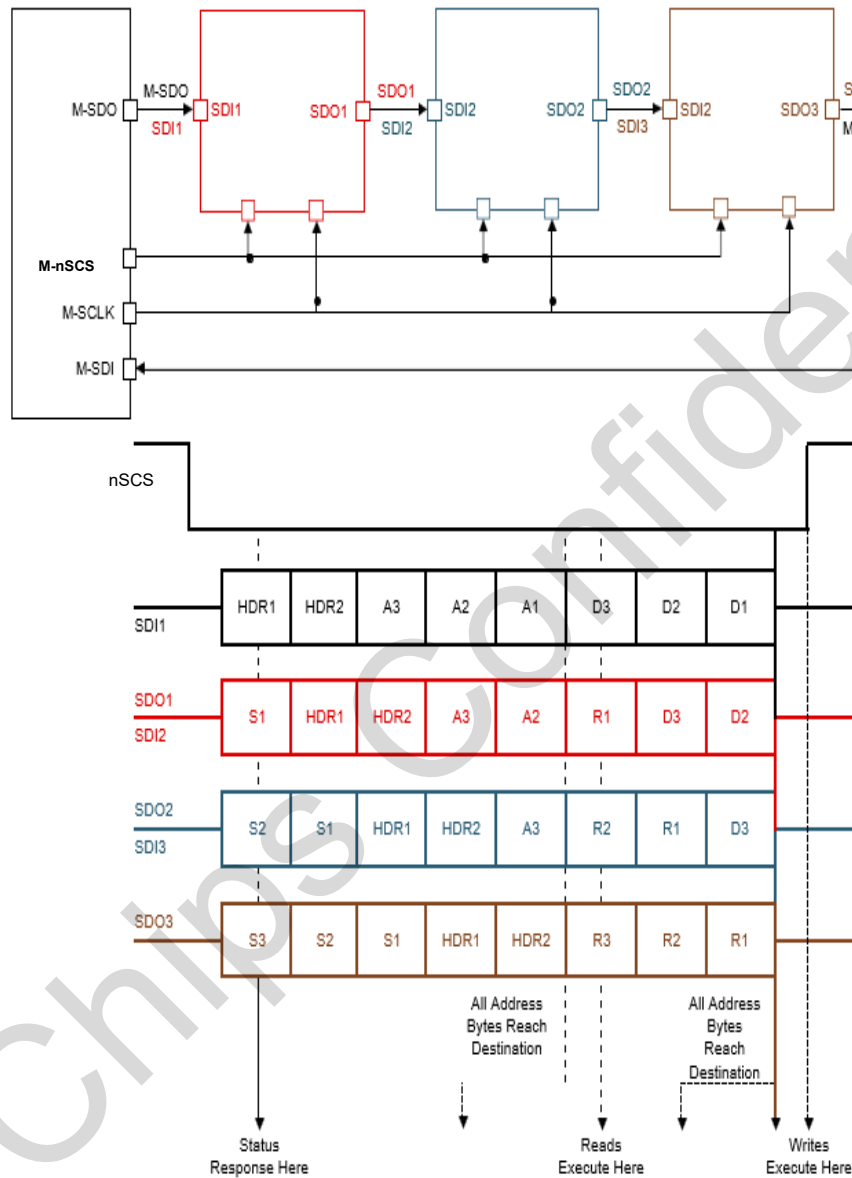


Figure 3-46 Daisy Chain SPI Operation

The first device in the chain shown above receives data from the master controller in the following format. See SDI1 in [Figure 3-46](#)

- 2 bytes of Header
- 3 bytes of Address
- 3 bytes of Data

After the data has been transmitted through the chain, the master controller receives it in the following format. See SDO3 in [Figure 3-46](#)

- 3 bytes of Status
- 2 bytes of Header (should be identical to the information controller sent)
- 3 bytes of Report

The Header bytes contain information of the number of devices connected in the chain, and a global clear fault command that will clear the fault registers of all the devices on the rising edge of the chip select (nSCS) signal. N5 through N0 are 6 bits dedicated to show the number of devices in the chain as shown in [Figure 3-47](#). Up to 63 devices can be connected in series per daisy chain connection.

The 5 LSBs of the HDR2 register are don't care bits that can be used by the MCU to determine integrity of the daisy chain connection. Header bytes must start with 1 and 0 for the two MSBs.

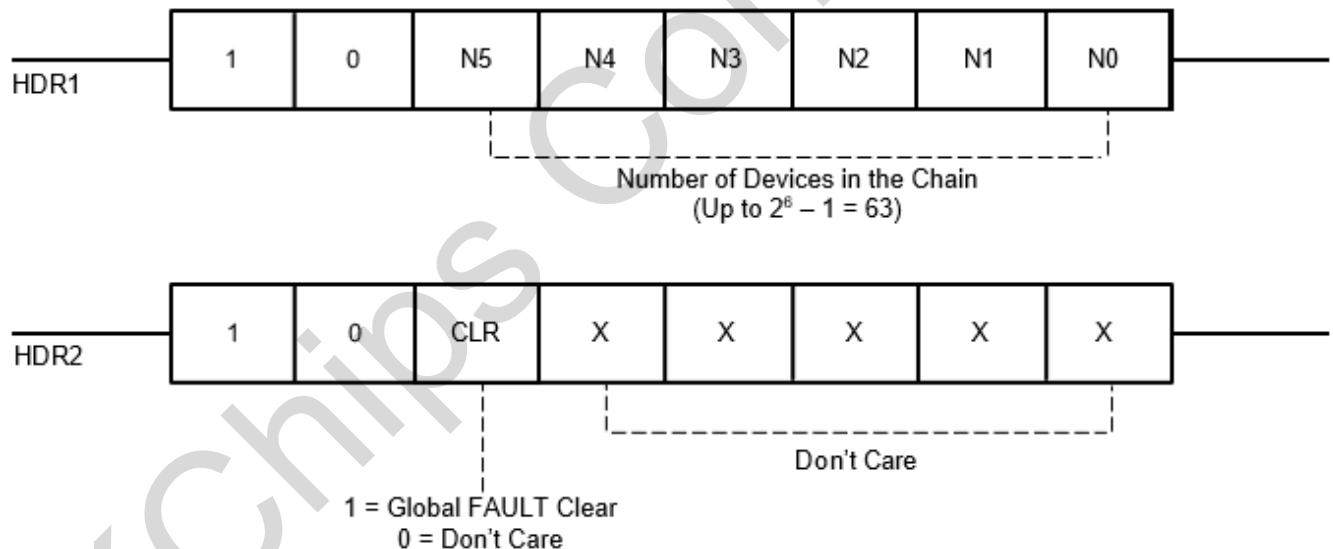


Figure 3-47 Header Bits

The Status byte provides information about the fault status register for each device in the daisy chain as shown in [Figure 3-48](#). That way the master controller does not have to initiate a read command to read the fault status from any particular device. This saves the controller additional read commands and makes the system more efficient to determine fault conditions flagged in a device.

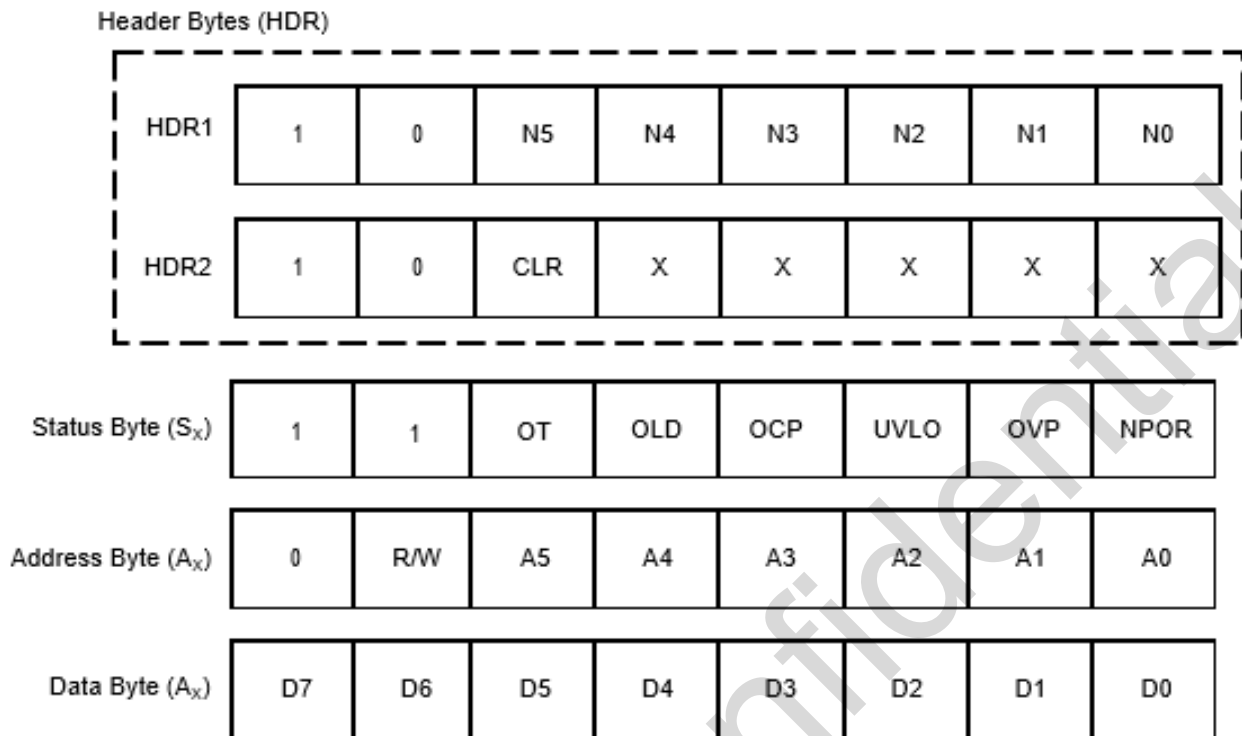


Figure 3-48 Daisy Chain Read Registers

When data passes through a device, it determines the position of itself in the chain by counting the number of Status bytes it receives following by the first Header byte. For example, in this 3 device configuration, device 2 in the chain will receive two Status bytes before receiving HDR1 byte, followed by HDR2 byte.

From the two Status bytes it knows that its position is second in the chain, and from HDR2 byte it knows how many devices are connected in the chain. That way it only loads the relevant address and data byte in its buffer and bypasses the other bits. This protocol allows for faster communication without adding latency to the system for up to 63 devices in the chain.

The address and data bytes remain the same with respect to a single device connection. The Report bytes (R1 through R3), as shown in the figure above, is the content of the register being accessed.

Detailed Description

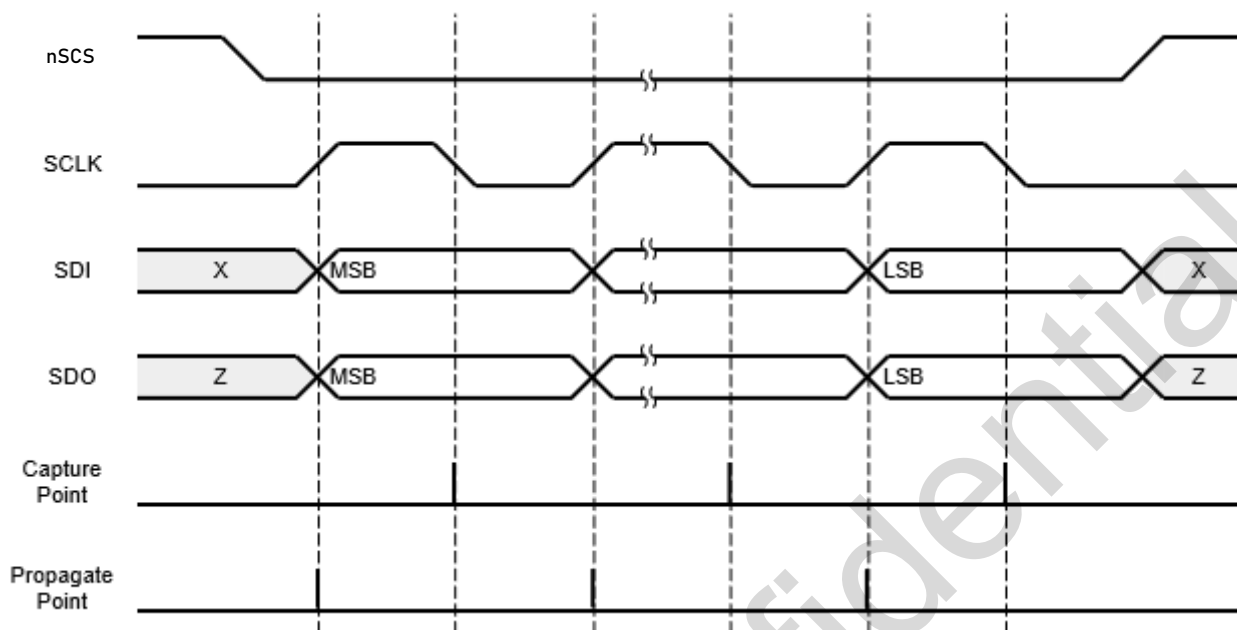


Figure 3-49 SPI Slave Timing Diagram

3.5 Register Map

This section contains the register maps and bit descriptions for all of the HE99MD9008 devices. [Table 3-16](#) summarizes the differences among the part numbers in the HE99MD9008

Table 3-16 Summary of HE99MD9008 Device Family

DEVICE	NUMBER OF HALF-BRIDGES	NUMBER OF PWM GENERATORS	OPEN-LOAD DETECTION SCHEMES	LINK TO REGISTER MAP
HE99MD9008	8	8	Passive OLD, Active OLD, Low-Current Active OLD, Negative-Current Active OLD	Table 2-18

Complex bit access types are encoded to fit into small table cells. [Table 3-17](#) shows the codes that are used for access types in this section.

Table 3-17 Control Registers Access Type Codes

Access Type	Code	Description
Read Type		
R	R	Read
Write Type		
W	W	Write
Reset or Default Value		
-n		Value after reset or the default value

3.5.1 HE99MD9008 Register Maps

Table 3-18 HE99MD9008 Register Map

Name	7	6	5	4	3	2	1	0	Type	Address
IC_STAT	Reserved	OTSD	OTW	OLD	OCP	UVLO	OVP	NPOR	R	00h
OCP_STAT_1	HB4_HS_O CP	HB4_LS_OC P	HB3_HS_O CP	HB3_LS_O CP	HB2_HS_O CP	HB2_LS_O CP	HB1_HS_O CP	HB1_LS_O CP	R	01h
OCP_STAT_2	HB8_HS_O CP	HB8_LS_OC P	HB7_HS_O CP	HB7_LS_O CP	HB6_HS_O CP	HB6_LS_O CP	HB5_HS_O CP	HB5_LS_O CP	R	02h
OCP_STAT_3	Reserved								R	03h
OLD_STAT_1	HB4_HS_O LD	HB4_LS_OL D	HB3_HS_O LD	HB3_LS_O LD	HB2_HS_OL D	HB2_LS_O LD	HB1_HS_O LD	HB1_LS_O LD	R	04h
OLD_STAT_2	HB8_HS_O LD	HB8_LS_OL D	HB7_HS_O LD	HB7_LS_O LD	HB6_HS_OL D	HB6_LS_O LD	HB5_HS_O LD	HB5_LS_O LD	R	05h
OLD_STAT_3	Reserved								R	06h
CONFIG_CTRL	POLD_EN	IC_ID			OCP_REP	OTW_REP	EXT_OVP	CLR_FLT	RW	07h
OP_CTRL_1	HB4_HS_E N	HB4_LS_E N	HB3_HS_E N	HB3_LS_E N	HB2_HS_E N	HB2_LS_E N	HB1_HS_E N	HB1_LS_E N	RW	08h
OP_CTRL_2	HB8_HS_E N	HB8_LS_E N	HB7_HS_E N	HB7_LS_E N	HB6_HS_E N	HB6_LS_E N	HB5_HS_E N	HB5_LS_E N	RW	09h
OP_CTRL_3	Reserved								RW	0Ah
PWM_CTRL_1	HB8_PWM	HB7_PWM	HB6_PWM	HB5_PWM	HB4_PWM	HB3_PWM	HB2_PWM	HB1_PWM	RW	0Bh
PWM_CTRL_2	PWM_CH8 DIS	PWM_CH7 DIS	PWM_CH6 DIS	PWM_CH5 DIS	PWM_CH4 DIS	PWM_CH3 DIS	PWM_CH2 DIS	PWM_CH1 DIS	RW	0Ch
FW_CTRL_1	HB8_FW	HB7_FW	HB6_FW	HB5_FW	HB4_FW	HB3_FW	HB2_FW	HB1_FW	RW	0Dh
FW_CTRL_2	Reserved								RW	0Eh
PWM_MAP_CT RL_1	Reserved		HB2_PWM_MAP			HB1_PWM_MAP			RW	0Fh
PWM_MAP_CTR L_2 ⁽¹⁾	Reserved		HB4_PWM_MAP			HB3_PWM_MAP			RW	10h ⁽¹⁾
PWM_MAP_CT RL_3	Reserved		HB6_PWM_MAP			HB5_PWM_MAP			RW	11h
PWM_MAP_CT RL_4	Reserved		HB8_PWM_MAP			HB7_PWM_MAP			RW	12h
PWM_FREQ_C TRL_1	PWM_CH4_FREQ		PWM_CH3_FREQ		PWM_CH2_FREQ		PWM_CH1_FREQ		RW	13h
PWM_FREQ_C TRL_2	PWM_CH8_FREQ		PWM_CH7_FREQ		PWM_CH6_FREQ		PWM_CH5_FREQ		RW	14h
PWM_DUTY_CT RL_1	PWM_DUTY_CH1								RW	15h
PWM_DUTY_CT RL_2	PWM_DUTY_CH2								RW	16h
PWM_DUTY_CT RL_3	PWM_DUTY_CH3								RW	17h
PWM_DUTY_CT RL_4	PWM_DUTY_CH4								RW	18h
PWM_DUTY_CT RL_5	PWM_DUTY_CH5								RW	19h
PWM_DUTY_CT RL_6	PWM_DUTY_CH6								RW	1Ah
PWM_DUTY_CT RL_7	PWM_DUTY_CH7								RW	1Bh

Detailed Description

Name	7	6	5	4	3	2	1	0	Type	Address
PWM_DUTY_CTL_8	PWM_DUTY_CH8								RW	1Ch
SR_CTRL_1	HB8_SR	HB7_SR	HB6_SR	HB5_SR	HB4_SR	HB3_SR	HB2_SR	HB1_SR	RW	1Dh
SR_CTRL_2	Reserved								RW	1Eh
OLD_CTRL_1	HB8_OLD_DIS	HB7_OLD_DIS	HB6_OLD_DIS	HB5_OLD_DIS	HB4_OLD_DIS	HB3_OLD_DIS	HB2_OLD_DIS	HB1_OLD_DIS	RW	1Fh
OLD_CTRL_2	OLD_REP	OLD_OP	PL_MODE_EN		Reserved				RW	20h
OLD_CTRL_3	OCP_DEG			OLD_NEG_EN	Reserved				RW	21h
OLD_CTRL_4	HB8_LOAD_EN	HB7_LOAD_EN	HB6_LOAD_EN	HB5_LOAD_EN	HB4_LOAD_EN	HB3_LOAD_EN	HB2_LOAD_EN	HB1_LOAD_EN	RW	22h
OLD_CTRL_5	HB8_POLD_EN	HB7_POLD_EN	HB6_POLD_EN	HB5_POLD_EN	HB4_POLD_EN	HB3_POLD_EN	HB2_POLD_EN	HB1_POLD_EN	RW	23h
OLD_CTRL_6	HB8_VM_P_OLD	HB7_VM_P_OLD	HB6_VM_P_OLD	HB5_VM_P_OLD	HB4_VM_P_OLD	HB3_VM_P_OLD	HB2_VM_P_OLD	HB1_VM_P_OLD	RW	24h
INx_MAP_CTRL_1	HB4_INx_MAP		HB3_INx_MAP		HB2_INx_MAP		HB1_INx_MAP		RW	25h
INx_MAP_CTRL_2	HB8_INx_MAP		HB7_INx_MAP		HB6_INx_MAP		HB5_INx_MAP		RW	26h
ADD_CONFIG_CTRL	Reserved						TSD_Cluster_EN	OVP_OP	RW	28h
ADD_STAT_1	Reserved	OUT1_HSLS		OUT2_HSLS		nSLEEP_ECHO	IN0_ECHO	IN1_ECHO	RW	30h
TC_STAT_1	CP_UV	OTSD					OTSD_CL1	OTW_CL1	R	31h
ADD_CTRL_EX	Reserved						SS_SELECT		RW	32h
ADD_OLD_CTRL_1	HS8_NEG_OLD_EN	HS7_NEG_OLD_EN	HS6_NEG_OLD_EN	HS5_NEG_OLD_EN	HS4_NEG_OLD_EN	HS3_NEG_OLD_EN	HS2_NEG_OLD_EN	HS1_NEG_OLD_EN	RW	33h
ADD_OLD_CTRL_2	LS8_NEG_OLD_EN	LS7_NEG_OLD_EN	LS6_NEG_OLD_EN	LS5_NEG_OLD_EN	LS4_NEG_OLD_EN	LS3_NEG_OLD_EN	LS2_NEG_OLD_EN	LS1_NEG_OLD_EN	RW	34h

3.5.1.1 Status Registers

The status registers are used to report warning and fault conditions. The status registers are read-only registers.

Table 3-19 lists the memory-mapped registers for the status registers. All register offset addresses not listed in Table 3-19 should be considered as reserved locations and the register contents should not be modified.

Table 3-19 Status Registers Summary Table

Address	Register Name	Section
0x00	IC Status	2.5.1.1.1
0x01	Overcurrent Protection (OCP) Status 1	2.5.1.1.2
0x02	Overcurrent Protection (OCP) Status 2	2.5.1.1.3
0x03	Overcurrent Protection (OCP) Status 3	2.5.1.1.4
0x04	Open-Load Detect (OLD) Status 1	2.5.1.1.5
0x05	Open-Load Detect (OLD) Status 2	2.5.1.1.6
0x06	Open-Load Detect (OLD) Status 3	2.5.1.1.7

3.5.1.1.1 IC Status (IC_STAT) Register (Address = 0x00) [reset = 0x00]

The IC status (IC_STAT) register is shown in [Table 3-20](#) and described in [Table 3-21](#).

Register access type: Read only

Table 3-20 IC Status Register

7	6	5	4	3	2	1	0
Reserved	OTSD	OTW	OLD	OCP	UVLO	OVP	NPOR
R-0b	R-0b	R-0b	R-0b	R-0b	R-0b	R-0b	R-0b

Table 3-21 IC Status Register Field Descriptions

Bit	Field	Type	Default	Description
7	Reserved	R	0b	Reserved
6	OTSD	R	0b	0b = No overtemperature shutdown is detected 1b = Overcurrent condition is detected
5	OTW	R	0b	0b = No overtemperature warning is detected 1b = Overcurrent condition is detected
4	OLD	R	0b	0b = No open-load condition is detected 1b = Open-load condition is detected
3	OCP	R	0b	0b = No overcurrent condition is detected 1b = Overcurrent condition is detected
2	UVLO	R	0b	0b = No undervoltage lock-out condition is detected 1b = Under-voltage lock-out condition is detected
1	OVP	R	0b	0b = No overvoltage condition is detected 1b = Overvoltage condition is detected
0	NPOR	R	1b	0b = Power-on-reset condition is detected 1b = No power-on-reset condition is detected

3.5.1.1.2 Overcurrent Protection (OCP) Status 1 (OCP_STAT_1) Register (Address = 0x01) [reset = 0x00]

The overcurrent protection (OCP) status 1 register is shown in [Table 3-22](#) and described in [Table 3-23](#).

Register access type: Read only

Table 3-22 Overcurrent Protection (OCP) Status 1 Register

7	6	5	4	3	2	1	0
HB4_HS_OC P	HB4_LS_OC P	HB3_HS_OC P	HB3_LS_OC P	HB2_HS_OC P	HB2_LS_OC P	HB1_HS_OC P	HB1_LS_OC P
R-0b	R-0b	R-0b	R-0b	R-0b	R-0b	R-0b	R-0b

Table 3-23 Overcurrent Protection (OCP) Status 1 Register Field Descriptions

Bit	Field	Type	Default	Description
7	HB4_HS_OC P	R	0b	0b = No overcurrent detected on high-side switch of half-bridge 4 1b = Overcurrent detected on high-side switch of half-bridge 4
6	HB4_LS_OC P	R	0b	0b = No overcurrent detected on low-side switch of half- bridge 4 1b = Overcurrent detected on low-side switch of half-bridge 4
5	HB3_HS_OC P	R	0b	0b = No overcurrent detected on high-side switch of half-bridge 3 1b = Overcurrent detected on high-side switch of half-bridge 3
4	HB3_LS_OC P	R	0b	0b = No overcurrent detected on low-side switch of half- bridge 3 1b = Overcurrent detected on low-side switch of half-bridge 3
3	HB2_HS_OC P	R	0b	0b = No overcurrent detected on high-side switch of half-bridge 2 1b = Overcurrent detected on high-side switch of half-bridge 2
2	HB2_LS_OC P	R	0b	0b = No overcurrent detected on low-side switch of half- bridge 2 1b = Overcurrent detected on low-side switch of half-bridge 2
1	HB1_HS_OC P	R	0b	0b = No overcurrent detected on high-side switch of half-bridge 1 1b = Overcurrent detected on high-side switch of half-bridge 1
0	HB1_LS_OC P	R	0b	0b = No overcurrent detected on low-side switch of half- bridge 1 1b = Overcurrent detected on low-side switch of half-bridge 1

3.5.1.1.3 Overcurrent Protection (OCP) Status 2 (OCP_STAT_2) Register (Address = 0x02) [reset = 0x00]

The overcurrent protection (OCP) status 2 register is shown in [Table 3-24](#) and described in [Table 3-25](#).

Register access type: Read only

Table 3-24 Overcurrent Protection (OCP) Status 2 Register

7	6	5	4	3	2	1	0
HB8_HS_OC P	HB8_LS_OC P	HB7_HS_OC P	HB7_LS_OC P	HB6_HS_OC P	HB6_LS_OC P	HB5_HS_OC P	HB5_LS_OC P
R-0b	R-0b	R-0b	R-0b	R-0b	R-0b	R-0b	R-0b

Table 3-25 Overcurrent Protection (OCP) Status 2 Register Field Descriptions

Bit	Field	Type	Default	Description
7	HB8_HS_OCP	R	0b	0b = No overcurrent detected on high-side switch of half- bridge 8 1b = Overcurrent detected on high-side switch of half-bridge 8
6	HB8_LS_OCP	R	0b	0b = No overcurrent detected on low-side switch of half- bridge 8 1b = Overcurrent detected on low-side switch of half-bridge 8
5	HB7_HS_OCP	R	0b	0b = No overcurrent detected on high-side switch of half- bridge 7 1b = Overcurrent detected on high-side switch of half-bridge 7
4	HB7_LS_OCP	R	0b	0b = No overcurrent detected on low-side switch of half- bridge 7 1b = Overcurrent detected on low-side switch of half-bridge 7
3	HB6_HS_OCP	R	0b	0b = No overcurrent detected on high-side switch of half- bridge 6 1b = Overcurrent detected on high-side switch of half-bridge 6
2	HB6_LS_OCP	R	0b	0b = No overcurrent detected on low-side switch of half- bridge 6 1b = Overcurrent detected on low-side switch of half-bridge 6
1	HB5_HS_OCP	R	0b	0b = No overcurrent detected on high-side switch of half- bridge 5 1b = Overcurrent detected on high-side switch of half-bridge 5
0	HB5_LS_OCP	R	0b	0b = No overcurrent detected on low-side switch of half- bridge 5 1b = Overcurrent detected on low-side switch of half-bridge 5

3.5.1.1.4 Overcurrent Protection (OCP) Status 3 (OCP_STAT_3) Register (Address = 0x03) [reset = 0x00]

The overcurrent protection (OCP) status 3 register is shown in Table 3-26 and described in Table 3-27.

Register access type: Read only

Table 3-26 Overcurrent Protection (OCP) Status 3 Register

7	6	5	4	3	2	1	0
Reserved							
R-0b	R-0b	R-0b	R-0b	R-0b	R-0b	R-0b	R-0b

Table 3-27 Overcurrent Protection (OCP) Status 3 Register Field Descriptions

Bit	Field	Type	Default	Description
7-0	Reserved	R	0b	Reserved.

3.5.1.1.5 Open-Load Detect (OLD) Status 1 (OLD_STAT_1) Register (Address = 0x04) [reset = 0x00]

The open-load detect (OLD) status 1 register is shown in Table 3-28 and described in Table 3-29.

Register access type: Read only

Table 3-28 Open-Load Detect (OLD) Status 1 Register

7	6	5	4	3	2	1	0
HB4_HS_OL D	HB4_LS_OL D	HB3_HS_OL D	HB3_LS_OL D	HB2_HS_OL D	HB2_LS_OL D	HB1_HS_OL D	HB1_LS_OL D
R-0b	R-0b	R-0b	R-0b	R-0b	R-0b	R-0b	R-0b

Table 3-29 Open-Load Detect (OLD) Status 1 Register Field Descriptions

Bit	Field	Type	Default	Description
7	HB4_HS_OLD	R	0b	0b = No open load detected on high-side switch of half- bridge 4 1b = Open load detected on high-side switch of half-bridge 4
6	HB4_LS_OLD	R	0b	0b = No open load detected on low-side switch of half- bridge 4 1b = Open load detected on low-side switch of half-bridge 4
5	HB3_HS_OLD	R	0b	0b = No open load detected on high-side switch of half- bridge 3 1b = Open load detected on high-side switch of half-bridge 3
4	HB3_LS_OLD	R	0b	0b = No open load detected on low-side switch of half- bridge 3 1b = Open load detected on low-side switch of half-bridge 3
3	HB2_HS_OLD	R	0b	0b = No open load detected on high-side switch of half- bridge 2 1b = Open load detected on high-side switch of half-bridge 2
2	HB2_LS_OLD	R	0b	0b = No open load detected on low-side switch of half- bridge 2 1b = Open load detected on low-side switch of half-bridge 2
1	HB1_HS_OLD	R	0b	0b = No open load detected on high-side switch of half- bridge 1 1b = Open load detected on high-side switch of half-bridge 1
0	HB1_LS_OLD	R	0b	0b = No open load detected on low-side switch of half- bridge 1 1b = Open load detected on low-side switch of half-bridge 1

3.5.1.1.6 Open-Load Detect (OLD) Status 2 (OLD_STAT_2) Register (Address = 0x05) [reset = 0x00]

The open-load detect (OLD) status 2 register is shown in [Table 3-30](#) and described in [Table 3-31](#).

Register access type: Read only

Table 3-30 Open-Load Detect (OLD) Status 2 Register

7	6	5	4	3	2	1	0
HB8_HS_O LD	HB8_LS_O LD	HB7_HS_O LD	HB7_LS_O LD	HB6_HS_O LD	HB6_LS_O LD	HB5_HS_O LD	HB5_LS_O LD
R-0b	R-0b	R-0b	R-0b	R-0b	R-0b	R-0b	R-0b

Table 3-31 Open-Load Detect (OLD) Status 2 Register Field Descriptions

Bit	Field	Type	Default	Description
7	HB8_HS_OLD	R	0b	0b = No open load detected on high-side switch of half- bridge 8 1b = Open load detected on high-side switch of half-bridge 8
6	HB8_LS_OLD	R	0b	0b = No open load detected on low-side switch of half- bridge 8 1b = Open load detected on low-side switch of half-bridge 8
5	HB7_HS_OLD	R	0b	0b = No open load detected on high-side switch of half- bridge 7 1b = Open load detected on high-side switch of half-bridge 7
4	HB7_LS_OLD	R	0b	0b = No open load detected on low-side switch of half- bridge 7 1b = Open load detected on low-side switch of half-bridge 7
3	HB6_HS_OLD	R	0b	0b = No open load detected on high-side switch of half- bridge 6 1b = Open load detected on high-side switch of half-bridge 6
2	HB6_LS_OLD	R	0b	0b = No open load detected on low-side switch of half- bridge 6 1b = Open load detected on low-side switch of half-bridge 6
1	HB5_HS_OLD	R	0b	0b = No open load detected on high-side switch of half- bridge 5 1b = Open load detected on high-side switch of half-bridge 5
0	HB5_LS_OLD	R	0b	0b = No open load detected on low-side switch of half- bridge 5 1b = Open load detected on low-side switch of half-bridge 5

3.5.1.1.7 Open-Load Detect (OLD) Status 3 (OLD_STAT_3) Register (Address = 0x06) [reset = 0x00]

The open-load detect (OLD) status 3 register is shown in [Table 3-32](#) and described in [Table 3-33](#).

Register access type: Read only

Table 3-32 Open-Load Detect (OLD) Status 3 Register

7	6	5	4	3	2	1	0
Reserved							
R-0b	R-0b	R-0b	R-0b	R-0b	R-0b	R-0b	R-0b

Table 3-33 Open-Load Detect (OLD) Status 3 Register Field Descriptions

Bit	Field	Type	Default	Description
7-0	Reserved	R	0b	Reserved.

3.5.1.2 Control Registers

The control registers are used to configure the device. The control registers are read and write capable.

Table 3-34 lists the memory-mapped registers for the control registers. All register offset addresses not listed in Table 3-34 should be considered as reserved locations and the register contents should not be modified.

Table 3-34 Control Registers Summary Table

Address	Register Name	Section
0x07	Configuration Register	2.5.1.2.1
0x08	Operation Control 1 Register	2.5.1.2.2
0x09	Operation Control 2 Register	2.5.1.2.3
0x0A	Operation Control 3 Register	2.5.1.2.4
0x0B	PWM Control 1 Register	2.5.1.2.5
0x0C	PWM Control 2 Register	2.5.1.2.6
0x0D	Free-Wheeling Control 1 Register	2.5.1.2.7
0x0E	Free-Wheeling Control 2 Register	2.5.1.2.8
0x0F	PWM Map Control 1 Register	2.5.1.2.9
0x10	PWM Map Control 2 Register	2.5.1.2.10
0x11	PWM Map Control 3 Register	2.5.1.2.11
0x12	PWM Map Control 4 Register	2.5.1.2.12
0x13	PWM Frequency Control 1 Register	2.5.1.2.13
0x14	PWM Frequency Control 2 Register	2.5.1.2.14
0x15	PWM Duty Control Channel 1 Register	2.5.1.2.15
0x16	PWM Duty Control Channel 2 Register	2.5.1.2.16
0x17	PWM Duty Control Channel 3 Register	2.5.1.2.17
0x18	PWM Duty Control Channel 4 Register	2.5.1.2.18
0x19	PWM Duty Control Channel 5 Register	2.5.2.2.19
0x1A	PWM Duty Control Channel 6 Register	2.5.2.2.20
0x1B	PWM Duty Control Channel 7 Register	2.5.2.2.21
0x1C	PWM Duty Control Channel 8 Register	2.5.2.2.22
0x1D	Slew Rate Control 1 Register	2.5.2.2.23
0x1E	Slew Rate Control 2 Register	2.5.2.2.24
0x1F	Open-Load Detect (OLD) Control 1 Register	2.5.2.2.25
0x20	Open-Load Detect (OLD) Control 2 Register	2.5.2.2.26
0x21	Open-Load Detect (OLD) Control 3 Register	2.5.2.2.27
0x22	Open-Load Detect (OLD) Control 4 Register	2.5.2.2.28

Detailed Description

Address	Register Name	Section
0x23	Open-Load Detect (OLD) Control 5 Register	2.5.2.2.29
0x24	Open-Load Detect (OLD) Control 6 Register	2.5.2.2.30
0x25	INx Pin Input Mapping 1 Register	2.5.2.2.31
0x26	INx Pin Input Mapping 2 Register	2.5.2.2.32
0x28	Additional Configuration Register	2.5.2.2.33
0x30	Additional Status Register	2.5.2.2.34
0x31	Thermal Cluster Status Register	2.5.2.2.35
0x32	Additional Control External Register	2.5.2.2.36
0x33	Additional Open-Load Detect Control 1 Register	2.5.2.2.37
0x34	Additional Open-Load Detect Control 2 Register	2.5.2.2.38

3.5.1.2.1 Configuration (CONFIG_CTRL) Register (Address = 0x07) [reset = 0x00]

The configuration register is shown in [Table 3-35](#) and described in [Table 3-36](#).

Register access type: Read/Write

Table 3-35 Configuration Register

7	6	5	4	3	2	1	0
POLD_EN	IC_ID			OCP_REP	OTW_REP	EXT_OVP	CLR_FLT
R/W-0b	R-Xb	R-Xb	R-Xb	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-36 Configuration Register Field Descriptions

Bit	Field	Type	Default	Description
7	POLD_EN	R/W	0b	0b = Passive OLD is disabled 1b = Passive OLD is enabled
6-4	IC_ID	R	XXXb	010b = Device connected is HE99MD9008 (8 Channel Device) 100b = Device connected is (4 Channel Device) 101b = Reserved 110b = Reserved 111b = Reserved
3	OCP_REP	R/W	0b	0b = Overcurrent condition is reported in nFAULT pin 1b = Overcurrent condition warning is not reported on the nFAULT pin
2	OTW_REP	R/W	0b	0b = Overtemperature warning is not reported in nFAULT pin 1b = Overtemperature warning is reported on the nFAULT pin
1	EXT_OVP	R/W	0b	0b = Overvoltage protection threshold is at 22.5 V (typ) 1b = Overvoltage protection threshold is at 31 V (typ)
0	CLR_FLT	R/W	0b	0b = Faults not cleared 1b = Clear all faults

Note: CLR_FLT bit is an auto-clear bit and will always read 0b.

3.5.1.2.2 Operation Control 1 (OP_CTRL_1) Register (Address = 0x08) [reset = 0x00]

The operation control 1 register is shown in [Table 3-37](#) and described in [Table 3-38](#).

Register access type: Read/Write

Table 3-37 Operation Control 1 Register

7	6	5	4	3	2	1	0
HB4_HS_EN	HB4_LS_EN	HB3_HS_EN	HB3_LS_EN	HB2_HS_EN	HB2_LS_EN	HB1_HS_EN	HB1_LS_EN
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-38 Operation Control 1 Register Field Descriptions

Bit	Field	Type	Default	Description
7	HB4_HS_EN	R/W	0b	0b = Half-bridge 4 high-side switch is disabled 1b = Half-bridge 4 high-side switch is enabled
6	HB4_LS_EN	R/W	0b	0b = Half-bridge 4 low-side switch is disabled 1b = Half-bridge 4 low-side switch is enabled
5	HB3_HS_EN	R/W	0b	0b = Half-bridge 3 high-side switch is disabled 1b = Half-bridge 3 high-side switch is enabled
4	HB3_LS_EN	R/W	0b	0b = Half-bridge 3 low-side switch is disabled 1b = Half-bridge 3 low-side switch is enabled
3	HB2_HS_EN	R/W	0b	0b = Half-bridge 2 high-side switch is disabled 1b = Half-bridge 2 high-side switch is enabled
2	HB2_LS_EN	R/W	0b	0b = Half-bridge 2 low-side switch is disabled 1b = Half-bridge 2 low-side switch is enabled
1	HB1_HS_EN	R/W	0b	0b = Half-bridge 1 high-side switch is disabled 1b = Half-bridge 1 high-side switch is enabled
0	HB1_LS_EN	R/W	0b	0b = Half-bridge 1 low-side switch is disabled 1b = Half-bridge 1 low-side switch is enabled

3.5.1.2.3 Operation Control 2 (OP_CTRL_2) Register (Address = 0x09) [reset = 0x00]

The operation control 2 register is shown in [Table 3-39](#) and described in [Table 3-40](#).

Register access type: Read/Write

Table 3-39 Operation Control 2 Register

7	6	5	4	3	2	1	0
HB8_HS_EN	HB8_LS_EN	HB7_HS_EN	HB7_LS_EN	HB6_HS_EN	HB6_LS_EN	HB5_HS_EN	HB5_LS_EN
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-40 Operation Control 2 Register Field Descriptions

Bit	Field	Type	Default	Description
7	HB8_HS_EN	R/W	0b	0b = Half-bridge 8 high-side switch is disabled 1b = Half-bridge 8 high-side switch is enabled
6	HB8_LS_EN	R/W	0b	0b = Half-bridge 8 low-side switch is disabled 1b = Half-bridge 8 low-side switch is enabled
5	HB7_HS_EN	R/W	0b	0b = Half-bridge 7 high-side switch is disabled 1b = Half-bridge 7 high-side switch is enabled
4	HB7_LS_EN	R/W	0b	0b = Half-bridge 7 low-side switch is disabled 1b = Half-bridge 7 low-side switch is enabled
3	HB6_HS_EN	R/W	0b	0b = Half-bridge 6 high-side switch is disabled 1b = Half-bridge 6 high-side switch is enabled
2	HB6_LS_EN	R/W	0b	0b = Half-bridge 6 low-side switch is disabled 1b = Half-bridge 6 low-side switch is enabled
1	HB5_HS_EN	R/W	0b	0b = Half-bridge 5 high-side switch is disabled 1b = Half-bridge 5 high-side switch is enabled
0	HB5_LS_EN	R/W	0b	0b = Half-bridge 5 low-side switch is disabled 1b = Half-bridge 5 low-side switch is enabled

3.5.1.2.4 Operation Control 3 (OP_CTRL_3) Register (Address = 0x0A) [reset = 0x00]

The operation control 3 register is shown in Table 3-41 and described in Table 3-42.

Register access type: Read/Write

Table 3-41 Operation Control 3 Register

7	6	5	4	3	2	1	0
Reserved							
R-0b	R-0b	R-0b	R-0b	R-0b	R-0b	R-0b	R-0b

Table 3-42 Operation Control 3 Register Field Descriptions

Bit	Field	Type	Default	Description
7-0	Reserved	R	0b	Reserved.

3.5.1.2.5 PWM Control 1 (PWM_CTRL_1) Register (Address = 0x0B) [reset = 0x00]

The PWM control 1 register is shown in [Table 3-43](#) and described in [Table 3-44](#).

Register access type: Read/Write

Table 3-43 PWM Control 1 Register

7	6	5	4	3	2	1	0
HB8_PWM	HB7_PWM	HB6_PWM	HB5_PWM	HB4_PWM	HB3_PWM	HB2_PWM	HB1_PWM
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-44 PWM Control 1 Register Field Descriptions

Bit	Field	Type	Default	Description
7	HB8_PWM	R/W	0b	0b = Half-bridge 8 is operating in continuous mode 1b = Half-bridge 8 is operating in PWM mode
6	HB7_PWM	R/W	0b	0b = Half-bridge 7 is operating in continuous mode 1b = Half-bridge 7 is operating in PWM mode
5	HB6_PWM	R/W	0b	0b = Half-bridge 6 is operating in continuous mode 1b = Half-bridge 6 is operating in PWM mode
4	HB5_PWM	R/W	0b	0b = Half-bridge 5 is operating in continuous mode 1b = Half-bridge 5 is operating in PWM mode
3	HB4_PWM	R/W	0b	0b = Half-bridge 4 is operating in continuous mode 1b = Half-bridge 4 is operating in PWM mode
2	HB3_PWM	R/W	0b	0b = Half-bridge 3 is operating in continuous mode 1b = Half-bridge 3 is operating in PWM mode
1	HB2_PWM	R/W	0b	0b = Half-bridge 2 is operating in continuous mode 1b = Half-bridge 2 is operating in PWM mode
0	HB1_PWM	R/W	0b	0b = Half-bridge 1 is operating in continuous mode 1b = Half-bridge 1 is operating in PWM mode

3.5.1.2.6 PWM Control 2 (PWM_CTRL_2) Register (Address = 0x0C) [reset = 0x00]

The PWM control 1 register is shown in [Table 3-45](#) and described in [Table 3-46](#).

Register access type: Read/Write

Table 3-45 PWM Control 2 Register

7	6	5	4	3	2	1	0
PWM_CH8_DIS	PWM_CH7_DIS	PWM_CH6_DIS	PWM_CH5_DIS	PWM_CH4_DIS	PWM_CH3_DIS	PWM_CH2_DIS	PWM_CH1_DIS
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-46 PWM Control 2 Register Field Descriptions

Bit	Field	Type	Default	Description
7	PWM_CH8_DIS	R/W	0b	0b = PWM Generator-8 is enabled 1b = PWM Generator-8 is disabled
6	PWM_CH7_DIS	R/W	0b	0b = PWM Generator-7 is enabled 1b = PWM Generator-7 is disabled
5	PWM_CH6_DIS	R/W	0b	0b = PWM Generator-6 is enabled 1b = PWM Generator-6 is disabled
4	PWM_CH5_DIS	R/W	0b	0b = PWM Generator-5 is enabled 1b = PWM Generator-5 is disabled
3	PWM_CH4_DIS	R/W	0b	0b = PWM Generator-4 is enabled 1b = PWM Generator-4 is disabled
2	PWM_CH3_DIS	R/W	0b	0b = PWM Generator-3 is enabled 1b = PWM Generator-3 is disabled
1	PWM_CH2_DIS	R/W	0b	0b = PWM Generator-2 is enabled 1b = PWM Generator-2 is disabled
0	PWM_CH1_DIS	R/W	0b	0b = PWM Generator-1 is enabled 1b = PWM Generator-1 is disabled

3.5.1.2.7 Free-Wheeling Control 1 (FW_CTRL_1) Register (Address = 0x0D) [reset = 0x00]

The Free-Wheeling Control 1 register is shown in Table 3-47 and described in Table 3-48.

Register access type: Read/Write

Table 3-47 Free-Wheeling Control 1 Register

7	6	5	4	3	2	1	0
HB8_FW	HB7_FW	HB6_FW	HB5_FW	HB4_FW	HB3_FW	HB2_FW	HB1_FW
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-48 Free-Wheeling Control 1 Register Field Descriptions

Bit	Field	Type	Default	Description
7	HB8_FW	R/W	0b	0b = Passive free-wheeling on half-bridge 8 is enabled 1b = Active free-wheeling on half-bridge 8 is enabled
6	HB7_FW	R/W	0b	0b = Passive free-wheeling on half-bridge 7 is enabled 1b = Active free-wheeling on half-bridge 7 is enabled
5	HB6_FW	R/W	0b	0b = Passive free-wheeling on half-bridge 6 is enabled 1b = Active free-wheeling on half-bridge 6 is enabled
4	HB5_FW	R/W	0b	0b = Passive free-wheeling on half-bridge 5 is enabled 1b = Active free-wheeling on half-bridge 5 is enabled
3	HB4_FW	R/W	0b	0b = Passive free-wheeling on half-bridge 4 is enabled 1b = Active free-wheeling on half-bridge 4 is enabled
2	HB3_FW	R/W	0b	0b = Passive free-wheeling on half-bridge 3 is enabled 1b = Active free-wheeling on half-bridge 3 is enabled
1	HB2_FW	R/W	0b	0b = Passive free-wheeling on half-bridge 2 is enabled 1b = Active free-wheeling on half-bridge 2 is enabled
0	HB1_FW	R/W	0b	0b = Passive free-wheeling on half-bridge 1 is enabled 1b = Active free-wheeling on half-bridge 1 is enabled

3.5.1.2.8 Free-Wheeling Control 2 (FW_CTRL_2) Register (Address = 0x0E) [reset = 0x00]

The Free-Wheeling Control 2 register is shown in [Table 3-49](#) and described in [Table 3-50](#).

Register access type: Read/Write

Table 3-49 Free-Wheeling Control 2 Register

7	6	5	4	3	2	1	0
Reserved							
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-50 Free-Wheeling Control 2 Register Field Descriptions

Bit	Field	Type	Default	Description
7-0	Reserved	R/W	0000b	Reserved.

3.5.1.2.9 PWM Map Control 1 (PWM_MAP_CTRL_1) Register (Address = 0x0F) [reset = 0x00]

The PWM Map Control 1 register is shown in [Table 3-51](#) and described in [Table 3-52](#).

Register access type: Read/Write

Table 3-51 PWM Map Control 1 Register

7	6	5	4	3	2	1	0
Reserved		HB2_PWM_MAP			HB1_PWM_MAP		
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-52 PWM Map Control 1 Register Field Descriptions

Bit	Field	Type	Default	Description
7-6	Reserved	R	00b	Reserved
5-3	HB2_PWM_MAP	R/W	000b	000b = HB2 mapped to PWM channel 1 001b = HB2 mapped to PWM channel 2 010b = HB2 mapped to PWM channel 3 011b = HB2 mapped to PWM channel 4 100b = HB2 mapped to PWM channel 5 101b = HB2 mapped to PWM channel 6 110b = HB mapped to PWM channel 7 111b = HB2 mapped to PWM channel 8
2-0	HB1_PWM_MAP	R/W	000b	000b = HB1 mapped to PWM channel 1 001b = HB1 mapped to PWM channel 2 010b = HB1 mapped to PWM channel 3 011b = HB1 mapped to PWM channel 4 100b = HB1 mapped to PWM channel 5 101b = HB1 mapped to PWM channel 6 110b = HB1 mapped to PWM channel 7 111b = HB1 mapped to PWM channel 8

3.5.1.2.10 PWM Map Control 2 (PWM_MAP_CTRL_2) Register (Address = 0x10) [reset = 0x00]

The PWM Map Control 2 register is shown in [Table 3-53](#) and described in [Table 3-54](#).

Register access type: Read/Write

Table 3-53 PWM Map Control 2 Register

7	6	5	4	3	2	1	0
Reserved		HB4_PWM_MAP			HB3_PWM_MAP		
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-54 PWM Map Control 2 Register Field Descriptions

Bit	Field	Type	Default	Description
7-6	Reserved	R	00b	Reserved
5-3	HB4_PWM_MAP	R/W	000b	000b = HB4 mapped to PWM channel 1 001b = HB4 mapped to PWM channel 2 010b = HB4 mapped to PWM channel 3 011b = HB4 mapped to PWM channel 4 100b = HB4 mapped to PWM channel 5 101b = HB4 mapped to PWM channel 6 110b = HB4 mapped to PWM channel 7 111b = HB4 mapped to PWM channel 8
2-0	HB3_PWM_MAP	R/W	000b	000b = HB3 mapped to PWM channel 1 001b = HB3 mapped to PWM channel 2 010b = HB3 mapped to PWM channel 3 011b = HB3 mapped to PWM channel 4 100b = HB3 mapped to PWM channel 5 101b = HB3 mapped to PWM channel 6 110b = HB3 mapped to PWM channel 7 111b = HB3 mapped to PWM channel 8

3.5.1.2.11 PWM Map Control 3 (PWM_MAP_CTRL_3) Register (Address = 0x11) [reset = 0x00]

The PWM Map Control 3 register is shown in [Table 3-55](#) and described in [Table 3-56](#).

Register access type: Read/Write

Table 3-55 PWM Map Control 3 Register

7	6	5	4	3	2	1	0
Reserved		HB6_PWM_MAP			HB5_PWM_MAP		
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-56 PWM Map Control 3 Register Field Descriptions

Bit	Field	Type	Default	Description
7-6	Reserved	R	00b	Reserved
5-3	HB6_PWM_MAP	R/W	000b	000b = HB6 mapped to PWM channel 1 001b = HB6 mapped to PWM channel 2 010b = HB6 mapped to PWM channel 3 011b = HB6 mapped to PWM channel 4 100b = HB6 mapped to PWM channel 5 101b = HB6 mapped to PWM channel 6 110b = HB6 mapped to PWM channel 7 111b = HB6 mapped to PWM channel 8
2-0	HB5_PWM_MAP	R/W	000b	000b = HB5 mapped to PWM channel 1 001b = HB5 mapped to PWM channel 2 010b = HB5 mapped to PWM channel 3 011b = HB5 mapped to PWM channel 4 100b = HB5 mapped to PWM channel 5 101b = HB5 mapped to PWM channel 6 110b = HB5 mapped to PWM channel 7 111b = HB5 mapped to PWM channel 8

3.5.1.2.12 PWM Map Control 4 (PWM_MAP_CTRL_4) Register (Address = 0x12) [reset = 0x00]

The PWM Map Control 4 register is shown in Table 3-57 and described in Table 3-58.

Register access type: Read/Write

Table 3-57 PWM Map Control 4 Register

7	6	5	4	3	2	1	0
Reserved		HB8_PWM_MAP			HB7_PWM_MAP		
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-58 PWM Map Control 4 Register Field Descriptions

Bit	Field	Type	Default	Description
7-6	Reserved	R	00b	Reserved
5-3	HB8_PWM_MAP	R/W	000b	000b = HB8 mapped to PWM channel 1 001b = HB8 mapped to PWM channel 2 010b = HB8 mapped to PWM channel 3 011b = HB8 mapped to PWM channel 4 100b = HB8 mapped to PWM channel 5 101b = HB8 mapped to PWM channel 6 110b = HB8 mapped to PWM channel 7 111b = HB8 mapped to PWM channel 8
2-0	HB7_PWM_MAP	R/W	000b	000b = HB7 mapped to PWM channel 1 001b = HB7 mapped to PWM channel 2 010b = HB7 mapped to PWM channel 3 011b = HB7 mapped to PWM channel 4 100b = HB7 mapped to PWM channel 5 101b = HB7 mapped to PWM channel 6 110b = HB7 mapped to PWM channel 7 111b = HB7 mapped to PWM channel 8

3.5.1.2.13 PWM Frequency Control 1 (PWM_FREQ_CTRL_1) Register (Address = 0x13 [reset = 0x00])

The PWM Frequency Control 1 register is shown in [Table 3-59](#) and described in [Table 3-60](#).

Register access type: Read/Write

Table 3-59 PWM Frequency Control 1 Register

7	6	5	4	3	2	1	0
PWM_CH4_FREQ	PWM_CH3_FREQ	PWM_CH2_FREQ	PWM_CH1_FREQ				
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-60 PWM Frequency Control 1 Register Field Descriptions

Bit	Field	Type	Default	Description
7-6	PWM_CH4_FREQ	R/W	00b	00b = PWM frequency is 80 Hz 01b = PWM frequency is 100 Hz 10b = PWM frequency is 200 Hz 11b = PWM frequency is 2000 Hz
5-4	PWM_CH3_FREQ	R/W	00b	00b = PWM frequency is 80 Hz 01b = PWM frequency is 100 Hz 10b = PWM frequency is 200 Hz 11b = PWM frequency is 2000 Hz
3-2	PWM_CH2_FREQ	R/W	00b	00b = PWM frequency is 80 Hz 01b = PWM frequency is 100 Hz 10b = PWM frequency is 200 Hz 11b = PWM frequency is 2000 Hz
1-0	PWM_CH1_FREQ	R/W	00b	00b = PWM frequency is 80 Hz 01b = PWM frequency is 100 Hz 10b = PWM frequency is 200 Hz 11b = PWM frequency is 2000 Hz

3.5.1.2.14 PWM Frequency Control 2 (PWM_FREQ_CTRL_2) Register (Address = 0x14 [reset = 0x00])

The PWM Frequency Control 2 register is shown in [Table 3-61](#) and described in [Table 3-62](#).

Register access type: Read/Write

Table 3-61 PWM Frequency Control 2 Register

7	6	5	4	3	2	1	0
PWM_CH8_FREQ	PWM_CH7_FREQ	PWM_CH6_FREQ	PWM_CH5_FREQ				
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-62 PWM Frequency Control 2 Register Field Descriptions

Bit	Field	Type	Default	Description
7-6	PWM_CH8_FREQ	R/W	00b	00b = PWM frequency is 80 Hz 01b = PWM frequency is 100 Hz 10b = PWM frequency is 200 Hz 11b = PWM frequency is 2000 Hz
5-4	PWM_CH7_FREQ	R/W	00b	00b = PWM frequency is 80 Hz 01b = PWM frequency is 100 Hz 10b = PWM frequency is 200 Hz 11b = PWM frequency is 2000 Hz
3-2	PWM_CH6_FREQ	R/W	00b	00b = PWM frequency is 80 Hz 01b = PWM frequency is 100 Hz 10b = PWM frequency is 200 Hz 11b = PWM frequency is 2000 Hz
1-0	PWM_CH5_FREQ	R/W	00b	00b = PWM frequency is 80 Hz 01b = PWM frequency is 100 Hz 10b = PWM frequency is 200 Hz 11b = PWM frequency is 2000 Hz

3.5.1.2.15 PWM Duty Control Channel 1 (PWM_DUTY_CH1) Register (Address = 0x15) [reset = 0x00]

The channel 1 PWM duty cycle control register is shown in Table 3-63 and described in Table 3-64.

Register access type: Read/Write

Table 3-63 PWM Duty Control 1 Register

7	6	5	4	3	2	1	0
PWM_DUTY_CH1							
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-64 PWM Duty Control 1 Register Field Descriptions

Bit	Field	Type	Default	Description
7-0	PWM_DUTY_CH1	R/W	00000000b	00000000b = 0 % PWM Duty 11111111b = 100 % PWM Duty Calculate duty as decimal (xxxxxxx) × 1/255

3.5.1.2.16 PWM Duty Control Channel 2 (PWM_DUTY_CH2) Register (Address = 0x16) [reset = 0x00]

The channel 2 PWM duty cycle control register is shown in Table 3-65 and described in Table 3-66.

Register access type: Read/Write

Table 3-65 PWM Duty Control 2 Register

7	6	5	4	3	2	1	0
PWM_DUTY_CH2							
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-66 PWM Duty Control 2 Register Field Descriptions

Bit	Field	Type	Default	Description
7-0	PWM_DUTY_CH2	R/W	00000000b	00000000b = 0 % PWM Duty 11111111b = 100 % PWM Duty Calculate duty as decimal (xxxxxxx) × 1/255

3.5.1.2.17 PWM Duty Control Channel 3 (PWM_DUTY_CH3) Register (Address = 0x17) [reset = 0x00]

The channel 3 PWM duty cycle control register is shown in Table 3-67 and described in Table 3-68.

Register access type: Read/Write

Table 3-67 PWM Duty Control 3 Register

7	6	5	4	3	2	1	0
PWM_DUTY_CH3							
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-68 PWM Duty Control 3 Register Field Descriptions

Bit	Field	Type	Default	Description
7-0	PWM_DUTY_CH3	R/W	00000000b	00000000b = 0 % PWM Duty 11111111b = 100 % PWM Duty Calculate duty as decimal (xxxxxxx) × 1/255

3.5.1.2.18 PWM Duty Control Channel 4 (PWM_DUTY_CH4) Register (Address = 0x18) [reset = 0x00]

The channel 4 PWM duty cycle control register is shown in Table 3-69 and described in Table 3-70.

Register access type: Read/Write

Table 3-69 PWM Duty Control 4 Register

7	6	5	4	3	2	1	0
PWM_DUTY_CH4							
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-70 PWM Duty Control 4 Register Field Descriptions

Bit	Field	Type	Default	Description
7-0	PWM_DUTY_CH4	R/W	00000000b	00000000b = 0 % PWM Duty 11111111b = 100 % PWM Duty Calculate duty as decimal (xxxxxxx) × 1/255

3.5.1.2.19 PWM Duty Control Channel 5 (PWM_DUTY_CH5) Register (Address = 0x19) [reset = 0x00]

The channel 5 PWM duty cycle control register is shown in Table 3-71 and described in Table 3-72.

Register access type: Read/Write

Table 3-71 PWM Duty Control 5 Register

7	6	5	4	3	2	1	0
PWM_DUTY_CH5							
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-72 PWM Duty Control 5 Register Field Descriptions

Bit	Field	Type	Default	Description
7-0	PWM_DUTY_CH5	R/W	00000000b	00000000b = 0 % PWM Duty 11111111b = 100 % PWM Duty Calculate duty as decimal (xxxxxxx) × 1/255

3.5.1.2.20 PWM Duty Control Channel 6 (PWM_DUTY_CH6) Register (Address = 0x1A) [reset = 0x00]

The channel 6 PWM duty cycle control register is shown in Table 3-73 and described in Table 3-74.

Register access type: Read/Write

Table 3-73 PWM Duty Control 6 Register

7	6	5	4	3	2	1	0
PWM_DUTY_CH6							
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-74 PWM Duty Control 6 Register Field Descriptions

Bit	Field	Type	Default	Description
7-0	PWM_DUTY_CH6	R/W	00000000b	00000000b = 0 % PWM Duty 11111111b = 100 % PWM Duty Calculate duty as decimal (xxxxxxx) × 1/255

3.5.1.2.21 PWM Duty Control Channel 7 (PWM_DUTY_CH7) Register (Address = 0x1B) [reset = 0x00]

The channel 7 PWM duty cycle control register is shown in [Table 3-75](#) and described in [Table 3-76](#).

Register access type: Read/Write

Table 3-75 PWM Duty Control 7 Register

7	6	5	4	3	2	1	0
PWM_DUTY_CH7							
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-76 PWM Duty Control 7 Register Field Descriptions

Bit	Field	Type	Default	Description
7-0	PWM_DUTY_CH7	R/W	00000000b	00000000b = 0 % PWM Duty 11111111b = 100 % PWM Duty Calculate duty as decimal (xxxxxxxb) × 1/255

3.5.1.2.22 PWM Duty Control Channel 8 (PWM_DUTY_CH8) Register (Address = 0x1C) [reset = 0x00]

The channel 8 PWM duty cycle control register is shown in [Table 3-77](#) and described in [Table 3-78](#).

Register access type: Read/Write

Table 3-77 PWM Duty Control 8 Register

7	6	5	4	3	2	1	0
PWM_DUTY_CH8							
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-78 PWM Duty Control 8 Register Field Descriptions

Bit	Field	Type	Default	Description
7-0	PWM_DUTY_CH8	R/W	00000000b	00000000b = 0 % PWM Duty 11111111b = 100 % PWM Duty Calculate duty as decimal (xxxxxxxb) × 1/255

3.5.1.2.23 Slew Rate Control 1 (SR_CTRL_1) Register (Address = 0x1D) [reset = 0x00]

The Slew Rate Control 1 register is shown in [Table 3-79](#) and described in [Table 3-80](#).

Register access type: Read/Write

Table 3-79 Slew Rate Control 1 Register

7	6	5	4	3	2	1	0
HB8_SR	HB7_SR	HB6_SR	HB5_SR	HB4_SR	HB3_SR	HB2_SR	HB1_SR
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-80 Slew Rate Control 1 Register Field Descriptions

Bit	Field	Type	Default	Description
7	HB8_SR	R/W	0b	0b = 0.6 V/ μ s 1b = 2.5 V/ μ s
6	HB7_SR	R/W	0b	0b = 0.6 V/ μ s 1b = 2.5 V/ μ s
5	HB6_SR	R/W	0b	0b = 0.6 V/ μ s 1b = 2.5 V/ μ s
4	HB5_SR	R/W	0b	0b = 0.6 V/ μ s 1b = 2.5 V/ μ s
3	HB4_SR	R/W	0b	0b = 0.6 V/ μ s 1b = 2.5 V/ μ s
2	HB3_SR	R/W	0b	0b = 0.6 V/ μ s 1b = 2.5 V/ μ s
1	HB2_SR	R/W	0b	0b = 0.6 V/ μ s 1b = 2.5 V/ μ s
0	HB1_SR	R/W	0b	0b = 0.6 V/ μ s 1b = 2.5 V/ μ s

3.5.1.2.24 Slew Rate Control 2 (SR_CTRL_2) Register (Address = 0x1E) [reset = 0x00]

The Slew Rate Control 2 register is shown in Table 3-81 and described in Table 3-82.

Register access type: Read/Write

Table 3-81 Slew Rate Control 2 Register

7	6	5	4				
Reserved							
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-82 Slew Rate Control 2 Register Field Descriptions

Bit	Field	Type	Default	Description
7-4	Reserved	R/W	0000b	Reserved

3.5.1.2.25 Open-Load Detect (OLD) Control 1 (OLD_CTRL_1) Register (Address = 0x1F) [reset = 0x00]

The Open-Load Detect (OLD) Control (OLD_CTRL_1) Register-1 is shown in [Table 3-83](#) and described in [Table 3-84](#).

Register access type: Read/Write

Table 3-83 Open-Load Detect (OLD) Control (OLD_CTRL_1) Register

7	6	5	4	3	2	1	0
HB8_OLD_D IS	HB7_OLD_D IS	HB6_OLD_D IS	HB5_OLD_D IS	HB4_OLD_D IS	HB3_OLD_D IS	HB2_OLD_D IS	HB1_OLD_D IS
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-84 Open-Load Detect (OLD) Control (OLD_CTRL_1) Register Field Descriptions

Bit	Field	Type	Default	Description
7	HB8_OLD_DIS	R/W	0b	0b = Open-load detection on half-bridge 8 is enabled 1b = Open-load on half-bridge 8 is disabled
6	HB7_OLD_DIS	R/W	0b	0b = Open-load detection on half-bridge 7 is enabled 1b = Open-load on half-bridge 7 is disabled
5	HB6_OLD_DIS	R/W	0b	0b = Open-load detection on half-bridge 6 is enabled 1b = Open-load on half-bridge 6 is disabled
4	HB5_OLD_DIS	R/W	0b	0b = Open-load detection on half-bridge 5 is enabled 1b = Open-load on half-bridge 5 is disabled
3	HB4_OLD_DIS	R/W	0b	0b = Open-load detection on half-bridge 4 is enabled 1b = Open-load on half-bridge 4 is disabled
2	HB3_OLD_DIS	R/W	0b	0b = Open-load detection on half-bridge 3 is enabled 1b = Open-load on half-bridge 3 is disabled
1	HB2_OLD_DIS	R/W	0b	0b = Open-load detection on half-bridge 2 is enabled 1b = Open-load on half-bridge 2 is disabled
0	HB1_OLD_DIS	R/W	0b	0b = Open-load detection on half-bridge 1 is enabled 1b = Open-load on half-bridge 1 is disabled

3.5.1.2.26 Open-Load Detect (OLD) Control 2 (OLD_CTRL_2) Register (Address = 0x20) [reset = 0x00]

The Open-Load Detect (OLD) Control (OLD_CTRL_2) Register-2 is shown in [Table 3-85](#) and described in [Table 3-86](#).

Register access type: Read/Write

Table 3-85 Open-Load Detect (OLD) Control (OLD_CTRL_2) Register

7	6	5	4	3	2	1	0
OLD REP	OLD OP	PL MODE EN		Reserved			
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-86 Open-Load Detect (OLD) Control (OLD_CTRL_2) Register Field Descriptions

Bit	Field	Type	Default	Description
7	OLD_REP	R/W	0b	0b = Report on nFAULT pin during OLD condition 1b = No report on nFAULT pin during OLD condition
6	OLD_OP	R/W	0b	0b = Half bridges are not active after OLD condition detect 1b = Half bridges are active after OLD condition detect
5-4	PL_MODE_EN	R/W	00b	00b = Parallel mode OCP fast turn-off slew is disabled 01b = Parallel mode OCP slow turn-off slew is enabled 10b = Invalid Setting 11b = Invalid Setting
3-0	Reserved	R	0b	Reserved

3.5.1.2.27 Open-Load Detect (OLD) Control 3 (OLD_CTRL_3) Register (Address = 0x21) [reset = 0x00]

The Open-Load Detect (OLD) Control (OLD_CTRL_3) Register-3 is shown in [Table 3-87](#) and described in [Table 3-88](#).

Register access type: Read/Write

Table 3-87 Open-Load Detect (OLD) Control (OLD_CTRL_3) Register

7	6	5	4	3	2	1	0
OCP_DEG			OLD_NEG_EN	Reserved			
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-88 Open-Load Detect (OLD) Control (OLD_CTRL_3) Register Field Descriptions

Bit	Field	Type	Default	Description
7-5	OCP_DEG	R/W	000b	000b = OCP deglitch time is 10 μ s 001b = OCP deglitch time is 5 μ s 010b = OCP deglitch time is 2.5 μ s 011b = OCP deglitch time is 1 μ s 100b = OCP deglitch time is 60 μ s 101b = OCP deglitch time is 40 μ s 110b = OCP deglitch time is 30 μ s 111b = OCP deglitch time is 20 μ s
4	OLD_NEG_EN	R/W	0b	0b = Negative-current OLD mode is disabled 1b = Negative-current OLD mode is enabled
3-0	Reserved	R/W	0b	Reserved

3.5.1.2.28 Open Load Detect (OLD) Control 4 (OLD_CTRL_4) Register (Address = 0x22) [reset = 0x00]

The Open-Load Detect (OLD) Control (OLD_CTRL_4) Register-4 is shown in [Table 3-89](#) and described in [Table 3-90](#).

Register access type: Read/Write

Table 3-89 Open-Load Detect (OLD) Control (OLD_CTRL_4) Register

7	6	5	4	3	2	1	0
HB8_LCOL D_EN	HB7_LOLD EN	HB6_LOLD EN	HB5_LOLD EN	HB4_LOLD EN	HB3_LOLD EN	HB2_LOLD EN	HB1_LOLD EN
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-90 Open-Load Detect (OLD) Control (OLD_CTRL_4) Register Field Descriptions

Bit	Field	Type	Default	Description
7	HB8_LOLD_EN	R/W	0b	0b = Low-current OLD on half-bridge 8 is disabled 1b = Low-current OLD on half-bridge 8 is enabled
6	HB7_LOLD_EN	R/W	0b	0b = Low-current OLD on half-bridge 7 is disabled 1b = Low-current OLD on half-bridge 7 is enabled
5	HB6_LOLD_EN	R/W	0b	0b = Low-current OLD on half-bridge 6 is disabled 1b = Low-current OLD on half-bridge 6 is enabled
4	HB5_LOLD_EN	R/W	0b	0b = Low-current OLD on half-bridge 5 is disabled 1b = Low-current OLD on half-bridge 5 is enabled
3	HB4_LOLD_EN	R/W	0b	0b = Low-current OLD on half-bridge 4 is disabled 1b = Low-current OLD on half-bridge 4 is enabled
2	HB3_LOLD_EN	R/W	0b	0b = Low-current OLD on half-bridge 3 is disabled 1b = Low-current OLD on half-bridge 3 is enabled
1	HB2_LOLD_EN	R/W	0b	0b = Low-current OLD on half-bridge 2 is disabled 1b = Low-current OLD on half-bridge 2 is enabled
0	HB1_LOLD_EN	R/W	0b	0b = Low-current OLD on half-bridge 1 is disabled 1b = Low-current OLD on half-bridge 1 is enabled

3.5.1.2.29 Open Load Detect (OLD) Control 5 (OLD_CTRL_5) Register (Address = 0x23) [reset = 0x00]

The Open-Load Detect (OLD) Control (OLD_CTRL_5) Register-5 is shown in [Table 3-91](#) and described in [Table 3-92](#).

Register access type: Read/Write

Table 3-91 Open-Load Detect (OLD) Control (OLD_CTRL_5) Register

7	6	5	4	3	2	1	0
HB8_POLD EN	HB7_POLD EN	HB6_POLD EN	HB5_POLD EN	HB4_POLD EN	HB3_POLD EN	HB2_POLD EN	HB1_POLD EN
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-92 Open-Load Detect (OLD) Control (OLD_CTRL_5) Register Field Descriptions

Bit	Field	Type	Default	Description
7	HB8_POLD_EN	R/W	0b	0b = Passive OLD operation of half-bridge 8 is disabled 1b = Passive OLD operation of half-bridge 8 is enabled
6	HB7_POLD_EN	R/W	0b	0b = Passive OLD operation of half-bridge 7 is disabled 1b = Passive OLD operation of half-bridge 7 is enabled
5	HB6_POLD_EN	R/W	0b	0b = Passive OLD operation of half-bridge 6 is disabled 1b = Passive OLD operation of half-bridge 6 is enabled
4	HB5_POLD_EN	R/W	0b	0b = Passive OLD operation of half-bridge 5 is disabled 1b = Passive OLD operation of half-bridge 5 is enabled
3	HB4_POLD_EN	R/W	0b	0b = Passive OLD operation of half-bridge 4 is disabled 1b = Passive OLD operation of half-bridge 4 is enabled
2	HB3_POLD_EN	R/W	0b	0b = Passive OLD operation of half-bridge 3 is disabled 1b = Passive OLD operation of half-bridge 3 is enabled
1	HB2_POLD_EN	R/W	0b	0b = Passive OLD operation of half-bridge 2 is disabled 1b = Passive OLD operation of half-bridge 2 is enabled
0	HB1_POLD_EN	R/W	0b	0b = Passive OLD operation of half-bridge 1 is disabled 1b = Passive OLD operation of half-bridge 1 is enabled

3.5.1.2.30 Open Load Detect (OLD) Control 6 (OLD_CTRL_6) Register (Address = 0x24) [reset = 0x00]

The Open-Load Detect (OLD) Control (OLD_CTRL_6) Register-6 is shown in Table 3-93 and described in Table 3-94.

Register access type: Read/Write

Table 3-93 Open-Load Detect (OLD) Control (OLD_CTRL_6) Register

7	6	5	4	3	2	1	0
HB8_VM_POLD	HB7_VM_POLD	HB6_VM_POLD	HB5_VM_POLD	HB4_VM_POLD	HB3_VM_POLD	HB2_VM_POLD	HB1_VM_POLD
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-94 Open-Load Detect (OLD) Control (OLD_CTRL_6) Register Field Descriptions

Bit	Field	Type	Default	Description
7	HB8_VM_POLD	R/W	0b	0b = Passive OLD operation for VM connected load of half- bridge 8 is disabled 1b = Passive OLD operation for VM connected load of half- bridge 8 is enabled
6	HB7_VM_POLD	R/W	0b	0b = Passive OLD operation for VM connected load of half- bridge 7 is disabled 1b = Passive OLD operation for VM connected load of half- bridge 7 is enabled
5	HB6_VM_POLD	R/W	0b	0b = Passive OLD operation for VM connected load of half- bridge 6 is disabled 1b = Passive OLD operation for VM connected load of half- bridge 6 is enabled
4	HB5_VM_POLD	R/W	0b	0b = Passive OLD operation for VM connected load of half- bridge 5 is disabled 1b = Passive OLD operation for VM connected load of half- bridge 5 is enabled

Detailed Description

Bit	Field	Type	Default	Description
3	HB4_VM_POLD	R/W	0b	0b = Passive OLD operation for VM connected load of half- bridge 4 is disabled 1b = Passive OLD operation for VM connected load of half- bridge 4 is enabled
2	HB3_VM_POLD	R/W	0b	0b = Passive OLD operation for VM connected load of half- bridge 3 is disabled 1b = Passive OLD operation for VM connected load of half- bridge 3 is enabled
1	HB2_VM_POLD	R/W	0b	0b = Passive OLD operation for VM connected load of half- bridge 2 is disabled 1b = Passive OLD operation for VM connected load of half- bridge 2 is enabled
0	HB1_VM_POLD	R/W	0b	0b = Passive OLD operation for VM connected load of half- bridge 1 is disabled 1b = Passive OLD operation for VM connected load of half- bridge 1 is enabled

Note: Add Digital Thermal Clusters, PWM and LH registers info

3.5.1.2.31 INx Pin Input Mapping (INx_MAP_CTRL_1) Register (Address = 0x25) [reset = 0x09]

The INx pin input mapping (INx_MAP_CTRL_1) register is shown in [Table 3-95](#) and described in [Table 3-96](#).

Table 3-95 INx Pin Input Mapping (INx_MAP_CTRL_1) Register

7	6	5	4	3	2	1	0
HB4 INx MAP		HB3 INx MAP		HB2 INx MAP		HB1 INx MAP	
R/W-00b		R/W-00b		R/W-00b		R/W-00b	

Table 3-96 INx Pin Input Mapping (INx_MAP_CTRL_1) Register Field Descriptions

Bit	Field	Type	Default	Description
7-6	HB4_INx_MAP	R/W	00b	00b = HB4 is not mapped to IN0 or IN1 01b = HB4 is mapped to IN0 10b = HB4 is mapped to IN1 11b = Invalid Setting
5-4	HB3_INx_MAP	R/W	00b	00b = HB3 is not mapped to IN0 or IN1 01b = HB3 is mapped to IN0 10b = HB3 is mapped to IN1 11b = Invalid Setting
3-2	HB2_INx_MAP	R/W	10b	00b = HB2 is not mapped to IN0 or IN1 01b = HB2 is mapped to IN0 10b = HB2 is mapped to IN1 11b = Invalid Setting
1-0	HB1_INx_MAP	R/W	01b	00b = HB1 is not mapped to IN0 or IN1 01b = HB1 is mapped to IN0 10b = HB1 is mapped to IN1 11b = Invalid Setting

3.5.1.2.32 INx Pin Input Mapping (INx_MAP_CTRL_2) Register (Address = 0x26) [reset = 0x00]

The INx pin input mapping (INx_MAP_CTRL_2) register is shown in [Table 3-97](#) and described in [Table 3-98](#).

Table 3-97 INx Pin Input Mapping (INx_MAP_CTRL_2) Register

7	6	5	4	3	2	1	0
HB8_INx_MAP		HB7_INx_MAP		HB6_INx_MAP		HB5_INx_MAP	
R/W-00b		R/W-00b		R/W-00b		R/W-00b	

Table 3-98 INx Pin Input Mapping (INx_MAP_CTRL_2) Register Field Descriptions

Bit	Field	Type	Default	Description
7-6	HB8_INx_MAP	R/W	00b	00b = HB8 is not mapped to IN0 or IN1 01b = HB8 is mapped to IN0 10b = HB8 is mapped to IN1 11b = Invalid Setting
5-4	HB7_INx_MAP	R/W	00b	00b = HB7 is not mapped to IN0 or IN1 01b = HB7 is mapped to IN0 10b = HB7 is mapped to IN1 11b = Invalid Setting
3-2	HB6_INx_MAP	R/W	00b	00b = HB6 is not mapped to IN0 or IN1 01b = HB6 is mapped to IN0 10b = HB6 is mapped to IN1 11b = Invalid Setting
1-0	HB5_INx_MAP	R/W	00b	00b = HB5 is not mapped to IN0 or IN1 01b = HB5 is mapped to IN0 10b = HB5 is mapped to IN1 11b = Invalid Setting

3.5.1.2.33 Additional Configuration (ADD_CONFIG_CTRL) Register (Address = 0x28) [reset = 0x00]

The additional configuration (ADD_CONFIG_CTRL) register is shown in [Table 3-99](#) and described in [Table 3-100](#).

Table 3-99 Additional Configuration (ADD_CONFIG_CTRL) Register

7	6	5	4	3	2	1	0
Reserved						TSD_Cluster_EN	OVP_OP
R/W-000000b	R/W-000000b	R/W-000000b	R/W-000000b	R/W-000000b	R/W-000000b	R/W-0b	R/W-0b

Table 3-100 Additional Configuration (ADD_CONFIG_CTRL) Register Field Descriptions

Bit	Field	Type	Default	Description
7-2	Reserved	R/W	000000b	Reserved
1	TSD_Cluster_EN	R/W	0b	0b = Standard Mode: as soon as any cluster reaches thermal threshold the device is switched off 1b = Cluster Mode: only the cluster that has reached shutdown temperature is switched off
0	OVP_OP	R/W	0b	0b = Half bridges are not active after OVP condition detect 1b = Half bridges are active after OVP condition detect

Add MAP_IN0 and MAP_IN1 input pin register info.

3.5.1.2.34 Additional Status (ADD_STAT_1) Register (Address = 0x30) [reset = 0x00]

The additional status (ADD_STAT_1) register is shown in [Table 3-101](#) and described in [Table 3-102](#).

Table 3-101 Additional Status (ADD_STAT_1) Register

7	6	5	4	3	2	1	0
Reserved	OUT1_HSLS		OUT2_HSLS		nSLEEP_ECHO	IN0_ECHO	IN1_ECHO
R-0b	RW-00b	RW-00b	RW-00b	RW-00b	RW-0b	RW-0b	RW-0b

Table 3-102 Additional Status (ADD_STAT_1) Register Field Descriptions

Bit	Field	Type	Default	Description
7	Reserved	R	0b	Reserved
6-5	OUT1_HSLS ⁽¹⁾	R/W	00b	00b = no detection occurred 01b = OUT1 work as LS in limp home mode 10b = OUT1 work as HS in limp home mode 11b = Load Error, OUT1 is kept off in limp home mode
4-3	OUT2_HSLS ⁽¹⁾	R/W	00b	00b = no detection occurred 01b = OUT2 work as LS in limp home mode 10b = OUT2 work as HS in limp home mode 11b = Load Error, OUT2 is kept off in limp home mode
2	nSLEEP_ECHO ⁽²⁾	R	0b	0b = nSLEEP pin is at low level 1b = nSLEEP pin is at high level
1	IN0_ECHO	R	0b	0b = IN0 pin is at low level 1b = IN0 pin is at high level
0	IN1_ECHO	R	0b	0b = IN1 pin is at low level 1b = IN1 pin is at high level

(1) Read values depend on OUTx voltage during power up. User can modify this configuration in normal operating mode.

(2) This bit can be used to identify device in limp home mode or normal operating mode

3.5.1.2.35 Thermal Cluster Status (TC_STAT_1) Register (Address = 0x31) [reset = 0x00]

Thermal Cluster Status (TC_STAT_1) register is shown in Table 3-103 and described in Table 3-104

Table 3-103 Thermal Cluster Status (TC_STAT_1) Register

7	6	5	4	3	2	1	0
CP_UV	OTSD					OTSD_CL1	OTW_CL1
R-0b	R-0b	R-0b	R-0b	R-0b	R-0b	R-0b	R-0b

Table 3-104 Thermal Cluster Status (TC_STAT_1) Register Field Descriptions

Bit	Field	Type	Default	Description
7	CP_UV	R	0b	0b = No Charge pump undervoltage condition is detected 1b = Charge pump undervoltage condition is detected
6	OTSD	R	0b	0b = No Overtemperature shutdown on thermal cluster 3 is detected 1b = Overtemperature shutdown on thermal cluster 3 is detected
5	OTSD	R	0b	0b = No Overtemperature shutdown on thermal cluster 4/6 is detected 1b = Overtemperature shutdown on thermal cluster 4/6 is detected
4	OTSD	R	0b	0b = No Overtemperature shutdown on thermal cluster 7 is detected 1b = Overtemperature shutdown on thermal cluster 7 is detected
3	OTSD	R	0b	0b = No Overtemperature shutdown on thermal cluster 2/8 is detected 1b = Overtemperature shutdown on thermal cluster 2/8 is detected
2	OTSD	R	0b	0b = No Overtemperature shutdown on thermal cluster 1/5 is detected 1b = Overtemperature shutdown on thermal cluster 1/5 is detected
1	OTSD_CL1	R	0b	0b = No Overtemperature shutdown on thermal cluster 1 is detected 1b = Overtemperature shutdown on thermal cluster 1 is detected
0	OTW_CL1	R	0b	0b = No overtemperature warning on thermal cluster 1 is detected 1b = Overtemperature warning on thermal cluster 1 is detected

3.5.1.2.36 Additional Control External (ADD_CTRL_EX) Register (Address = 0x32) [reset = 0x00]

Additional Control External (ADD_CTRL_EX) register is shown in [Table 3-105](#) and described in [Table 3-106](#).

Table 3-105 Additional Control External(ADD_CTRL_EX) Register

7	6	5	4	3	2	1	0
Reserved						SS_SELECT	
R/W-000000b	R/W-000000b	R/W-000000b	R/W-000000b	R/W-000000b	R/W-000000b	R/W-0b	R/W-0b

Table 3-106 Additional Control External (ADD_CTRL_EX) Register Field Descriptions

Bit	Field	Type	Default	Description
7-2	Reserved	R	00000b	Reserved
1-0	SS_SELECT	R/W	00b	00b = spread spectrum mode($\pm 4\%$) 01b = spread spectrum mode ($\pm 12\%$) others = spread spectrum mode($\pm 4\%$)

3.5.1.2.37 Additional Open-Load Detect Control (ADD_OLD_CTRL_1) Register (Address = 0x33) [reset = 0x00]

Additional Open-Load Detect Control (ADD_OLD_CTRL_1) Register is shown in [Table 3-107](#) and described in [Table 3-108](#).

Table 3-107 Additional Open-Load Detect Control (ADD_OLD_CTRL_1) Register

7	6	5	4	3	2	1	0
HS8_NEG_OLD_EN	HS7_NEG_OLD_EN	HS6_NEG_OLD_EN	HS5_NEG_OLD_EN	HS4_NEG_OLD_EN	HS3_NEG_OLD_EN	HS2_NEG_OLD_EN	HS1_NEG_OLD_EN
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-108 Additional Open-Load Detect Control (ADD_OLD_CTRL_1) Register Field Descriptions

Bit	Field	Type	Default	Description
7	HS8_NEG_OLD_EN	R/W	0b	0b=Negative-current OLD on high-side 8 is disabled 1b= Negative-current OLD on high-side 8 is enabled
6	HS7_NEG_OLD_EN	R/W	0b	0b=Negative-current OLD on high-side 7 is disabled 1b= Negative-current OLD on high-side 7 is enabled
5	HS6_NEG_OLD_EN	R/W	0b	0b=Negative-current OLD on high-side 6 is disabled 1b= Negative-current OLD on high-side 6 is enabled
4	HS5_NEG_OLD_EN	R/W	0b	0b=Negative-current OLD on high-side 5 is disabled 1b= Negative-current OLD on high-side 5 is enabled
3	HS4_NEG_OLD_EN	R/W	0b	0b=Negative-current OLD on high-side 4 is disabled 1b= Negative-current OLD on high-side 4 is enabled
2	HS3_NEG_OLD_EN	R/W	0b	0b=Negative-current OLD on high-side 3 is disabled 1b= Negative-current OLD on high-side 3 is enabled
1	HS2_NEG_OLD_EN	R/W	0b	0b=Negative-current OLD on high-side 2 is disabled 1b= Negative-current OLD on high-side 2 is enabled
0	HS1_NEG_OLD_EN	R/W	0b	0b=Negative-current OLD on high-side 1 is disabled 1b= Negative-current OLD on high-side 1 is enabled

3.5.1.2.38 Additional Open-Load Detect Control (ADD_OLD_CTRL_2) Register (Address = 0x34) [reset = 0x00]

Additional Open-Load Detect Control (ADD_OLD_CTRL_2) Register is shown in Table 3-109 and described in Table 3-110.

Table 3-109 Additional Open-Load Detect Control (ADD_OLD_CTRL_12 Register

7	6	5	4	3	2	1	0
LS8_NEG_OLD_EN	LS7_NEG_OLD_EN	LS6_NEG_OLD_EN	LS5_NEG_OLD_EN	LS4_NEG_OLD_EN	LS3_NEG_OLD_EN	LS2_NEG_OLD_EN	LS1_NEG_OLD_EN
R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b	R/W-0b

Table 3-110 Additional Open-Load Detect Control (ADD_OLD_CTRL_2) Register Field Descriptions

Bit	Field	Type	Default	Description
7	LS8_NEG_OLD_EN	R/W	0b	0b=Negative-current OLD on low-side 8 is disabled 1b= Negative-current OLD on low-side 8 is enabled
6	LS7_NEG_OLD_EN	R/W	0b	0b=Negative-current OLD on low-side 7 is disabled 1b= Negative-current OLD on low-side 7 is enabled
5	LS6_NEG_OLD_EN	R/W	0b	0b=Negative-current OLD on low-side 6 is disabled 1b= Negative-current OLD on low-side 6 is enabled
4	LS5_NEG_OLD_EN	R/W	0b	0b=Negative-current OLD on low-side 5 is disabled 1b= Negative-current OLD on low-side 5 is enabled
3	LS4_NEG_OLD_EN	R/W	0b	0b=Negative-current OLD on low-side 4 is disabled 1b= Negative-current OLD on low-side 4 is enabled
2	LS3_NEG_OLD_EN	R/W	0b	0b=Negative-current OLD on low-side 3 is disabled 1b= Negative-current OLD on low-side 3 is enabled
1	LS2_NEG_OLD_EN	R/W	0b	0b=Negative-current OLD on low-side 2 is disabled 1b= Negative-current OLD on low-side 2 is enabled
0	LS1_NEG_OLD_EN	R/W	0b	0b=Negative-current OLD on low-side 1 is disabled 1b= Negative-current OLD on low-side 1 is enabled

4 Typical Application

4.1 Primary Application

The HE99MD9008 is primarily used for the control of multiple brushed DC motors which can be connected in independent-type, sequential-type or the parallel-type motor connection as shown in Figure 4-1.

An automotive battery powers the device to power supply pin (VM). A 3.3-V regulated power supply is generated for the supplying power to the digital core (VDD) of the device. A micro-controller is connected to the HE99MD9008 device via the SPI interface (4-lines) for control, configuration and diagnostics. The device operating or sleep state is controlled by the nSLEEP pin and nFAULT pins is used as an additional hardware diagnostics.

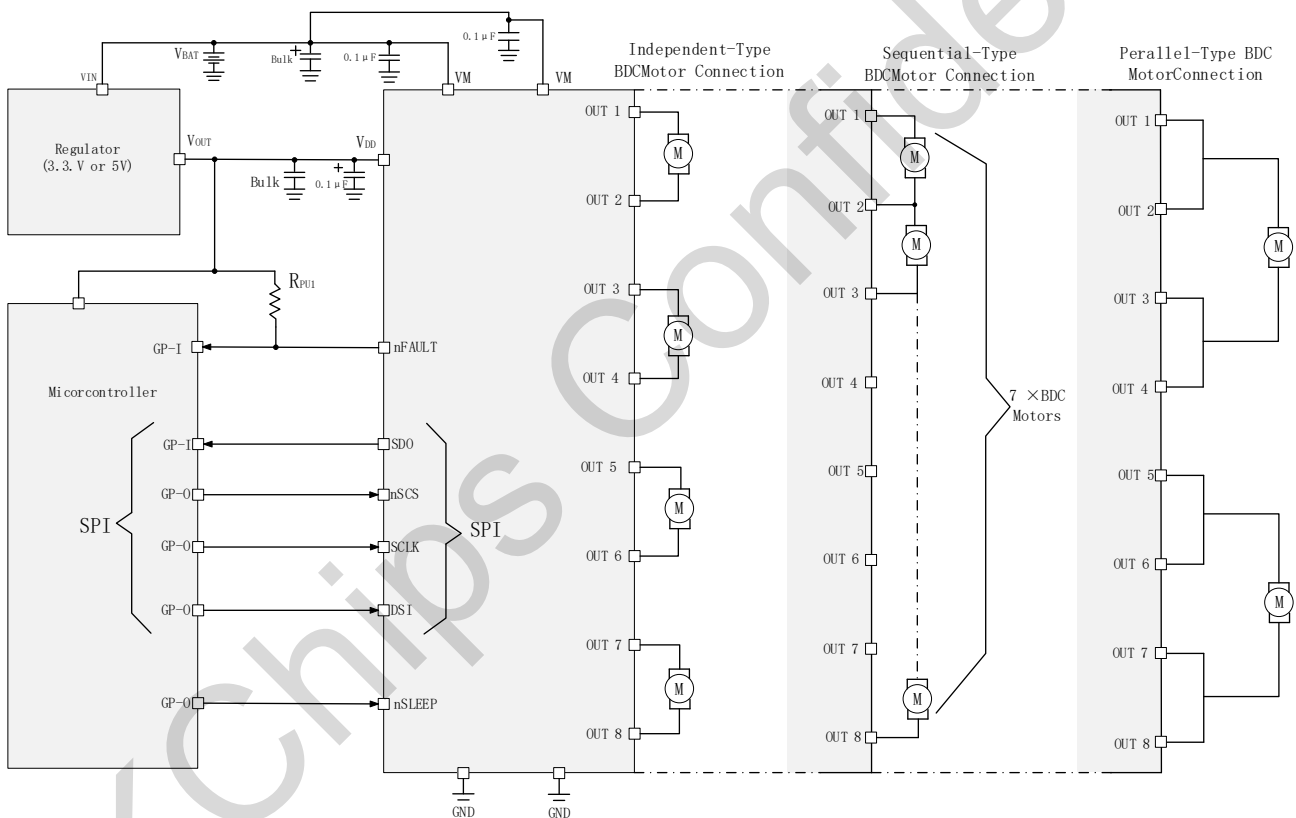


Figure 4-1 Primary Application Schematic (Automotive HVAC Application)

4.1.1 Design Requirements

Table 4-1 lists example input parameters for the system design.

Table 4-1 Design Parameters

DESIGN PARAMETERS	REFERENCE	EXAMPLE VALUE
Supply voltage	V _{VM}	13.5-V
Supply digital voltage	V _{VDD}	3.3-V
Number of motor connected	N	6 motors
Number of motor operating in normal operation	N _F	4 motors
Number of motor operating in stall condition	N _S	2 motors
Motor RMS current	I _{RMS}	200-mA
Motor peak current	I _{PEAK}	800-mA
Motor resistance	R _{MOTOR}	16.9-Ω
Motor inductance	L _{MOTOR}	10-mH
PWM Frequency	f _{PWM}	2-kHz (Internal)
Rise and fall time for continuous mode (SR = 0)	t _{RISE_CONT} , t _{FALL_CONT}	22.5-μs
Rise and fall time for PWM mode (SR = 1)	t _{RISE_PWM} , t _{FALL_PWM}	5.4-μs

4.1.2 Detailed Design Procedure

The design procedure includes the selection of motor current rating the power dissipation to meet the desired thermal performance.

4.1.2.1 Motor Current Rating

Motor specification selection is the most importance criteria for the design. Each half-bridge (OUTx) of the HE99MD9008 device is designed to handle RMS current of 1 A and the peak current is limited by the minimum over-current (OCP) limit of 1.3-A. Therefore, a motor with peak starting current higher than 1.3-A is expected to hit OCP limit. For higher peak current motors (starting current higher than 1.3-A), following methods can be implemented:

1. Current Chopping: During starting, if supply voltage is connected directly to the motor, then due to low back-emf (when speed is zero or low), a huge peak current is demanded by the motor. This peak current is only limited by the motor's winding resistance (R_{MOTOR}). This peak current of motor can be limited by starting the motor with low-duty PWM switching operation and then gradually increasing (duty-ramping) the duty with speed to 100% PWM operation (equivalent to motor operating in continuous mode). This duty-ramping provides enough time to ramp motor speed and build sufficient back-emf which limits the peak current. The HE99MD9008 device implements a 2-kHz PWM switching operation which is suitable for the HVAC damper motors.

2. OCP Deglitch Time Adjustments: This method is applicable if the motor inertia is low and the motor can quickly pick up the speed. For this method, the motor starting current should settle to lower than minimum over-current limit (I_{OCP}) before OCP deglitch time (t_{OCP}) is over. The device provides multiple (8 settings) OCP deglitch time settings with a default deglitch time of 10-μs and can be increased to a maximum value of 60-μs.

Typical Application

Note: For multiple motor connection, it has to be ensured that the total device current should be lower than the maximum current-carrying capability of the power-supply (VM/GND) pins i.e. 6-A (maximum).

4.2 Alternative Application

The HE99MD9008 can alternatively be used for the mirrors targeting the mirror-fold, mirror x-y direction control and side indicator LED's as shown in [Figure 4-2](#).

The half-bridges are connected in parallel to support the higher current requirement of the mirror fold application. Whereas, single half-bridges can be used for driving the low-current motors used for the mirror X and Y positioning. Moreover, the LED's used in side indicators, puddle lamp is lower current which can be easily driven by single half-bridges.

The driver is powered by the automotive battery with a 3.3-V regulated power supply generated for the supplying power to the digital pin (VDD). A micro-controller is connected to the HE99MD9008 device with the SPI interface (4-lines) for control, configuration and diagnostics. The device operating or sleep state is controlled by the nSLEEP pin and nFAULT pins is used as an additional hardware diagnostics.

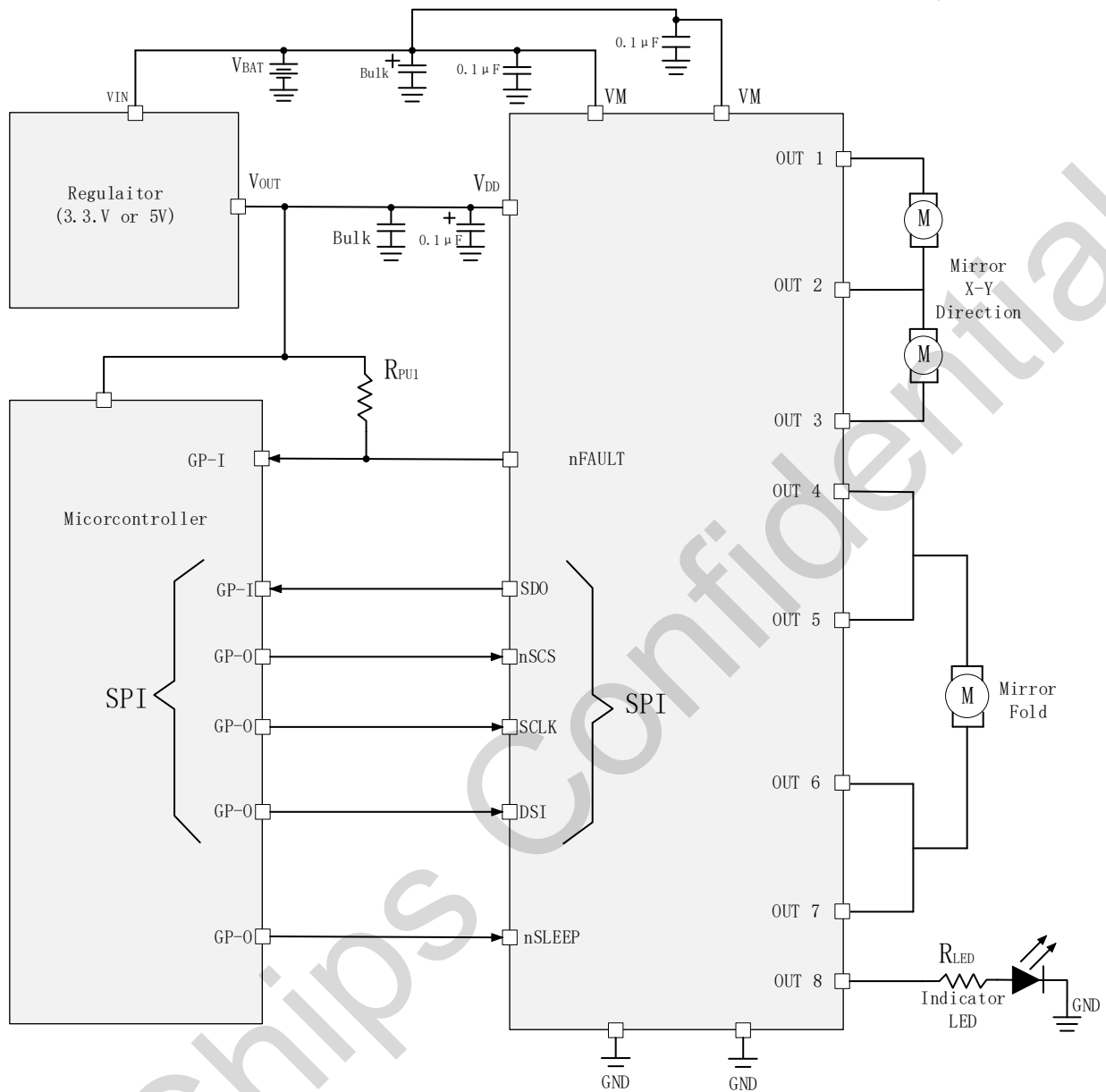
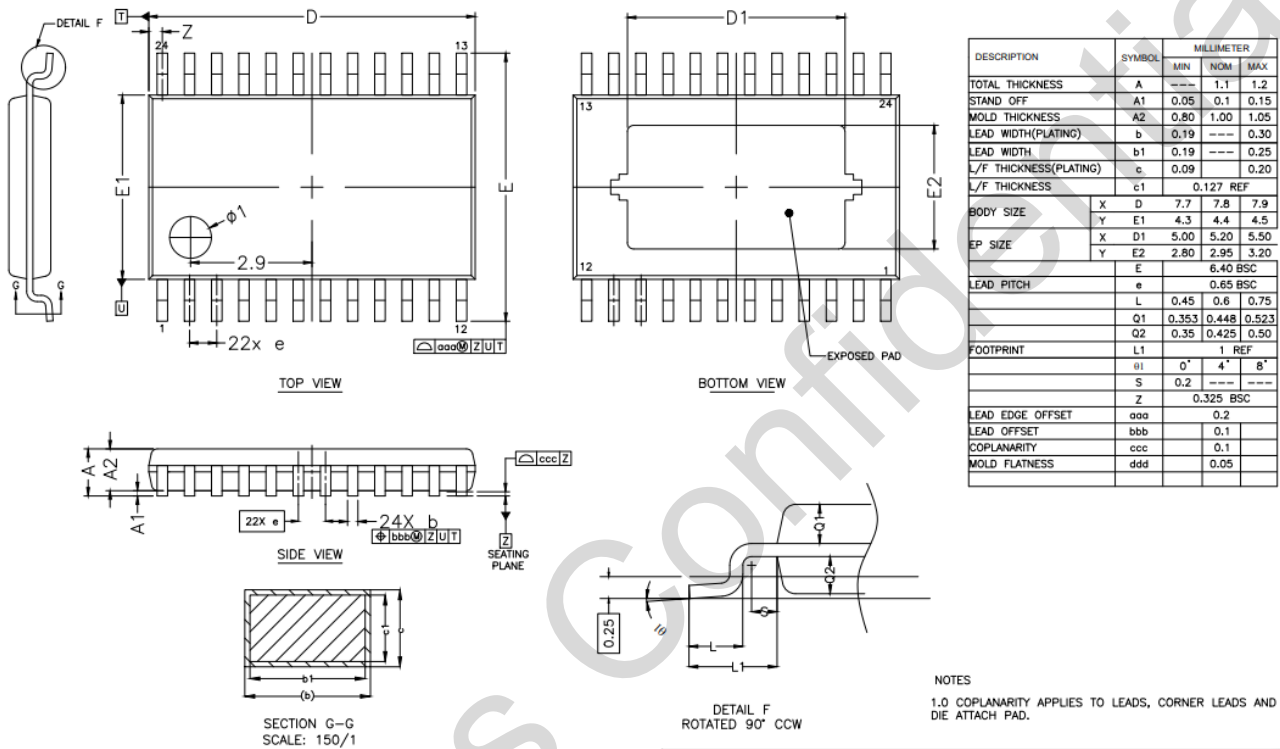
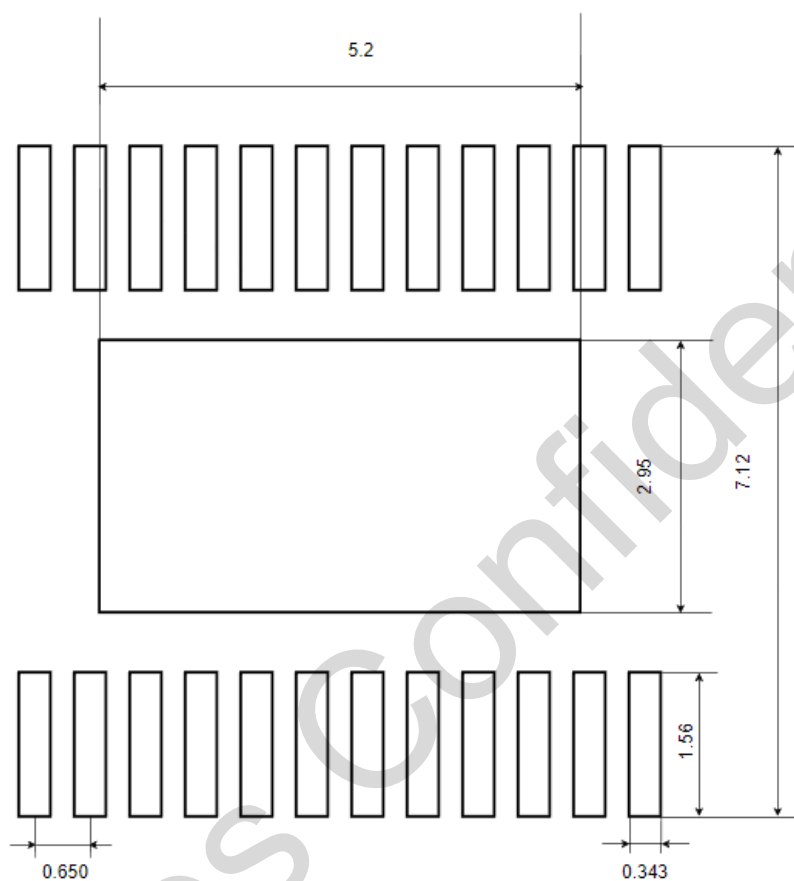


Figure 4-2 Alternative Application Schematic (Automotive Side-Mirror Application)

5 Package and information

5.1 ETSSOP mechanical data



6 Recommended pad size

Unit: mm

Figure 6-1 Recommended pad size

7 Customer Service

KKChips

9th Floor, Grandtek Building B2, No. 18 Kexue Avenue, Huangpu District, Guangzhou.

Revision history

Date	Revision	Changes
2023/7/6	1.0	1 . Product model change
2023/07/31	2.0	1. Format updated to 2.0
2024/6/11	1.0	1 . Add recommended pad size. 2 . Format updated to 1.0.0
2024/8/16	1.1	1 . Update part number
2024/12/6	1.2	1 . Format updated to 1.2
2025/4/7	1.3	1. Optimized parameters. 2. Optimized description in register. 3. Add application instructions. 4. Format updated to 1.3.
2025/9/12	1.4	1. parameter optimization. 2. Update of thermal resistance coefficient. 3. Typical Characteristics Chart Update. 4. Optimize the description of the spread spectrum function.